



Observatory
of Support
for Entrepreneurship

Region
Gulf

Chapter
Qatar

Qatar's Entrepreneurial Ecosystem

"What gets measured
gets managed"

Peter Drucker

2024 Report

Structural and Configurational Analysis

National Support



**QATAR SCIENCE &
TECHNOLOGY PARK**

وزارة الاتصالات وتكنولوجيا المعلومات
Ministry of Communications and Information Technology
دولة قطر • State of Qatar



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Edited by **ADALIA** Institute



Special Issue Report

1. Executive Summary

1.1. Why It Matters for Policymakers

This report provides a comprehensive analysis of Qatar's entrepreneurial ecosystem for 2018–2023, offering actionable insights to policymakers for fostering innovation, inclusivity, and economic growth. The study evaluates connectivity, resource distribution, actor roles, and collaboration dynamics while identifying key gaps and opportunities. By addressing regional disparities, enhancing sectoral diversity, and strengthening synergies among stakeholders, policymakers can create a more resilient and globally competitive ecosystem. The findings align with Qatar's Vision 2030, supporting economic diversification, innovation-driven growth, and sustainable entrepreneurship.

1.2. Key Findings: Actionable Insights and Strategic Highlights

Qatar's ecosystem showcases strong leadership, connectivity, and collaboration, driven by key actors the Qatar Foundation, MCIT and QDB. Financing and innovation hubs play vital roles, but disparities in regional development and underrepresented sectors (e.g., FinTech, AgriTech) constrain inclusivity. Limited academia-investor synergy and low redundancy in critical services (e.g., humanboost) hinder ecosystem optimization. Despite these gaps, the ecosystem fosters robust entrepreneurial opportunities, leveraging strong governance, resource availability, and innovative frameworks to drive sustainable economic growth.

1.3. Strategic Recommendations: Guiding the Path Forward

Policymakers should focus on enhancing inclusivity by investing in underrepresented cities like Lusail and Al-Khor. Support niche sectors through targeted incentives and cross-sector partnerships. Bridge gaps between academia and investors to commercialize research and strengthen funding for startups. Increase redundancy in underrepresented services and optimize resource distribution to foster resilience. Develop digital platforms to improve connectivity and resource access. By implementing these strategies, Qatar can achieve a dynamic, innovative, and globally competitive entrepreneurial ecosystem, aligned with its long-term vision.



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2. Opening Credits

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Our heartfelt thanks go to the report team for their meticulous effort and to our strategic partners—Ministry of Communications and Information Technology State Qatar, HEC Paris Doha, Qatar University, ADALIA Institute, Invest Qatar, QSTP, the ETI Chair at Sorbonne University, the Academy of Entrepreneurship and Innovation, ICSB, and Gitex Africa—whose collaboration and shared vision have been critical in driving this initiative forward.

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Finally, we honor Dr. Grégory Guéneau, founder and president of OSE in Africa and the author of the OSE methodology, whose visionary leadership has been the cornerstone of this endeavor. This report is a testament to his unwavering commitment to fostering entrepreneurship and innovation in Africa and beyond. Thank you all for making this achievement possible.

2.3. Partners of This Report

Strategic Partners

وزارة الاتصالات وتكنولوجيا المعلومات
Ministry of Communications and Information Technology
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This report offers significant value to Qatar's Ministry of Communications and Information Technology (MCIT) as it provides a data-driven analysis of the entrepreneurial ecosystem, highlighting opportunities and gaps across various communities. By leveraging advanced methodologies such as VRIO, Louvain, and Infomap, the report identifies key clusters and strategic actors, enabling MCIT to understand which sectors and communities contribute most to innovation, digital transformation, and economic diversification.

The insights are particularly relevant for MCIT's goals of fostering a knowledge-based economy under Qatar National Vision 2030. The detailed evaluations of value, rarity, and inimitability of resources help prioritize investments in digital infrastructure and innovation hubs. Moreover, the ecosystem's structural and functional analyses provide actionable recommendations to strengthen inter-organizational collaborations, encourage resource-sharing, and optimize policy frameworks.

By aligning these insights with MCIT's strategic priorities, the report ensures informed decision-making, boosting Qatar's global competitiveness in the communications and information technology sectors.

HEC Paris, Doha has a vested interest in becoming a strategic partner of the OSE's report on the Qatari Entrepreneurial Ecosystem due to its commitment to fostering

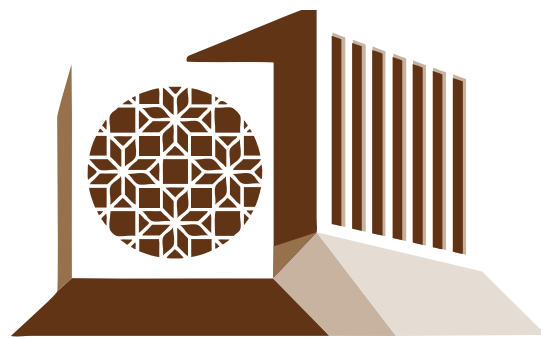


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innovation and leadership in Qatar's rapidly evolving economic landscape. As a leading academic institution with a strong focus on executive education and entrepreneurship, HEC Paris aligns perfectly with the goals of the OSE report, which emphasizes ecosystem dynamics, actor interconnectivity, and entrepreneurial opportunities. Partnering with OSE offers HEC Paris unparalleled insights into ecosystem metrics, enabling it to tailor its programs to meet the real-world needs of entrepreneurs and policymakers. Furthermore, this collaboration strengthens HEC Paris' position as a thought

leader, reinforcing its role in shaping Qatar's entrepreneurial future. By contributing to the report, HEC Paris enhances its academic impact, supports sustainable development, and bolsters its network within Qatar's entrepreneurial community, creating value for stakeholders across the ecosystem.

Qatar University (QU) has a strategic interest in partnering with OSE's report on the Qatari Entrepreneurial Ecosystem due to its pivotal role as a leading academic



جامعة قطر
QATAR UNIVERSITY

institution and research hub in Qatar. As the national university, QU is deeply committed to advancing Qatar's Vision 2030 goals of economic diversification and sustainable development. Collaborating with OSE enables QU to leverage cutting-edge insights into ecosystem dynamics, fostering a better understanding of entrepreneurial opportunities, resource distribution, and innovation gaps. This partnership would enhance QU's ability to align its research and educational programs with the needs of the entrepreneurial ecosystem, producing graduates

who are well-equipped to drive economic growth. Moreover, this collaboration reinforces QU's leadership in shaping evidence-based policy recommendations and strengthening ties with key stakeholders in government, industry, and academia. By contributing to this report, Qatar University solidifies its position as a cornerstone of Qatar's knowledge economy and a catalyst for entrepreneurial development.

Scientific Partners



ADALIA Institute | School of Business and Management, is a ranked business school located in Casablanca, Morocco. ADALIA Institute is the funding organization of OSE in Africa and provide resources and support to OSE in Africa



The ETI Chair at Paris 1 Sorbonne is a scientific partner of the Observatory of Support for Entrepreneurship (OSE) in Africa. This collaboration focuses on advancing research in entrepreneurship, innovation, and economic development, providing valuable insights and frameworks that inform policy-making and enhance the effectiveness of African entrepreneurial ecosystems.



The Académie de l'Entrepreneuriat et de l'Innovation (AEI) collaborates with the Observatory of Support for Entrepreneurship (OSE) in Africa as a scientific partner. This partnership focuses on advancing research and education in entrepreneurship and innovation, aiming to enhance entrepreneurial capacities and drive sustainable economic development within the region.

Support Partners



This ICSB-supported report strategically benefits Qatar by offering a globally recognized framework for fostering entrepreneurship. It emphasizes actionable insights into ecosystem dynamics, resource optimization, and policy alignment. By integrating global best practices with localized analysis, it empowers stakeholders to drive innovation, support SMEs, and strengthen Qatar's entrepreneurial competitiveness internationally.



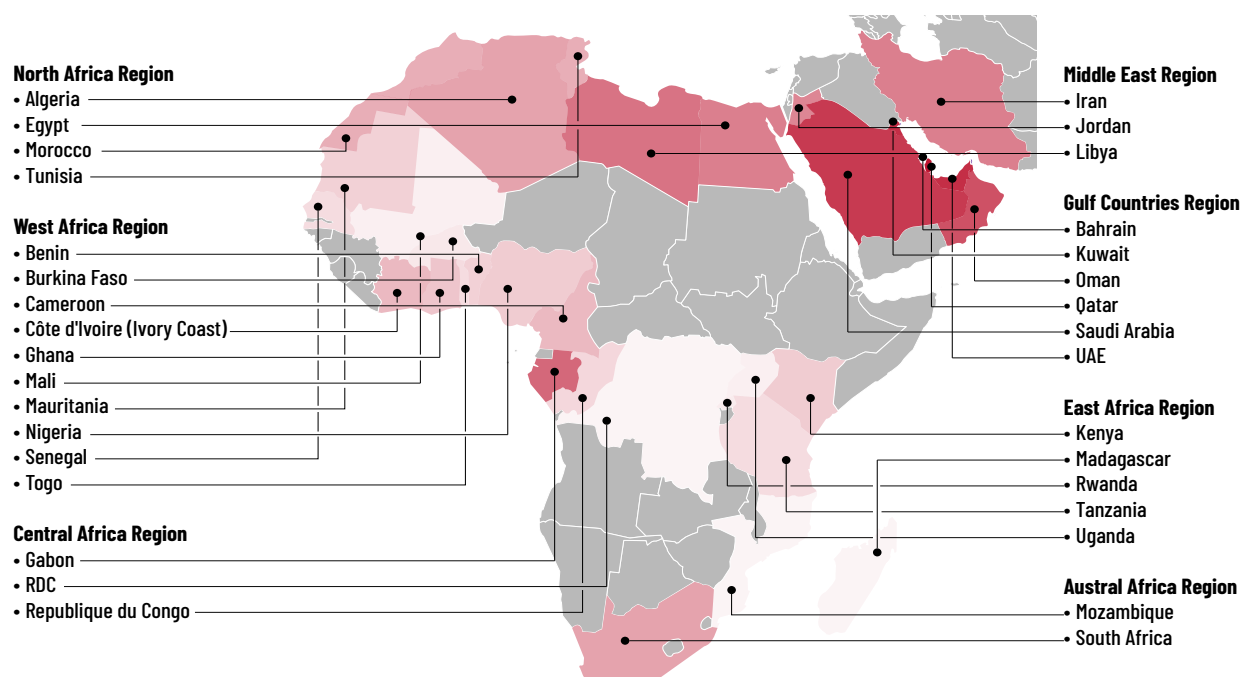
Supported by GITEX Africa, this report highlights Qatar’s entrepreneurial ecosystem within a global technology and innovation context. It fosters strategic partnerships, showcases Qatar’s progress in fostering innovation, and aligns with Africa’s tech-driven economic growth. This collaboration enhances Qatar’s visibility, strengthens cross-regional ties, and attracts investment and talent to its ecosystem.

3. OSE in Qatar

3.1. OSE in Africa: Overview and Vision

We are dedicated to investigating and developing African entrepreneurial ecosystems through a unique technique that combines the network method with deep learning.

Figure 1: OSE in Africa Report editions in 33 countries



Our objective is to conduct a thorough examination of the mechanisms that drive entrepreneurial development on the African continent. To that purpose, we have chosen 33 national entrepreneurial ecosystems that reflect Africa's diversity and depth. We map these ecosystems' connections, interactions, communities, and flows in order to better understand the success factors and problems that entrepreneurs face.

OSE in Africa covers 33 countries (figure 1.), separated in 7 regions :

Austral Africa Chapter including Mozambique and South Africa

Austral African countries are integrated through regional bodies like the Southern African Development Community (SADC). They share many common economic and development issues.

Central Africa Chapter including Gabon, RDC, République du Congo

Central African countries often face similar challenges related to economic development and entrepreneurship, and there are efforts to integrate economically and politically within the region.

East Africa Chapter including Kenya, Madagascar, Rwanda, Tanzania, Uganda

East African countries are often grouped together due to their regional integration efforts, such as the East African Community (EAC). They share similar issues related to trade, development, and entrepreneurial activities.

Gulf Countries Chapter including , Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, United Arab Emirates (UAE)

Although the Gulf countries are not part of Africa, including them in the analysis provides valuable insights into their role as major economic partners and investors in the African entrepreneurial ecosystem. Their economic influence, investments, and development aid in Africa make them relevant to understanding broader regional dynamics and opportunities. Moreover, many African countries have significant economic and trade relationships with Gulf nations, influencing entrepreneurial activities and opportunities.

Middle East Chapter including Iran, Jordan and Libya

The Middle East Chapter includes countries that have unique geopolitical and economic interactions with both Africa and the Gulf region. Iran, Jordan, and Libya are strategically important in regional dynamics, and their inclusion helps to understand their influence on African entrepreneurial ecosystems. Iran's regional influence and trade relations, Jordan's role as a trade and investment hub, and Libya's historical connections with neighboring African countries provide a comprehensive view of how these countries impact and interact with the African entrepreneurial landscape. This chapter helps to explore cross-regional connections and opportunities that might not be captured within the confines of strictly African or Gulf-focused analyses.

North Africa Chapter including Algeria, Egypt, Morocco, Tunisia

These countries share historical, cultural, and economic ties, and their entrepreneurial ecosystems often have similar challenges and opportunities due to their geographic proximity and similar socio-economic contexts.

West Africa Chapter including : Benin, Burkina Faso, Cameroon, Côte d'Ivoire (Ivory Coast), Ghana, Mali, Mauritania, Nigeria, Senegal, Togo

West African countries generally share linguistic, cultural, and economic characteristics. Many of them are part of the West African Economic and Monetary Union (UEMOA) and have similar entrepreneurial dynamics and challenges.

The segmentation of OSE in Africa into seven chapters—Austral Africa, Central Africa, East Africa, Gulf Countries, Middle East, North Africa, and West Africa—provides a comprehensive framework for understanding the diverse and interconnected entrepreneurial ecosystems across these regions.

Scientific Committee of OSE in Africa

At the heart of OSE in Africa, including for its Qatar's chapter, lies a prestigious Scientific Committee, composed of leading scholars and practitioners renowned for their expertise in entrepreneurship and economic development. This committee is dedicated to guiding our research initiatives and ensuring the highest standards of academic rigor.



David B. Audretsch

A distinguished professor and authority in entrepreneurship, David has made significant contributions to understanding the interplay between innovation, entrepreneurship, and economic growth. His work emphasizes the critical role of entrepreneurial ecosystems in shaping competitive advantages.



Erik Stam

As a prominent researcher in the field of entrepreneurship, Erik focuses on the dynamics of entrepreneurial ecosystems and their impact on economic performance. His insights into policy frameworks and ecosystem development are instrumental in shaping our strategies.



Saras Sarasvathy

An influential figure in entrepreneurship education and research, Saras is best known for her groundbreaking work on effectuation theory, which explores how entrepreneurs navigate uncertainty. Her perspective is vital for understanding entrepreneurial decision-making processes.



Didier Chabaud

With extensive experience in economic development and entrepreneurship, Didier brings a wealth of knowledge in public policy and the fostering of entrepreneurial initiatives. His expertise helps align our objectives with broader economic goals.



Karim Messeghem

A leading academic in entrepreneurship, Karim specializes in innovation and technology management. His research provides critical insights into how technological advancements can be leveraged to enhance entrepreneurial success across regions.

Together, this esteemed Scientific Committee ensures that OSE in Qatar not only advances the understanding of entrepreneurial ecosystems but also contributes to meaningful and impactful developments in the region. Their collective expertise strengthens our mission to foster innovation and entrepreneurship in Qatar and beyond.

3.3. OSE in Gulf: Regional Focus and Contributions

Entrepreneurship-led studies are vital in the Gulf region, where economic diversification and innovation are strategic priorities. Understanding the entrepreneurial ecosystem's value chains provides insights into how startups, SMEs, and large enterprises interconnect to drive regional growth. These studies illuminate key drivers such as funding access, talent development, market opportunities, and regulatory frameworks.

In the Gulf, with its distinct mix of global aspirations and local traditions, identifying the strengths and gaps within value chains helps optimize resource allocation and foster innovation clusters. Such research enables policymakers and stakeholders to align initiatives with economic visions, like Qatar's Vision 2030 or Saudi Arabia's Vision 2030, to create globally competitive ecosystems.

By mapping and analyzing these networks, OSE contributes to building resilient, scalable, and sustainable entrepreneurial landscapes that position the Gulf as a hub for innovation and economic leadership.

3.2. OSE in Qatar: National Engagement and Impact

Qatar occupies a pivotal role in the Gulf as a hub for innovation, economic diversification, and entrepreneurship. Its strategic location, substantial natural resources, and forward-looking policies have positioned it as a key player in the region's entrepreneurial landscape. Qatar's Vision 2030 emphasizes knowledge-based economic growth, fostering initiatives to support startups, SMEs, and technological innovation.

In this context, a detailed report on Qatar's entrepreneurial ecosystem is crucial. By analyzing its health, structure, configuration, and strengths and weaknesses, the report provides actionable insights for policymakers, investors, and entrepreneurs. It identifies opportunities for innovation, gaps in support systems, and potential areas for international collaboration. Moreover, it highlights Qatar's readiness to adapt to global economic trends and strengthens its regional leadership in entrepreneurship.

Such an analysis ensures the alignment of efforts with national and Gulf-wide economic goals, enhancing Qatar's role as a model for sustainable entrepreneurial ecosystems in the Middle East.

4. Introduction

4.1. Purpose of the Report: Framing the Policy Context

This first chapter lays the groundwork for a thorough examination of Africa's entrepreneurial ecosystem, with a particular focus using the Observatory of Support for Entrepreneurship (OSE) dataset. In this document we present the vision, ambition, missions, and values that OSE represents in Africa, highlighting its critical role in the transformation and development of entrepreneurial ecosystems on the continent. By examining the societal, economic, and practical issues that motivate this study, we highlight the fundamental issues to which OSE seeks to respond in order to stimulate creativity, collaboration, and long-term growth.

Furthermore, we describe the research methodology that will lead this study. We demonstrate the multidimensional approach used to understand entrepreneurial ecosystems in their whole by emphasizing the relevance of structural and configurational analysis, analysis of support communities, and analysis of activity topics. This integrated approach provides a holistic perspective, allowing for the understanding of complex inter-actor interactions, collaborative dynamics, emerging trends, and innovation opportunities.

This introductory chapter established the conceptual and methodological framework necessary for our study, laying the groundwork for a thorough and comprehensive analysis of African entrepreneurial ecosystems, highlighting the importance of an integrated approach to understanding the complexities of these dynamics.

4.2. Scope and Methodology: Simplified Overview

Introduction to the Methodology of OSE in Africa

The research methodology employed by the Observatory of Support for Entrepreneurship (OSE) combines network theory, Natural Language Processing (NLP), and advanced ecosystem diagnostics. This innovative approach is unique in its application to entrepreneurial ecosystems, offering unprecedented insights into the dynamics of

entrepreneurship in Africa. First scientifically validated and published in 2021, this method is systematically deployed across the continent, establishing OSE as a global leader in entrepreneurial ecosystem analysis. No other research institution has matched the breadth, depth, and longitudinal scope of our studies, which enable policymakers, researchers, and practitioners to access and leverage actionable data for ecosystem development.

4.2.1. Structural and Configurational Analysis

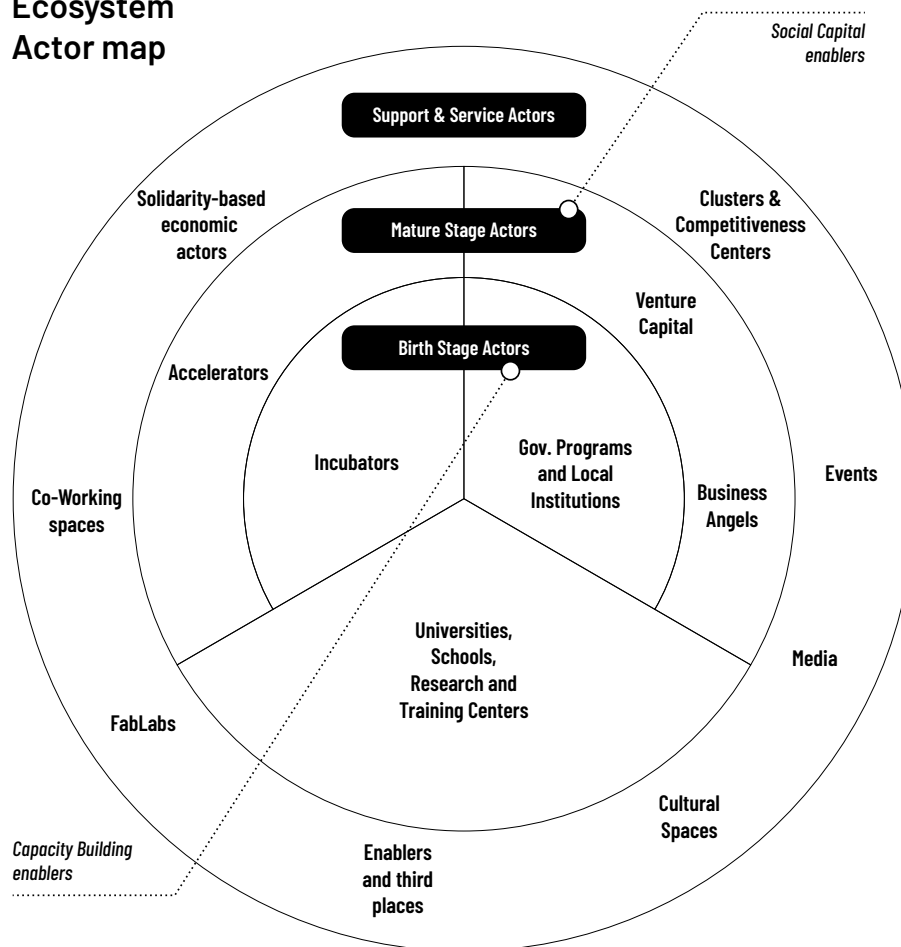
Interorganizational Networks: Unveiling Ecosystem Dynamics

Interorganizational network analysis is a cornerstone of our methodology. This approach uncovers the structure, interactions, and dynamics among actors (figure 2) within entrepreneurial ecosystems. By analyzing network structures, we:

- 1. Identify Key Roles and Intermediaries:** Understanding the roles of actors, from central connectors to peripheral players, helps in mapping critical pathways of influence and collaboration.
- 2. Measure Connectivity and Cohesion:** Metrics such as network density, proximity centrality, and eigenvector centrality reveal how actors share resources, knowledge, and opportunities, fostering innovation and growth.
- 3. Evaluating Resource Complementarities :** A “value chain” framework of measure to understand how actors align their resources in the ecosystem to deliver value to entrepreneurs.
- 4. Enhance Strategic Decision-Making:** Insights from network structures guide stakeholders in forming partnerships, optimizing resource allocation, and driving systemic improvements.

Figure 2: Mapping the Types of Actors in Entrepreneurial Ecosystems

Entrepreneurial Ecosystem Actor map



Metrics and Indicators

Key network metrics provide actionable insights:

Network Density: Reflects the overall level of connectivity, indicating collaboration potential.

Proximity Centrality: Highlights actors with quick access to resources and information.

Eigenvector Centrality: Identifies influential actors with strong connections to other key players.

Clustering Coefficient: Measures the degree of actor grouping, signifying niche or cluster formation.

These metrics are not only descriptive but also diagnostic, helping to pinpoint areas where the ecosystem can improve inclusivity, efficiency, and resilience.

4.2.2. Community Analysis Using the Louvain Method

The Louvain method enables the detection of natural communities within networks, providing a granular view of collaboration dynamics:

Cluster Identification: Reveals groups of actors with shared interests, goals, or characteristics.

Interaction Insights: Maps intra- and inter-community relationships, highlighting potential synergies and collaboration opportunities.

Resource Targeting: Facilitates precise allocation of support programs to community specific needs.

Leadership Identification: Pinpoints key influencers and decision-makers within communities.

This community-focused analysis helps in crafting tailored strategies to strengthen ecosystem cohesion and foster innovation.

4.2.3. Topic and Thematic Analysis with NLP

By employing advanced NLP techniques, including deep learning models such as Latent Dirichlet Allocation (LDA) and Non-Negative Matrix Factorization (NMF), OSE extracts meaningful themes and trends from vast textual datasets. These techniques are used to:

Discover Emerging Topics: Identify key themes driving ecosystem conversations, such as funding, policy, or technology trends.

Understand Actor Priorities: Decode the concerns and focus areas of ecosystem participants.

Guide Strategic Focus: Inform decision-makers on actionable insights based on real-time data.

NLP-driven topic modeling bridges the gap between qualitative ecosystem narratives and quantitative analyses, offering a balanced and data-rich perspective.

4.2.4. Integrated Multidimensional Analysis

The integration of structural, community, and thematic analyses provides a holistic understanding of entrepreneurial ecosystems. This multidimensional approach enables:

Synergy Across Dimensions: Correlating network structures, community dynamics, and thematic trends to uncover deeper insights.

Balanced Diagnostics: Capturing interactions, resource flows, and innovation hotspots in a unified framework.

Comprehensive Policy Recommendations: Informing stakeholders with evidence-based strategies that address the unique characteristics of their ecosystems.

4.2.5. Resource-based Ecosystem Diagnostics

The updated methodology further incorporates:

Actor Network Functionality: Evaluating roles in resource flows and specialization, ensuring balanced development.

Territorial Analysis: Addressing regional disparities by mapping actor distributions and identifying gaps.

Leadership and Keystone Nodes: Highlighting influential actors and ensuring ecosystem stability through robust centrality measures.

Dynamic and Functional Measures: Assessing systemic dependencies, interaction strength, and redundancy gaps to enhance ecosystem robustness.

4.2.6. Conclusion

OSE's methodology represents a significant leap forward in understanding and shaping entrepreneurial ecosystems. By combining cutting-edge techniques with a long-term, systematic approach, this framework not only generates unparalleled insights but also establishes a roadmap for actionable development. Through structural analysis, community mapping, and thematic exploration, OSE delivers tools and recommendations that empower Africa's entrepreneurial ecosystems to thrive on a global scale.

4.3. Key Concepts and Definitions: Breaking Down Complexity

Table 1: key concepts and definitions

Term	Definition
PESTEL Analysis	A framework analyzing Political, Economic, Social, Technological, Environmental, and Legal factors to assess external influences on entrepreneurial ecosystems.
Nodes	Entities belonging to an entrepreneurial ecosystem.
Ties	Connections or relationships between nodes, representing collaboration, communication, or resource flow.
Average Weight	The mean strength of ties in a network, indicating the average intensity of relationships between actors.

Term	Definition
Graph Density	A measure of how interconnected actors are in a graph, calculated as the ratio of actual ties to possible ties.
Network Density	A similar metric to graph density, reflecting the overall level of connectivity in an entrepreneurial network.
Proximity Centrality	A measure of how close an actor is to all others, indicating its ability to quickly access resources or information.
Eigenvector Centrality	A metric assigning importance to actors based on their connections to other influential actors, identifying key players in the ecosystem.
Clustering Coefficient	The degree to which actors form tightly-knit clusters, representing local collaboration or niche groups in the ecosystem.
Average Shortest Path	The average number of steps needed to connect any two actors, reflecting network efficiency and accessibility.
Average Length	Synonymous with Average Shortest Path in the context of network analysis.
Network Diameter	The longest shortest path between any two actors, representing the maximum distance within the network.
Network Eccentricity	The maximum distance from an actor to all others, reflecting how far an actor is from the least accessible parts of the network.
Modularity	A measure of the structure of a network, indicating how well it divides into distinct communities or modules.
Degree Assortativity	A measure of the tendency for actors to connect with others that have similar degrees, indicating network homophily or diversity.
Louvain Communities: Cluster	Groups of actors identified using the Louvain algorithm, highlighting areas of dense interconnectivity and shared activity.
Louvain Communities: Infomap	A clustering method focusing on optimizing information flow within communities, identifying similar clusters as Louvain but with different emphasis.
Community Size, Density, and Activity	Metrics analyzing the number of actors, interconnectedness, and level of engagement within identified communities.
NLP	Natural Language Processing techniques analyze textual data to extract themes, trends, and insights about entrepreneurial ecosystems.
Alluvial Diagram	A visualization showing flows and changes in categories over time, often used to map transitions in actor roles or resources.

Term	Definition
Heatmap	A graphical representation of data using color gradients to show intensity, often used for performance metrics or resource distribution.
MutualInfo	A measure of shared information or dependencies between variables or entities, reflecting correlations or relationships in the ecosystem.
Keystone Nodes	Critical actors that hold the network together by maintaining connectivity and enabling resource flow.
Structural Accessibility Index	A metric evaluating the inclusivity and reachability of resources and opportunities in the ecosystem.
Dendrogram	A tree-like diagram used to show the hierarchical clustering of actors, revealing nested community structures.
Synergy Index	A metric assessing the combined effectiveness of collaborations or resource integrations, highlighting areas of high collective impact.
Topic Analysis	Identifying and categorizing key themes within textual data using statistical and machine learning techniques.
Thematic Analysis	A qualitative method for identifying and interpreting patterns or themes in textual data, providing contextual ecosystem insights.
Trophic Levels	Layers of roles in the network based on resource dependencies, inspired by ecological food chains but applied to ecosystem actors.
Coverage	A measure of the completeness of resources or actors represented in the network, indicating gaps or overlaps.
Flow Diversity	The variety and distribution of resource or information flows within the ecosystem, highlighting adaptability and inclusiveness.
Interaction Strength	The intensity of relationships or collaborations between actors, reflecting the robustness of connections.
Risk Diagram	A visual representation of vulnerabilities or constraints in resource or information flows, identifying critical points of failure.
Sankey Diagram	A flow diagram showing the distribution and magnitude of resources, relationships, or activity across different ecosystem components.
Chord Plot	A circular diagram illustrating connections and interactions between actors or categories, emphasizing relational strength.

Term	Definition
Cohesiveness	The degree to which actors in the ecosystem are interconnected, promoting collaboration and shared goals.
Interconnectedness	The extent of linkages between actors in the ecosystem, fostering exchange of resources, knowledge, and opportunities.
Proximity	A measure of closeness between actors or resources, indicating ease of access and interaction within the ecosystem.

5. Country Context

This chapter explores Qatar's entrepreneurial landscape from 2018 to 2023. By examining the ecosystem during this pivotal period, we emphasize the importance of network analysis in understanding the mechanisms that have fostered business development and entrepreneurial support. Additionally, we identify key players within Qatar's entrepreneurial ecosystem, setting the stage for a deeper exploration in subsequent chapters. The Observatory of Support for Entrepreneurship (OSE) in Africa highlights Qatar as a unique case, given its significant economic development alongside a relatively low entrepreneurial dynamism, as reflected in findings from the *Global Entrepreneurship Monitor*.

5.1. Macro-Level Challenges in Qatar

From 2018 until 2023, the context

This section explores the dynamic developments and strategic initiatives that shaped Qatar's entrepreneurial environment from 2018 to 2023, illustrating the country's ongoing commitment to fostering a robust and innovative business ecosystem.

2018: Implementing Diversification and Technological Integration

In 2018, Qatar solidified its approach to economic diversification away from hydrocarbon dependence by implementing policies that enhance technological integration and infrastructural development. Laws such as the excise tax (Law No. 25) and income tax (Law No. 24) were notable for streamlining the fiscal framework to support business ventures and entrepreneurial activities, as part of Qatar's Vision 2030 to foster a diversified economy (Alkhateeb, 2019).

2019 : Global Collaboration and Regulatory Enhancements

Qatar expanded its international collaborations and continued to improve its regulatory environment in 2019. Initiatives included new laws to facilitate foreign investment and enhance the legal framework for trade and entrepreneurship, notably through the establishment of the Mohammed bin Ghanem Al-Ghanim Maritime Academy, which further positioned Qatar as a favorable destination for global business operations (World Development Indicators, 2024).

2020 : Strategic Adjustments Amidst Global Challenges

In response to global economic disruptions, 2020 saw Qatar making strategic adjustments to support its economy and entrepreneurial sector. Laws aimed at improving human resources management and enhancing business operational flexibility played a critical

role in maintaining economic stability and fostering a resilient entrepreneurial ecosystem during this turbulent period (World Development Indicators, 2024).

2021 : Strengthening Cybersecurity and Dispute Resolution

The formation of the National Cybersecurity Agency and the establishment of the Investment and Trade Court in 2021 marked significant enhancements in protecting digital transactions and resolving business disputes efficiently. These advancements ensured a secure and supportive environment for entrepreneurs to innovate and grow (Yosri, 2021).

2022 : Emphasizing Sustainable Development and Economic Diversification

Continuing its commitment to sustainable development, 2022 was characterized by legislative actions to enhance environmental protections and manage economic diversification. Laws to regulate national product competitiveness and support environmental sustainability initiatives illustrated Qatar's strategic focus on creating a sustainable and diversified economic landscape (Al-Housani et al., 2023).

2023 : Innovation and Expansion of Entrepreneurial Support Systems

In 2023, Qatar advanced its infrastructure and regulatory frameworks to further support entrepreneurial activities. The reorganization of key governmental bodies to better facilitate business operations and the emphasis on digital and technological innovation underscored Qatar's ongoing efforts to create an optimal environment for business growth and innovation (Al-Housani et al., 2024).

From 2018 to 2023, this network study has particular interest.

This report provides a comprehensive analysis of Qatar's entrepreneurial ecosystem during a transformative period (2018-2023), highlighting strategic policy shifts, regulatory advancements, and sustainability efforts. It offers critical insights into how Qatar tackled global challenges, diversified its economy, and fostered innovation to support entrepreneurship in alignment with Vision 2030.

Holistic Evaluation of Economic and Entrepreneurial Trends in Qatar (2018-2023)

Between 2018 and 2023, Qatar experienced significant transformations in its economic and entrepreneurial landscapes, guided by strategic policy reforms, international collaborations, and a strong commitment to sustainability in alignment with Vision 2030. These years saw a mix of challenges and opportunities as Qatar worked to diversify its economy and strengthen its entrepreneurial ecosystem.

The country's GDP per capita showed considerable fluctuations during this period. It declined from \$66,264 in 2018 to \$52,316 in 2020, reflecting the impacts of global economic disruptions such as the COVID-19 pandemic. However, a remarkable recovery followed, with GDP per capita reaching \$87,661 in 2022, highlighting the success of diversification efforts and strategic investments in infrastructure and non-hydrocarbon sectors. Foreign direct investment (FDI) also exhibited volatility, peaking at \$7.26 billion in 2019 before sharply declining to \$1.25 billion in 2021, mirroring global uncertainties while showcasing Qatar's need to enhance its investment attraction strategies.

Unemployment rates remained relatively high, reflecting the pandemic's impact on labor markets. However, a slight improvement in 2022 suggests the effectiveness of job creation initiatives and entrepreneurial opportunities supported by the government. Despite limited data on new business density, the robust figure of 62.63 registrations per 1,000 people in 2018 demonstrates a strong entrepreneurial culture.

Qatar's entrepreneurial ecosystem expanded significantly, supported by initiatives such as the National Cybersecurity Agency and the Investment and Trade Court. These developments improved infrastructure, dispute resolution, and digital security, creating a supportive environment for startups and SMEs. While the ecosystem aligned well with key growth areas like digital transformation, innovation, and sustainability, challenges such as regional disparities, limited academia-investor collaboration, and inefficiencies in resource distribution remain. Addressing these gaps will further strengthen Qatar's entrepreneurial ecosystem and foster sustainable, inclusive growth.

5.2. Overview of the Entrepreneurial Ecosystem: Challenges and Opportunities

Between 2018 and 2023, Qatar underwent transformative changes aimed at establishing a diversified, innovative, and sustainable economy aligned with its QNV 2030 goals. While notable progress was made, the entrepreneurial ecosystem faced several macro-level challenges that required strategic responses to sustain growth and resilience. These challenges were shaped by the evolving global economic landscape, the nation's ambition to shift away from hydrocarbon dependency, and the increasing demand for robust digital infrastructure and sustainability practices.

This section identifies and explores five main challenges detected through the macro-level analysis of Qatar during this period. These challenges—economic diversification, regulatory adaptability, global economic shocks, cybersecurity and digital infrastructure, and sustainability integration—offer critical insights into the complexities of fostering entrepreneurship in a rapidly evolving socio-economic environment. By addressing these challenges, Qatar demonstrates its commitment to creating a thriving ecosystem that supports innovation, investment, and sustainable growth.

Economic Diversification

Transitioning from a hydrocarbon-dependent economy to a knowledge-based one requires significant investment in infrastructure, innovation, and human capital development.

Regulatory Adaptability

While improvements were made, aligning regulations with global standards and ensuring their consistent implementation remain critical to sustaining entrepreneurial momentum.

Global Economic Shocks

Events like the COVID-19 pandemic tested the resilience of Qatar's entrepreneurial ecosystem, requiring adaptive strategies to mitigate disruptions.

Cybersecurity and Digital Infrastructure

As Qatar advances in digital innovation, ensuring robust cybersecurity frameworks and protecting digital transactions remain ongoing challenges.

Sustainability Integration

Balancing rapid economic development with environmental sustainability and resource management poses a critical challenge for long-term ecosystem viability.

From 2018 to 2023 (figure 3.), Qatar's entrepreneurial ecosystem evolved through economic diversification, regulatory reforms, and digital innovation, aligning with QNV 2030. Key challenges included transitioning from hydrocarbon dependency, adapting regulations, managing global shocks, strengthening cybersecurity, and balancing rapid development with sustainability, showcasing resilience and commitment to fostering a robust business environment.

Figure 3: Expansion of Data for Qatar and Positioning of the Current Report

2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
2018 Report												
	2019 Report											
		2020 Report										
			2021 Report									
				2022 Report								
					2023 Report							
						2024 Report						
							2025 Report					

Conclusion

From 2018 to 2023, Qatar's entrepreneurial ecosystem experienced significant transformation as the nation advanced its Vision 2030 agenda. The period was marked by ambitious efforts to transition from hydrocarbon dependency toward a knowledge-based economy, emphasizing economic diversification, technological integration, and sustainable development. Qatar implemented strategic policies, enhanced regulatory frameworks, and invested in critical infrastructure to foster innovation and entrepreneurship, positioning itself as a competitive global business hub.

However, this evolution came with challenges that shaped the ecosystem's trajectory. Economic diversification required substantial investments in human capital and innovation, while regulatory adaptability needed alignment with global standards to sustain momentum. Global economic shocks, such as the COVID-19 pandemic, underscored the necessity for resilience and adaptability in navigating external pressures. The rise of digital innovation highlighted the importance of robust cybersecurity measures to protect transactions and foster trust within the ecosystem. Meanwhile, balancing rapid economic growth with environmental sustainability remained a critical concern, demanding careful policy integration and resource management.

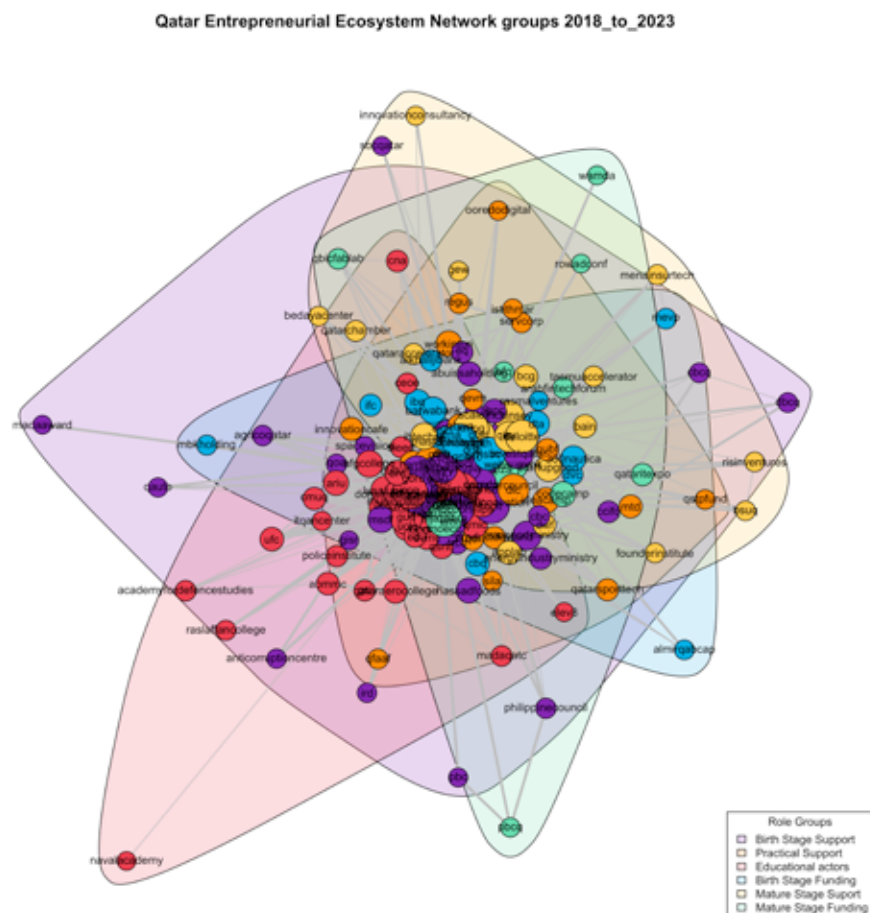
Despite these challenges, Qatar demonstrated resilience and adaptability, leveraging its strategic position, international collaborations, and progressive policies to build a supportive entrepreneurial environment. The advancements achieved during this period underscore Qatar's commitment to fostering a thriving, innovative, and sustainable business ecosystem. The lessons drawn from this journey offer valuable insights for other nations pursuing similar transitions and serve as a testament to the potential of strategic foresight in shaping a dynamic and resilient entrepreneurial landscape.

6. Entrepreneurial Ecosystem Diagnosis

6.1. Actor Network Analysis: Measuring Connectivity and Collaboration

We begin our investigation by assessing the dimension of Qatar's entrepreneurial ecosystem using significant measures. By identifying the support actors, their typology, specialization, and function, we are able to create a captivating portrayal of the diversity of Qatar's entrepreneurial landscape. This typology helps us understand how each Actor contributes to the overall dynamic of the ecosystem. This allows us to present a plot of this ecosystem in this particular period (figure 4).

Figure 4: Representation of the Qatar's Entrepreneurial Ecosystem Network from 2018 to 2023.



6.1.1. Network Measures: Mapping Relationships and Ecosystem Structure

This section analyzes key network metrics to map the relationships and structural dynamics of Qatar's entrepreneurial ecosystem from 2018 to 2023. Each metric reflects critical dimensions of ecosystem connectivity, resilience, and functionality.

Number of Nodes & Ties

Metric: Total number of actors (nodes) is **223**, and collaborations (ties) are **2158**.

Meaning: Nodes represent active entities in the ecosystem; ties represent collaborative relationships.

Analysis: A significant increase in both nodes and ties indicates ecosystem expansion and stronger collaborative networks compared to the prior period.

Average Weight

Metric: **33.022** (+0.270 compared to the prior period).

Meaning: Reflects the average strength of ties or frequency of collaborations.

Analysis: Increased average weight shows that collaborations became slightly more impactful or frequent, suggesting a qualitative enhancement in actor relationships.

Graph Density

Metric: **0.1615** (-0.0071).

Meaning: Measures the proportion of actual ties relative to all possible ties.

Analysis: The slight reduction suggests a larger network with slightly less overall connectivity, typical of growing ecosystems.

Centrality Measures

Metric: Centrality is **0.021** (no change from the prior period).

Meaning: Highlights key actors' influence within the network.

Analysis: Stability in centrality suggests the role of influential actors remains unchanged, with a dispersed yet connected structure.

Distance Measures (Average Shortest Path & Diameter)

Metric: Average Shortest Path is **4.0592** (-0.2096).

Meaning: Reflects how efficiently information or resources flow through the network.

Analysis: A decrease indicates improved accessibility and reduced effort to connect across the ecosystem.

Network Diameter

Metric: 4.0592 (-0.2096).

Meaning: The longest distance between any two nodes in the network.

Analysis: A smaller diameter suggests improved reachability and compactness in the ecosystem structure.

Clustering Coefficient

Metric: 0.4680 (-0.0070).

Meaning: Measures the likelihood of forming clusters or tightly-knit groups.

Analysis: A small decline indicates marginally fewer localized connections, though clustering remains relatively strong.

Interconnectedness Factors

Metric: 24.29 (+1.67).

Meaning: Combines interconnectivity and average weight to assess overall network robustness.

Analysis: Increased interconnectedness reflects stronger and more dynamic relationships within the ecosystem.

Proximity Factor

Metric: 16.48 (-1.75).

Meaning: Combines efficiency (short path lengths) and reachability (network diameter).

Analysis: A slight decline in proximity suggests reduced integration at macro levels but remains functionally robust.

Clustering Coefficient and Modularity

Metric: 0.4680 clustering (-0.0070 modularity).

Meaning: Reflects community cohesiveness and modular separation.

Analysis: Minor decreases indicate that while modular structures exist, they may have slightly less tight-knit group activity.

Table 2. Qatar's Entrepreneurial Ecosystem Measures 2018 to 2023

Metric	2018 to 2023	vs previous Period	Interpretation
Total Actors (Nodes)	223	10	Ecosystem expansion
Collaborations (Ties)	2158	172	Increased collaborations
Average Weight	33.022	+0.270	Strengthened ties
Graph Density	0.1615	-0.0071	Reduced connectivity
Centrality	0.021	0.000	Stable key actor influence
Average Shortest Path	4.0592	-0.2096	Improved accessibility
Network Diameter	4.0592	-0.2096	Better reachability
Clustering Coefficient	0.4680	-0.0070	Marginally fewer clusters
Interconnectedness Factor	24.29	+1.67	Stronger relationships
Proximity Factor	16.48	-1.75	Slightly reduced efficiency

6.1.2. Actor Types: Categorization and Diversity Insights

The ecosystem includes 18 distinct actor types, with national institutions (35), universities (28), and programs (15) leading in numbers. The relatively stable presence of critical actors underscores consistent institutional support.

The entrepreneurial ecosystem in Qatar from 2018 to 2023 showcases diverse actor types, with notable representation from universities, national institutions, and venture capital entities. This distribution reflects the ecosystem's strong emphasis on institutional support, education, and funding mechanisms. Despite minimal growth in new actor categories compared to the prior period, the presence of accelerators, incubators, and co-working spaces highlights a growing infrastructure to nurture startups. The ecosystem's limited diversity in categories such as business angels, FabLabs, and cultural spaces suggests opportunities to further enhance its innovation landscape by fostering grassroots and creative ventures. Overall, the distribution reveals a balanced yet evolving ecosystem, signaling progress in creating a supportive environment for entrepreneurship.

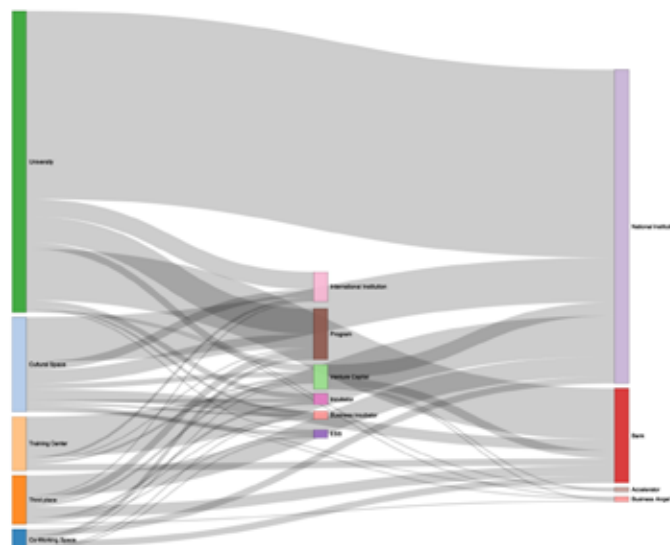
Table 3. Type of Active Actors in the period

Type of Actor	2018 to 2023	vs previous Period	Interpretation
Accelerator	4	0	Stable support for startups
Bank	12	0	Financial backbone consistent
Business Incubator	2	0	Minimal incubation opportunities

Co-Working Space	4	0	Small but stable infrastructure
Consulting	12	0	Steady advisory availability
Cultural Space	3	0	Limited creative hubs
ESS	2	0	Niche presence
Event	6	0	Moderate event-based engagement
Incubator	2	1	Emerging growth in incubation
International Institution	14	0	Strong global connectivity
Media	1	0	Sparse media representation
National Institution	35	0	Dominant institutional presence
Platform	3	1	Growing digital infrastructure
Program	15	0	Stable structured programs
Third Place	4	0	Consistent alternative spaces
Training Center	11	0	Stable skill development hubs
University	28	1	Strengthened academic presence
Venture Capital	7	0	Stable funding availability
Business Angel	1	0	Limited individual investors
Fablab	1	0	Sparse innovation labs

The following Sankey diagram showcases the flows and connections between actor types, highlighting the volume and direction of interactions. Thicker lines indicate stronger collaborations, making it easier to identify key relationships and understand ecosystem dynamics (see Figure 5).

Figure 5. Sankey diagram between Actors Roles in the period



Policy should prioritize fostering diversity by supporting underrepresented actor types like FabLabs, cultural spaces, and business angels, ensuring a balanced ecosystem that promotes innovation, grassroots entrepreneurship, and creative industries.

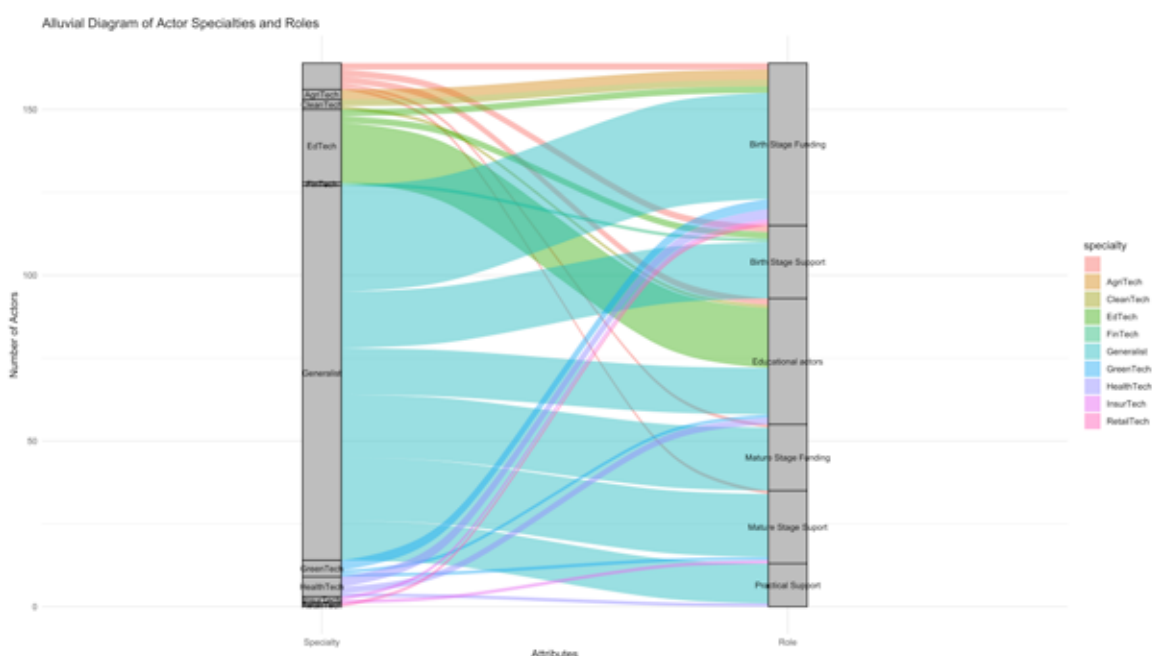
Qatar's entrepreneurial ecosystem demonstrates strong institutional support but limited diversity in creative and grassroots actors. Strengthening categories like FabLabs and cultural spaces, while expanding funding access through business angels, could enhance innovation and inclusivity. Policies encouraging ecosystem diversity will ensure a more dynamic and resilient entrepreneurial landscape aligned with long-term goals.

6.1.3. Actor Specializations: Mapping Expertise and Niche Roles

Understanding actor specializations reveals ecosystem strengths and gaps, enabling policymakers to align strategies with dominant industries while fostering growth in underrepresented niches for balanced economic development.

An Alluvial diagram illustrates the connections between specialties and roles over time, emphasizing how contributions shift or remain consistent across the ecosystem. The width of the flows represents the strength or volume of these relationships, enabling insights into the alignment between specialties and their roles within the entrepreneurial landscape (see Figure 6).

Figure 6. Alluvial diagram between specialties and roles in the period



Qatar's entrepreneurial ecosystem from 2018 to 2023 is dominated by generalist actors (116), highlighting a broad focus on diverse sectors. Specialized fields like EdTech (22) and HealthTech (6) indicate growing interest in knowledge and health-driven innovation, though niche industries like AgriTech, CleanTech, and InsurTech remain underdeveloped. While

specialization diversity has remained stable, targeted policy initiatives are necessary to support emerging markets such as GreenTech (5) and FinTech (1), fostering innovation in key economic and sustainable domains.

Table 4. Specialization of Active Actors in the Period

Specialization	2018 to 2023	vs previous Period	Interpretation
Generalist	116	3	Broad ecosystem focus
AgriTech	3	0	Limited focus on agricultural tech
CleanTech	3	0	Underdeveloped sustainability tech
EdTech	22	0	Growing education innovation
GreenTech	5	0	Emerging sustainability efforts
HealthTech	6	0	Moderate health sector interest
InsurTech	2	0	Limited insurance innovation
RetailTech	1	0	Sparse retail technology presence
FinTech	1	0	Minimal financial tech focus

Policies should nurture niche industries like CleanTech, GreenTech, and FinTech, providing incentives for innovation, investment, and ecosystem development while leveraging the existing strengths in EdTech and generalist actor versatility.

Qatar's ecosystem remains dominated by generalists, with EdTech and HealthTech showing promise. However, niche sectors like Fintech in Qatar, for example, began being regulated in 2022, starting with payment service providers, and it was not until 2023 that the Financial Sector Strategy Plan included a Fintech Framework to foster the sector.

6.1.4. Actor Services: Evaluating Services in Resource Flows

Analyzing actor services reveals the availability and alignment of resources to ecosystem needs, highlighting areas of strength, gaps, and opportunities for strategic resource allocation and support.

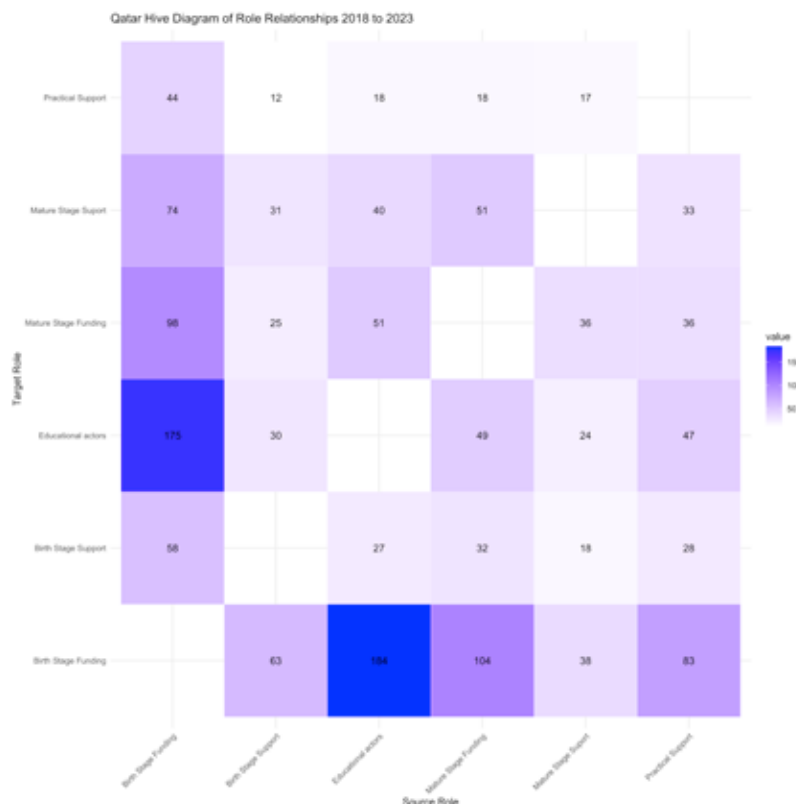
The entrepreneurial ecosystem in Qatar from 2018 to 2023 demonstrates robust service offerings, with notable strengths in **InnovBoost** (63) and **FundBoost** (41), which reflect the focus on innovation development and financial support. Moderate growth in **SalesBoost** (42) and **MarketBoost** (35) underscores efforts to enhance market entry and sales capabilities. However, underrepresented areas like **HumanBoost** (3) and **TestBoost** (6) indicate limited support in skill development and product testing, suggesting opportunities for targeted policy and service improvements. Strengthening these areas can ensure balanced resource flows and a more inclusive ecosystem.

Table 5. Services of Active Actors in the Period

Type	2018 to 2023	vs previous Period	Interpretation
BuizPlanBoost	19	0	Stable business planning support
FundBoost	41	3	Increased financial resource access
HumanBoost	3	0	Limited skill development services
InnovBoost	63	1	Strong innovation support focus
LawBoost	9	0	Stable legal resource availability
MarketBoost	35	1	Growing market entry support
ProtoBoost	7	0	Minimal prototype development support
SalesBoost	42	1	Expanded sales enablement services
TestBoost	6	0	Limited product testing services

A Heatmap Hive Diagram visualizes the relationships between services within the ecosystem, using color intensity to represent the strength or frequency of connections. The hive structure highlights clusters and patterns, revealing how services interact and contribute to the overall network. This provides a clear view of service interdependencies and collaboration dynamics (see Figure 7).

Figure 7. Hive Diagram between actors services in the period



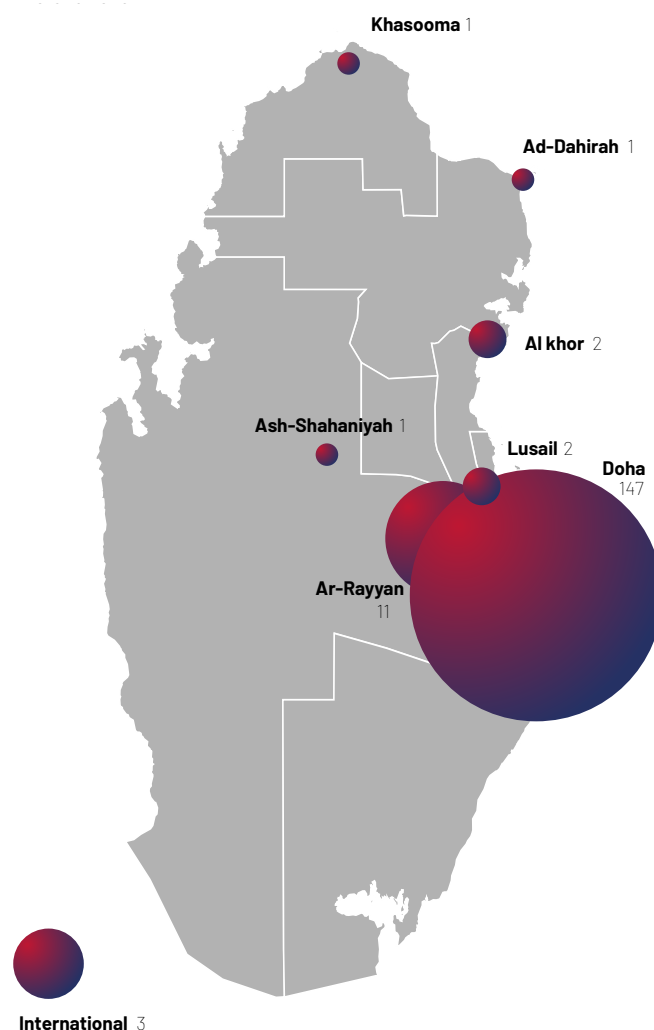
Qatar's ecosystem excels in innovation and financial support but **lacks adequate services in skill development, prototype creation, and product testing**. Targeted policy initiatives can strengthen these underrepresented areas, ensuring more comprehensive resource flows and enabling broader entrepreneurial success. Balanced service offerings are essential to fostering a resilient and dynamic ecosystem.

6.2. Territorial Analysis: Addressing Regional Disparities

Qatar's entrepreneurial ecosystem exhibits a concentrated distribution of active cities, with Doha emerging as the dominant hub. This territorial analysis highlights regional strengths and disparities, aiming to balance opportunities, foster inclusivity, and leverage the potential of underrepresented areas for ecosystem-wide development.

6.2.1. Regional Distribution of Actors: Strengths and Gaps

Figure 8: Map of Actor Concentration in Qatar



The concentration of actors in Doha underscores its pivotal role as Qatar’s entrepreneurial hub. Cities like Al-Rayyan and Lusail show moderate activity but lack the momentum seen in Doha. Addressing gaps in smaller cities like Ad-Dahirah and Al-Shahaniya is crucial for fostering regional inclusivity and development.

Table 6. Geographic Division of Actors in Qatar

City	Active Actors (2019-2024)	Gap vs Previous Period	Interpretation
Doha	147	3	Major hub with growing influence.
Al-Rayyan	11	0	Moderate activity, potential to scale.
Lusail	2	0	Emerging city with untapped potential.
Al-Khor	2	0	Minimal activity, needs ecosystem support.
Ad-Dahirah	1	0	Underrepresented, requires focused efforts.
Al-Shahaniya	1	0	Limited participation, potential for growth.
Khasooma	1	0	Niche engagement, needs targeted outreach.
Out of Qatar	3	0	Regional partnership with marginal impact.

6.2.2. Sub-Ecosystem Metrics: Community Size, and Density

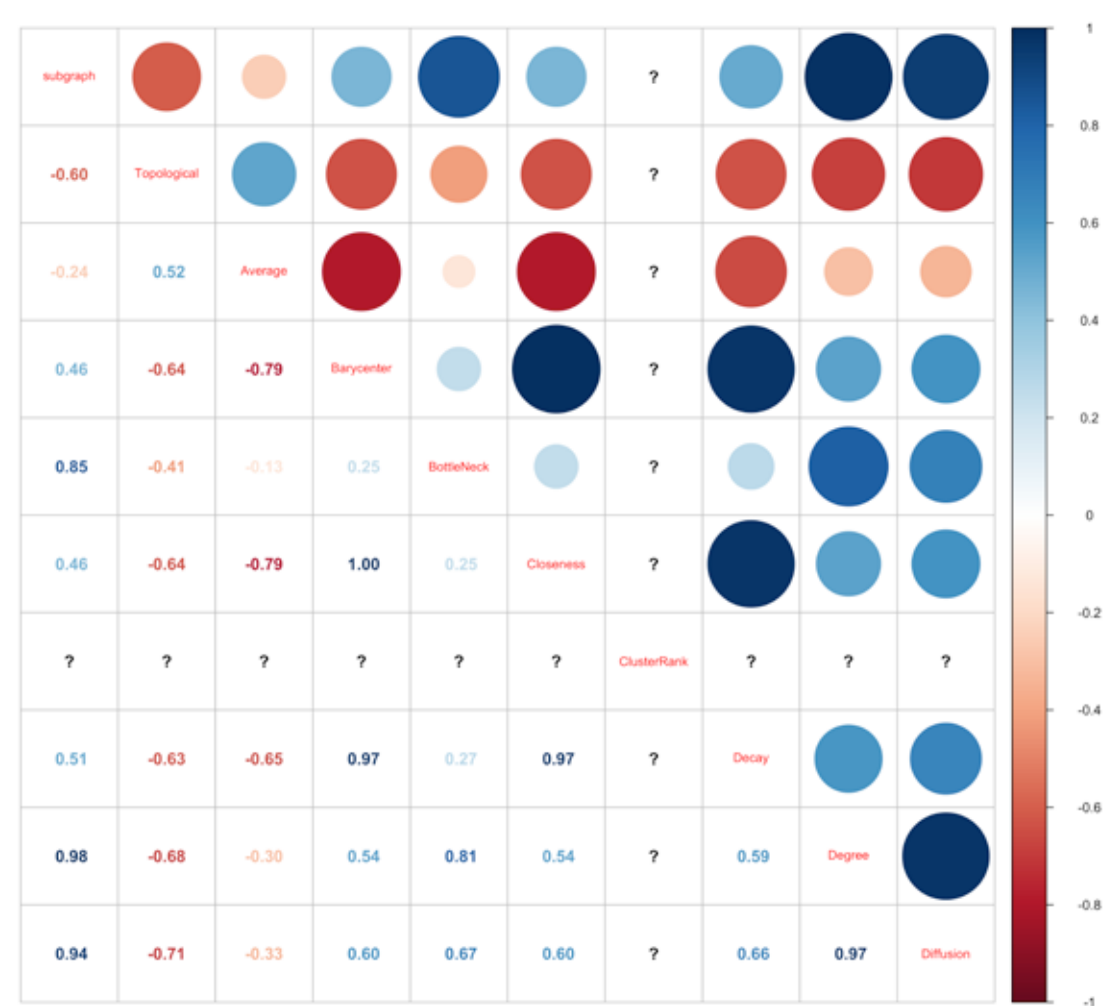
Qatar’s entrepreneurial ecosystem comprises five distinct communities varying in size and density. Community 1, with 66 members and a low density(0.1324), reflects a broad network of innovation hubs, accelerators, and policy actors. Community 2 (38 members, density 0.3158) highlights educational and institutional collaboration. Community 3 (25 members, density 0.4367) emphasizes financial and consulting expertise. Community 4 (32 members, density 0.3629) includes research and academic actors. Community 5 (3 members, density 1.0) signifies cohesive, specialized collaboration.

Smaller communities exhibit higher cohesion, while larger networks show diversity but lower density. Strategic integration enhances ecosystem efficiency.

6.3. Leadership and Centrality: Identifying Influential Actors

Figure 9 illustrates a Pearson Correlation heatmap between centrality measures, showcasing the strength and direction of relationships among key metrics. Positive correlations are marked by stronger, warmer colors, while negative correlations appear cooler. This visualization helps identify patterns and interdependencies between centrality measures, offering deeper insights into the structural dynamics of the ecosystem.

Figure 9: Pearson Correlation between centrality measures

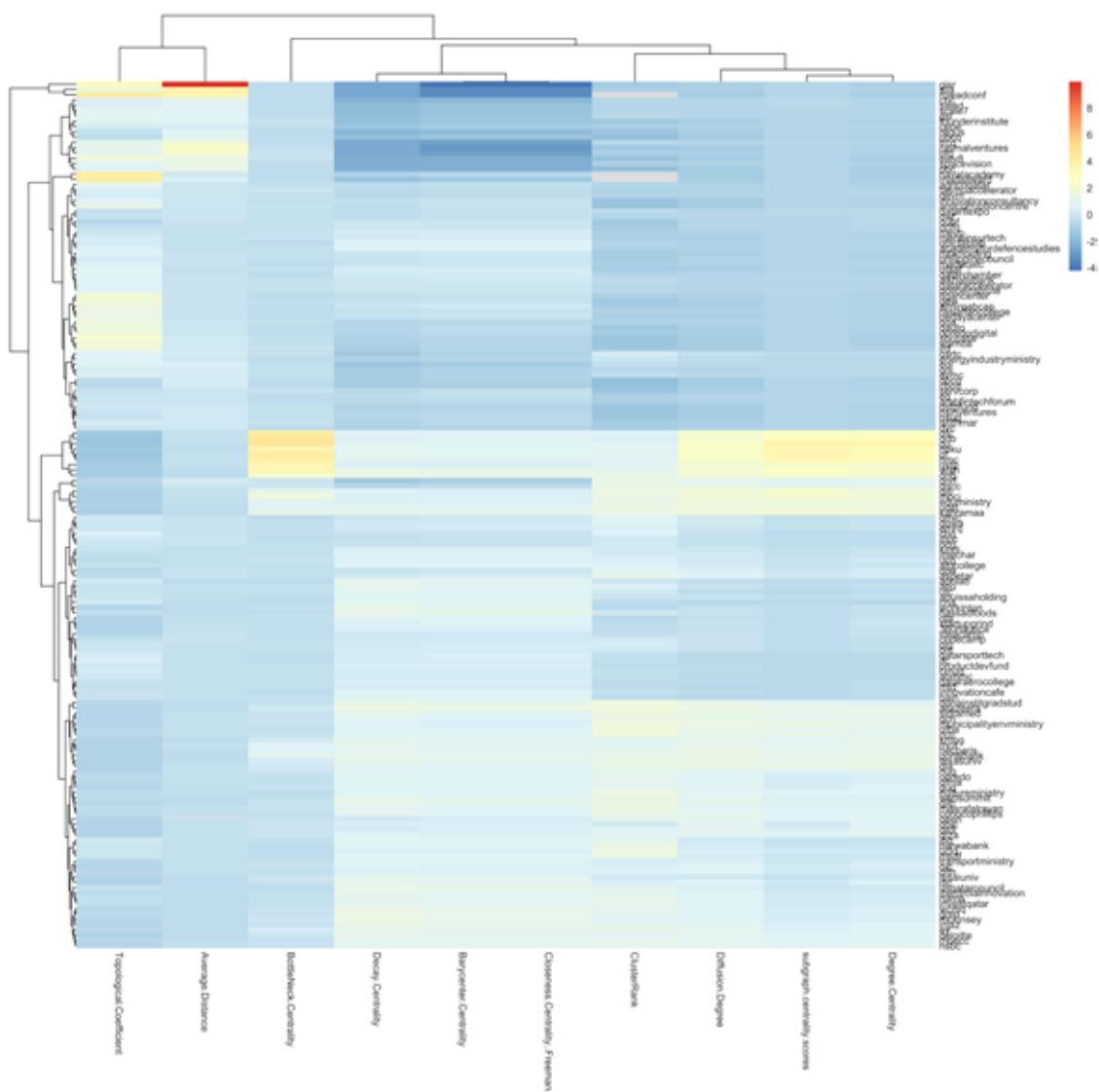


6.3.1. Leadership Measures: Spotlighting Key Ecosystem Drivers

Figure 10 presents a Heatmap of Network Actors Centrality, where color intensity represents the centrality level of each actor within the ecosystem. This visualization highlights the

most influential actors based on their centrality measures, revealing key players and their roles in driving the network's dynamics and connectivity during the analyzed period.

Figure 10: Heatmap of Network Actors Centrality

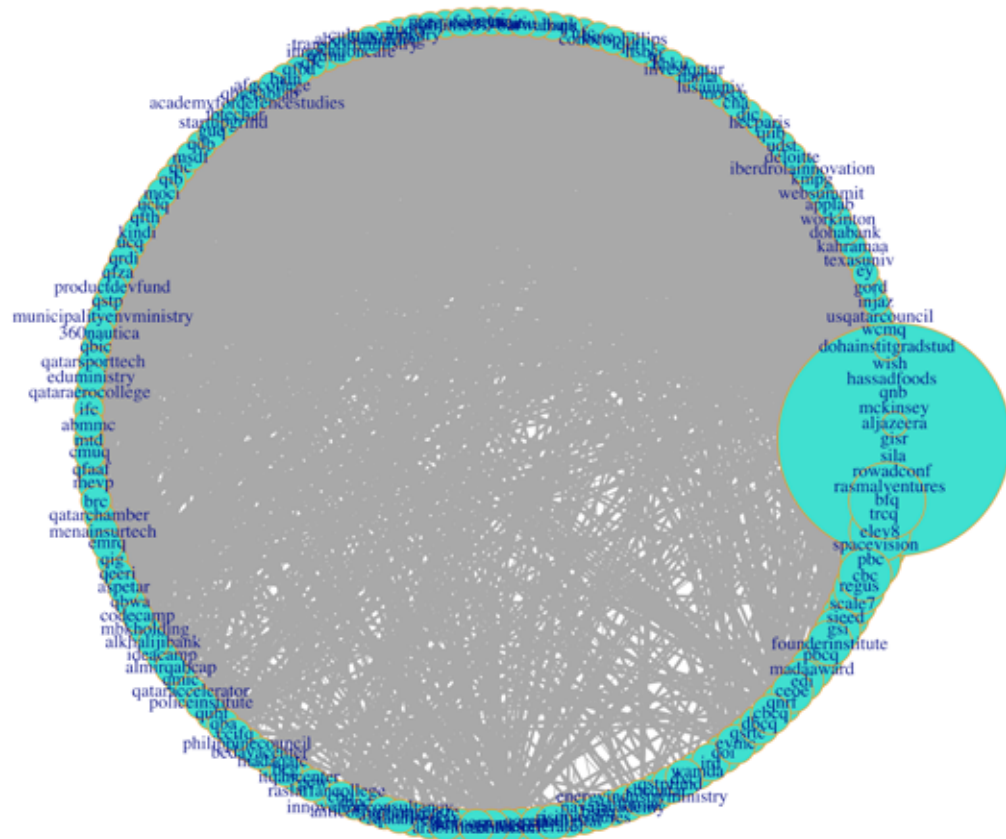


Introduction

Centrality power, or leadership, reflects an actor's influence within an ecosystem, shaping its dynamics and strategic direction. High-centrality actors drive growth and innovation, while lower-tier contributors provide specialized support. Together, these roles create a balanced structure fostering collaboration and ecosystem resilience.

Figure 11 displays a Network Circle Plot generated using the Wiener method, arranging actors based on their centrality within the ecosystem. The circular layout emphasizes relationships and interactions, with proximity indicating stronger connections. This visualization provides a clear and organized view of the network's structure, highlighting influential actors and their roles in maintaining ecosystem cohesion.

Figure 11: Network plot using the Wiener method



Detailed Leadership Analysis

Very High Leadership:

Actors: Qatar Islamic Bank, Qatar University, Qatar Foundation, Hamad Medical Corporation.

Significance: These organizations act as the ecosystem's core. Their influence spans policy-making, financial empowerment, and institutional leadership.

Role: Critical for strategic funding, infrastructure support, and fostering innovation.

High Leadership:

Actors: Qatar Development Bank, Qatar Financial Centre.

Significance: They provide robust support for entrepreneurial growth through funding, incubation, and regulatory frameworks.

Role: Enabling startups and established players alike to thrive.

Average Leadership:

Actors: Qatar Science & Technology Park, Sidra Medicine.

Significance: These actors balance innovation and specialization. They operate at the intersection of research and commercial viability.

Role: Supporting innovation clusters and bridging academia with industry.

Low Leadership:

Actors: Masraf Al Rayan, Qatar Investment Authority.

Significance: These actors have limited but focused roles, often tied to specific sectors or initiatives.

Role: Catalyzing targeted growth and creating niche opportunities.

Very Low Leadership:

Actors: Niche players like Founder Institute, Bedaya Center.

Significance: Contributing through specialized or community-driven initiatives.

Role: Limited but valuable ecosystem impact.

Table 7. Centrality of Active Actors in the Period

Actor	Centrality Power	Interpretation
Qatar Islamic Bank, Qatar University, Qatar Foundation, Hamad Bin Khalifa University, Hamad Medical Corporation, Aspetar Orthopaedic and Sports Medicine Hospital	Very High (>250)	Central figures, ecosystem's backbone.
Qatar Development Bank, Qatar International Islamic Bank, Qatari Businessmen Association, Qatar Financial Centre, Qatar National Bank	High (100–250)	Significant influence, strategic players.
Qatar Science & Technology Park, Sidra Medicine, QEERI, HSBC, University of Doha for Science and Technology, Community College of Qatar	Average (50–100)	Steady contributors, visible ecosystem role.
Qatar Investment Authority, Qatar Computing Research Institute, Qatar Financial Markets Authority, QNRF, Doha Bank, Masraf Al Rayan	Low (10–50)	Specialized roles, targeted influence.
Various smaller organizations (e.g., Founder Institute, Bedaya Center, etc.)	Very Low (<10)	Marginal influence, niche contributions.

Conclusion

The centrality analysis highlights a hierarchy of leadership roles, from foundational institutions to niche contributors. Very high and high leadership actors dominate the landscape, shaping its strategic direction, while smaller players provide complementary and focused support to enrich the ecosystem.

Takeaways

Qatar's entrepreneurial ecosystem thrives through its leadership² hierarchy, anchored by robust institutions like Qatar Islamic Bank and Qatar University. High and medium-tier actors enhance this by targeting specific needs. The ecosystem's vibrancy stems from a balanced mix of strategic, operational, and niche players.

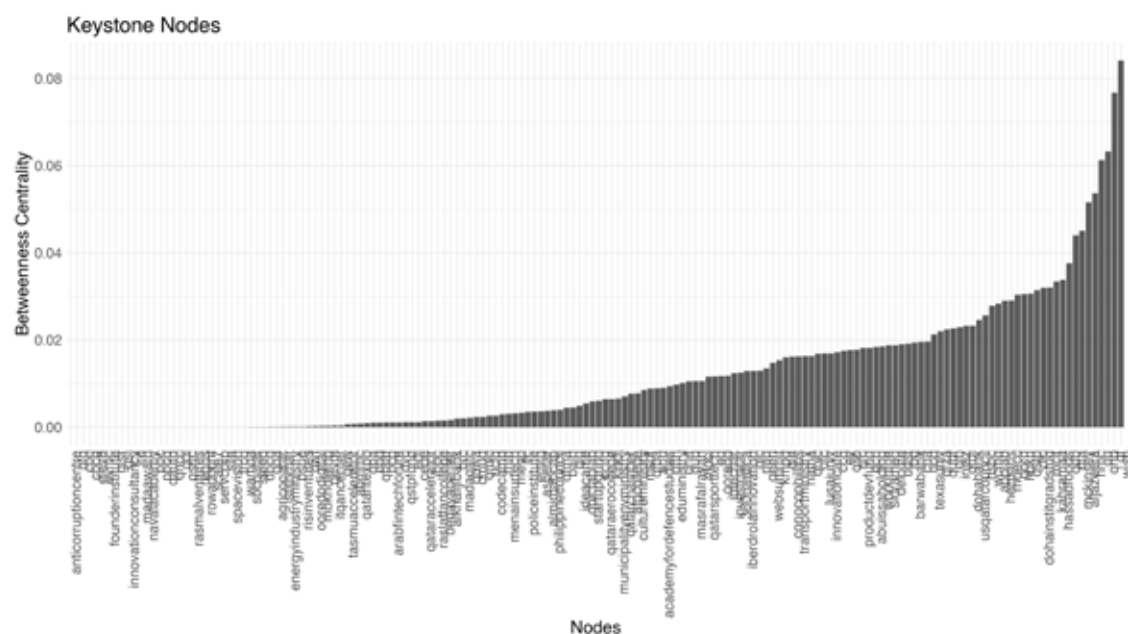
² through centrality measures

6.3.2. Keystone Nodes: Ensuring Ecosystem Stability

Qatar's entrepreneurial ecosystem thrives through keystone nodes that maintain stability and innovation. Key actors, like institutions and businesses, anchor the ecosystem, fostering collaboration, resilience, and growth. Their influence ensures sustained progress across diverse sectors, enabling a robust foundation for entrepreneurial success.

Figure 12 highlights Keystone Nodes among actors, showcasing the most critical players within the network. These nodes are identified based on their unique ability to connect and sustain the ecosystem. The visualization emphasizes their roles in fostering collaboration, bridging gaps, and ensuring the stability and resilience of the entrepreneurial ecosystem.

Figure 12: Keystone Nodes among actors



Detailed Stability Analysis

Table 8. Keystone Actors analysis

Actors	Keystone Node level	Interpretation
World Innovation Summit for Health, Qatar Foundation	Very High	Anchor institutions driving stability.
Qatar National Bank, Qatar University, Hamad Medical Corporation, Al Jazeera	High	Critical contributors ensuring resilience.
McKinsey & Company, Doha Tech Angels, Qatar Development Bank, Hassad Foods, HEC Paris	Average	Influential but sector-specific impact.
Qatar Financial Centre, Qatar Islamic Bank, Ministry of Communications and Information Technology	Low	Limited reach but steady contribution.
Smaller councils, niche ventures, and emerging initiatives	Very Low	Peripheral actors with minimal impact.

Conclusion

Qatar's ecosystem is upheld by keystone actors spanning various leadership levels. Core institutions ensure continuity, while emerging entities contribute to niche areas. This hierarchy bolsters innovation and system-wide resilience.

Takeaways

Strong keystone actors ensure stability in critical sectors.

Collaboration among varying leadership levels drives overall ecosystem growth.

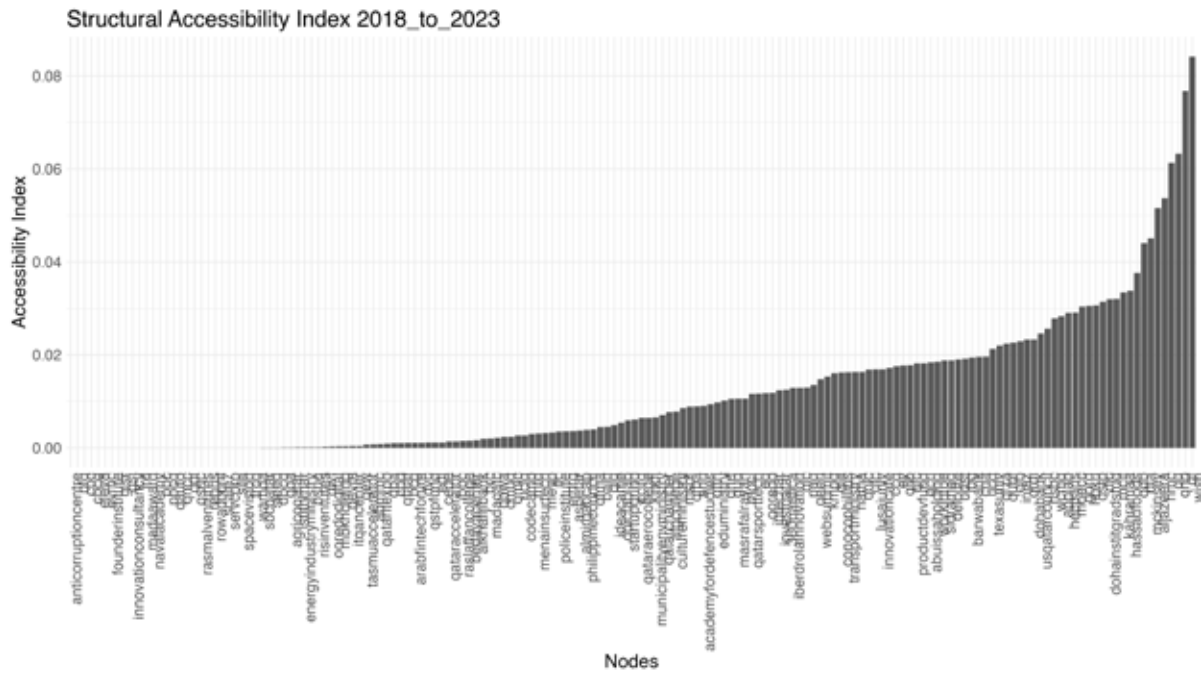
6.3.3. Structural Accessibility: Enhancing Inclusivity and Reach

Qatar's entrepreneurial ecosystem reflects varied accessibility levels, with key actors fostering inclusivity and facilitating ecosystem engagement. This accessibility enhances the ecosystem's reach and supports equitable participation across sectors, creating opportunities for collaboration, innovation, and sustainable development.

Figure 13 presents a Structural Accessibility Diagram of actors, illustrating how easily different nodes can be reached within the network. The diagram highlights pathways and barriers to connectivity, emphasizing actors with the highest accessibility. This

visualization provides insights into the flow of resources, information, and opportunities across the ecosystem.

Figure 13: Structural Accessibility of actors



Detailed Accessibility Analysis

Table 8 highlights the accessibility levels of key actors within Qatar’s entrepreneurial ecosystem, categorized by their Accessibility Index. Actors with a **Very High Index**, such as the World Innovation Summit for Health and Qatar Foundation, demonstrate wide accessibility and play critical roles as enablers in multiple sectors. Entities with a **High Index**, such as Qatar National Bank and Qatar University, exhibit strong cross-domain influence, reflecting their active participation in fostering entrepreneurship.

Meanwhile, organizations like McKinsey & Company and Doha Tech Angels, with an **Average Index**, show moderate inclusivity, often focused on specific sectors. Actors with a **Low Index**, such as Qatar Islamic Bank and the Ministry of Communications and Information Technology, have limited outreach, catering to niche audiences. Lastly, those with a **Very Low Index**, including smaller councils and under-resourced ventures, face challenges in network inclusion, reflecting structural barriers to participation. These findings emphasize the ecosystem’s hierarchical connectivity and areas for improvement.

Table 9. Accessibility Index analysis

Actors	Accessibility Index	Interpretation
World Innovation Summit for Health, Qatar Foundation	Very High	Widely accessible and critical enablers.
Qatar National Bank, Qatar University, Hamad Medical Corporation, Al Jazeera	High	Strong accessibility across key domains.
McKinsey & Company, Doha Tech Angels, Qatar Development Bank, Digital Incubation Center	Average	Moderate inclusivity with sectoral focus.
Qatar Islamic Bank, Qatar Financial Centre, Ministry of Communications and Information Technology	Low	Limited outreach and specialized access.
Smaller councils, niche ventures, and under-resourced entities	Very Low	Minimal reach with restricted inclusion.

Conclusion

Accessibility varies significantly among Qatar's ecosystem actors. While leading institutions drive inclusivity, smaller organizations often struggle with outreach. Strategic improvements in communication and resource allocation can enhance ecosystem accessibility and reach.

Takeaways

Strengthen communication channels: Promote ecosystem awareness across all sectors.

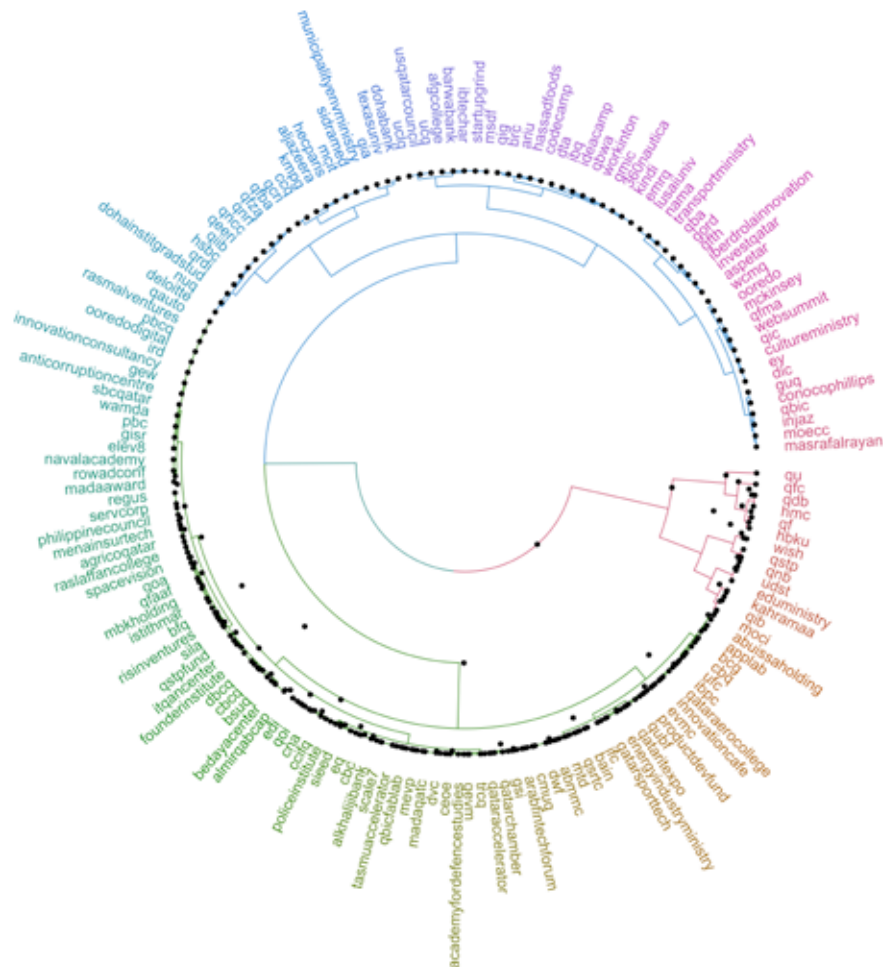
Support smaller actors: Equip niche entities with tools to increase their accessibility.

6.4. Community Dynamics: Unlocking Collaboration Opportunities

This chapter delves into the depths of the support communities that fuel Qatar's entrepreneurial ecosystem. We identified and analyzed these critical communities, revealing their power, interconnection, and impact on innovation, entrepreneurship, and national economy through rigorous methodology.

Figure 14 illustrates a **Cluster Representation Dendrogram**, visually mapping the hierarchical relationships among ecosystem actors. Each leaf represents an actor, while branches indicate clusters formed based on connectivity or similarity. The diagram reveals how actors group together, with shorter branch lengths denoting stronger connections, providing insights into the network's structure and identifying key collaborative clusters.

Figure 14: Cluster Representation Dendrogram



6.4.1. Louvain and Infomap Clusters: Fostering Innovation

The Louvain method uncovers natural groupings of actors within Qatar's Entrepreneurial Ecosystem. Each cluster highlights distinct subsets and focus areas, emphasizing specialization, collaboration, and interconnections among entities. This analysis sheds light on how clusters function and contribute to fostering innovation and entrepreneurship across the ecosystem.

Cluster Communities

The cluster communities identified through the Louvain method for Qatar's Entrepreneurial Ecosystem illustrate the segmentation and focus areas within the network. Each cluster represents a unique subset of the ecosystem, highlighting specialized functions and interrelationships among various entities. Here's a breakdown and analysis of each cluster community:

Cluster Community #1: Business and Startup Network

Characteristics: This cluster comprises coworking spaces, incubators, accelerators, and startup-focused organizations like Workinton, StartupGrind, and TASMU Accelerator. It also includes international entities fostering collaboration, such as the US-Qatar Business Council and the Dutch Business Council.

Role in Ecosystem: Central to supporting startups, fostering collaboration, and accelerating innovation through shared resources and international partnerships.

Cluster Community #2: Academic and Research Institutions

Characteristics: This group includes universities and research centers like Qatar University, University of Doha for Science and Technology, and Hamad Medical Corporation. It emphasizes knowledge creation, capacity building, and R&D.

Role in Ecosystem: Drives innovation through research, talent development, and partnerships with industry.

Cluster Community #3: Advanced Technology and R&D Hubs

Characteristics: This cluster features high-tech hubs and global research entities, such as Weill Cornell Medicine-Qatar, Texas A&M University at Qatar, and Qatar Science & Technology Park. These entities focus on cutting-edge technology and interdisciplinary innovation.

Role in Ecosystem: Acts as the innovation engine, connecting research to practical applications and fostering technological advancements.

Cluster Community #4: Policy and Infrastructure Drivers

Characteristics: This group includes ministries and infrastructure-focused entities like the Ministry of Energy and Industry and Qatar General Electricity & Water Corporation. It drives policy-making and infrastructure development essential for ecosystem growth.

Role in Ecosystem: Shapes the entrepreneurial environment by ensuring infrastructure, regulations, and energy support.

Cluster Community #5: Financial Powerhouses

Characteristics: Banks and financial institutions dominate this cluster, including Qatar National Bank, Qatar Islamic Bank, and Masraf Al Rayan. It also features international players like HSBC and KPMG.

Role in Ecosystem: Provides capital, financial advisory, and stability for entrepreneurial ventures.

Cluster Community #6: Regional and Sectoral Councils

Characteristics: This small but vital group includes sector-specific councils like the Philippine Business Council and Pakistan Business Council-Qatar, which focus on regional and cultural entrepreneurial opportunities.

Role in Ecosystem: Fosters inclusivity and cross-regional collaboration to diversify the ecosystem.

Conclusion about Cluster Communities

The cluster communities in Qatar's entrepreneurial ecosystem reveal a dynamic and interconnected landscape. Each cluster plays a unique role in driving innovation, nurturing startups, fostering collaboration, and shaping policies. Together, they create a robust ecosystem with complementary strengths and synergies, paving the way for sustained growth.

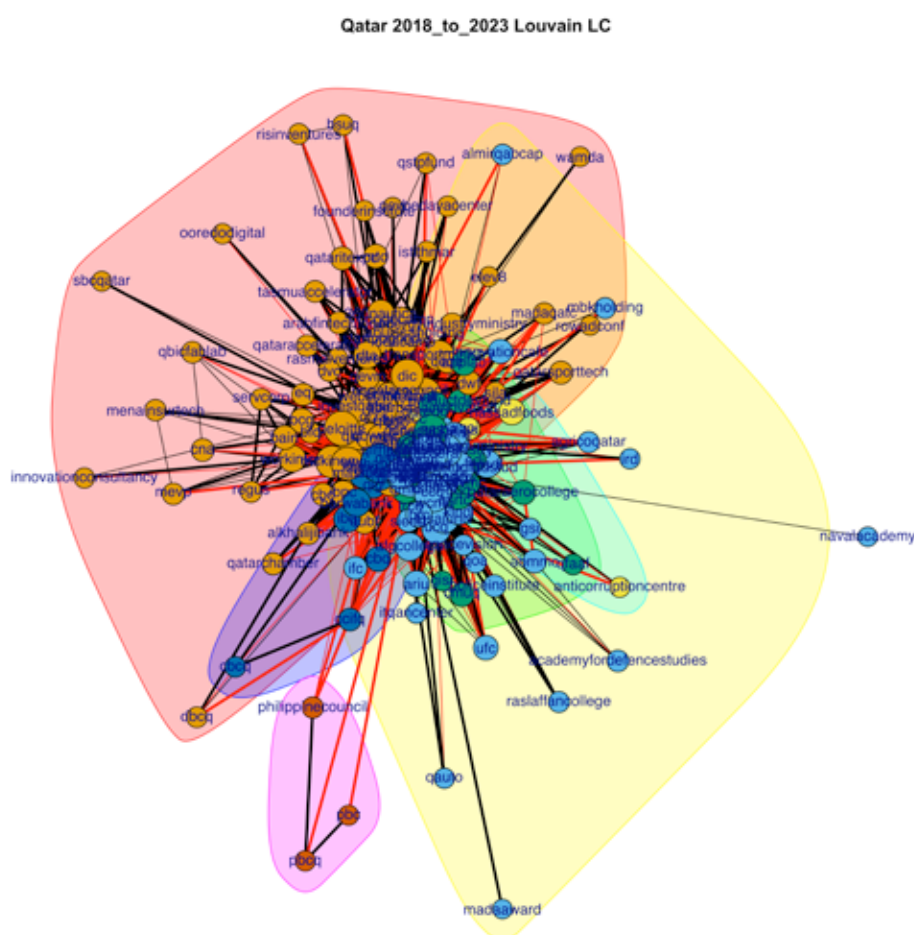
Table 10: Cluster Community Composition

Cluster Communities	Members
1	Workinton, Web Summit Qatar, Wamda, US-Qatar Business Council, The Ministry of Transport, TASMU Accelerator, StartupGrind, SILA, ServCorp, Scale7, Singapore Business Council Qatar, Rowad Conference, Risin Ventures, Regus, Rasmal Ventures, Qatar-Ukraine Business Forum, QSTP Tech Venture Fund, Qatar Insurance Group, Qatar Investment Authority, Business Innovation Park, Center of Entrepreneurship Qatar FinTech Hub, Qatar Financial Centre, Qatar Finance and Business Academy, Qatar Exchange Venture Market, Qatar Development Bank, Qatari Businesswomen Association, QBIC Fab Lab, Qatar Business Incubation Center QBIC, Qatari Businessmen Association, Qatar Sports Tech, Qatar IT Conference and Exhibition, Accelerator Qatar Business District, Qatar Chamber of Commerce and Industry, Ooredoo Digital & Beyond, Make the Deal, Ministry of Commerce and Industry, Microsoft for Startups Founders Hub, Middle East Venture Partners, MENA Insurtech, McKinsey & Company, Ministry of Communications and Information Technology, Mada - Qatar Assistive Technology Center, ISTITHMAR (SME EQ UITY), Invest Qatar, Innovation Consultancy, International Finance Corporation, Idea Camp, Ibtechar, Indian Business & Professional Council, Hyperthink, Global Entrepreneurship Week Network, Founder Institute, Ernst & Young, elev8, Doha Women Forum, Doha Venture Capital, Doha Tech Angels, Digital Incubation Center, Deloitte, Dutch Business Council Qatar, Code Camp, College of the North Atlantic Qatar, Center for Entrepreneurship & Organizational Excellence, Croatian Business Council, Business Start Up Qatar, Bangladesh Forum Qatar, Bedaya Center for Entrepreneurship and Career Development, Boston Consulting Group, Bain & Company Doha, Arab Fintech Forum, Al Mirqab Capital, Abu Issa Holding, 360 Nautica

- 2 World Innovation Summit for Health, Virginia Commonwealth University School of the Arts Qatar, University Foundation College, University of Doha for Science and Technology, University of Calgary in Qatar, Spacevision, Strategic Innovation, Entrepreneurship & Economic Development, Sidra Medicine, Ras Laffan Emergency and Safety College, Qatar University, Qatar Olympic Academy, Centre National De Convention Du Qatar, Q Auto, Police Institute, Mohammed Bin Ghanem Al Ghanem Naval Academy, Nama, Ministry of Municipality and Environment, Ministry of Social Development and Family, Ministry of Environment and Climate Change, MBK Holdings, Mada Innovation Award, Lusail University, KINDI Center For Computing Research, Itqan Clinical Simulation and Innovation Centre, Infrastructure Research and Development, Innovation Cafe, Hamad Medical Corporation, Global Studies Institute, Enterprise Qatar, ExxonMobil Research Qatar, Ministry of Education and Higher Education, Doha Institute for Graduate Studies, Ministry of Culture, Community College of Qatar, Biomedical Research Center, Aspetar Orthopaedic and Sports Medicine Hospital, Al Rayyan International University College, Al Jazeera, Agrico for Agricultural Development, Al Faleh Group College with the University of Aberdeen, Joaan Bin Jassim Academy for Defence Studies, Ahmed Bin Mohammed Military College
- 3 Weill Cornell Medicine-Qatar, University College London, TotalEnergies Research Center, Texas A&M University at Qatar, Qatar Science & Technology Park, Qatar Shell Research and Technology Center, Qatar National Research Fund, Qatar Mobility Innovation Center, QF's annual Alumni Forum, Qatar Foundation, Qatar Environment and Energy Research Institute, Qatar Computing Research Institute, Qatar Aeronautical College, Product Development Fund Tasmu Digital Valley, Northwestern University in Qatar, Iberdrola Innovation Middle East, HEC Paris in Qatar, Hamad Bin Khalifa University, Georgetown University in Qatar, Global Institute for Strategic Research, Equine Veterinary Medical Center, Education Development Institute, ConocoPhillips, Carnegie Mellon University Qatar, CCI France Qatar, AppLab
- 4 Qatar Research Development and Innovation Council, Qatar Open Innovation, Ooredoo Qatar, Qatar General Electricity & Water Corporation, Hassad Foods, Ministry of Energy and Industry, Rule of Law and Anti-Corruption Centre
- 5 Qatar National Bank, Qatar International Islamic Bank, Qatari Investors Group, Qatar Islamic Bank, Qatar Financial Markets Authority, Masraf Al Rayan, KPMG, Injaz Qatar, International Bank of Qatar, Hongkong and Shanghai Banking Corporation, Gulf Organisation for Research & Development, Doha Bank, Commercial Bank of Qatar, Canadian Business Council Qatar, Barwa Bank, Al Khaliji Commercial Bank
- 6 Philippine Business Council, Pakistan Business Council-Qatar, Portuguese Business Council

Figure 15 visualizes the **Qatar Cluster Communities Plot** using the **Louvain method**, a modularity-based algorithm for community detection. Each node represents an actor, and nodes within the same cluster share strong intra-group connections. Distinct clusters reveal tightly-knit communities within the ecosystem, highlighting collaborative networks and areas of influence, offering insights into how actors contribute to entrepreneurial dynamics.

Figure 15: Qatar Cluster communities plot



Communities of Interest

The Infomap method has segmented Qatar's Entrepreneurial Ecosystem into 11 distinct communities, each reflecting specific sectors and functions. Here is an analysis of each community:

Community of Interest #1: Innovation and Co-Working Spaces

Characteristics: This cluster includes co-working spaces (e.g., Workinton), global networks (ServCorp, Regus), and academic innovation hubs (College of the North Atlantic Qatar), focusing on early-stage innovation.

Role in Ecosystem: Provides foundational support and space for startups and small businesses, fostering collaboration and resource sharing.

Community of Interest #2: Academic and Research Institutions

Characteristics: Dominated by universities (Qatar University, Texas A&M at Qatar) and research entities (QSTP, QNRF), alongside global educational institutions like HEC Paris.

Role in Ecosystem: Drives research, education, and advanced knowledge creation, acting as a backbone for innovation and human capital development.

Community of Interest #3: Entrepreneurial Networks and Financing Hubs

Characteristics: Includes financial actors (Qatar Financial Centre, QDB) and entrepreneurship-support entities (QSTP Tech Venture Fund, Qatar Business Incubation Center QBIC).

Role in Ecosystem: Facilitates access to funding, mentorship, and networking, vital for entrepreneurial ecosystem growth.

Community of Interest #4: Transport and Tech Integration

Characteristics: Focused on mobility and technology integration, including entities like The Ministry of Transport and Qatar IT Conference.

Role in Ecosystem: Advances technological infrastructure and policies critical to entrepreneurial development in tech and transport.

Community of Interest #5: Startups and Micro-Enterprises

Characteristics: Focused on grassroots startups, this cluster includes 360 Nautica and Risin Ventures, supporting small businesses and innovative ventures.

Role in Ecosystem: Acts as an incubator for emerging businesses, accelerating entrepreneurial activity.

Community of Interest #6: Financial Institutions

Characteristics: Features major financial institutions (QNB, QIB, HSBC) providing essential capital and investment infrastructure.

Role in Ecosystem: Backbone of economic stability, enabling financial access for entrepreneurs.

Community of Interest #7: Professional Consulting and Services

Characteristics: Includes McKinsey, Deloitte, and KPMG, offering consulting, financial advisory, and strategy services.

Role in Ecosystem: Guides businesses with expertise and strategic planning, enhancing operational efficiency.

Community of Interest #8: Multinational Business Councils

Characteristics: Composed of international councils like the Philippine and Portuguese Business Councils, emphasizing global trade and collaboration.

Role in Ecosystem: Builds bridges for international business connections and partnerships.

Community of Interest #9: Tech Support and Incubation

Characteristics: Features digital incubators (Digital Incubation Center, elev8) and innovation-focused programs like Mada - Qatar Assistive Technology Center.

Role in Ecosystem: Promotes digital transformation and technology entrepreneurship across various industries.

Community of Interest #10: Niche Financial Services

Characteristics: Small cluster of niche financial entities like International Bank of Qatar and Barwa Bank, providing specialized services.

Role in Ecosystem: Supports financial innovation with tailored banking and investment solutions.

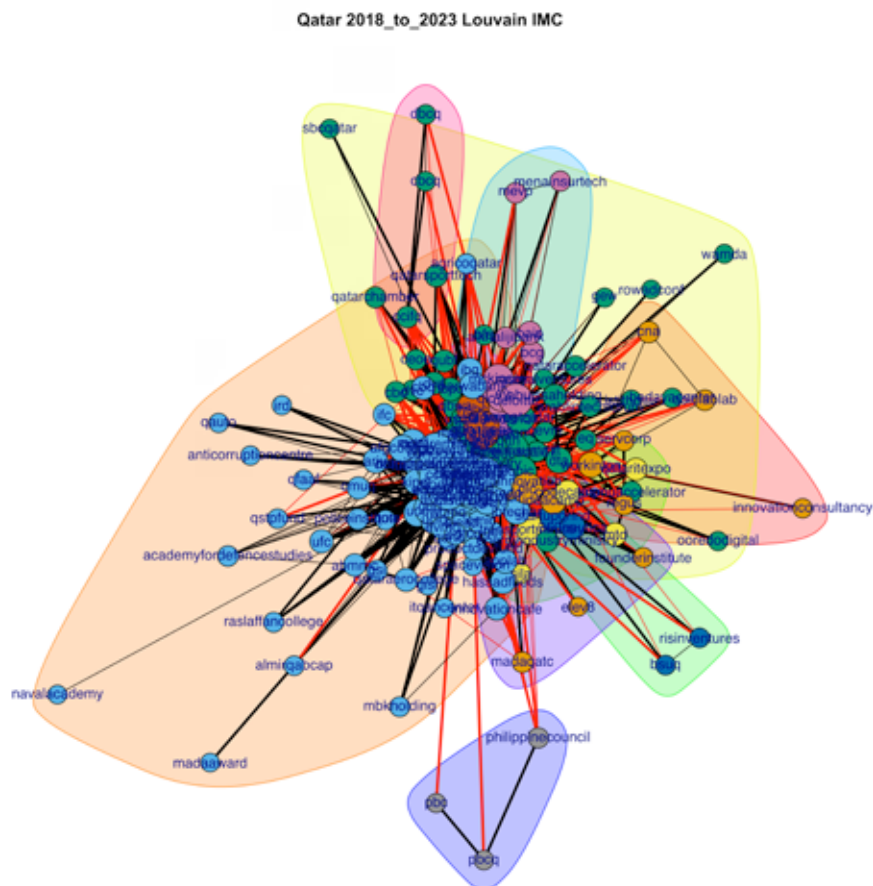
Community of Interest #11: Regional Business Councils

Characteristics: Includes Dutch, Canadian, and French business councils, fostering regional business development and foreign investments.

Role in Ecosystem: Facilitates regional economic integration and knowledge sharing.

Figure 16 presents the **Qatar Communities Plot** generated using the **Infomap method**, a flow-based algorithm for community detection. Nodes represent ecosystem actors, with clusters reflecting groups of actors that frequently interact. This visualization emphasizes the flow of information and collaboration within the ecosystem, uncovering key community structures and identifying influential actors driving connectivity and knowledge exchange.

Figure 16: Qatar Infomap communities plot



Conclusion About Infomap Communities

The Infomap segmentation reveals the ecosystem's diversity, highlighting specialized communities that address innovation, education, finance, and global integration. Each cluster's role aligns with Qatar's vision of a diversified economy, underscoring its strength in fostering collaboration, innovation, and entrepreneurship at local and global levels.

Table 11: Community of Interest Composition

Infomap Communities	Members
1	Workinton, ServCorp, Regus, QBIC Fab Lab, Innovation Consultancy, College of the North Atlantic Qatar
2	World Innovation Summit for Health, Weill Cornell Medicine-Qatar, University Foundation College, University of Doha for Science and Technology, University of Calgary in Qatar, University College London, TotalEnergies Research Center, Texas A&M University at Qatar, Spacevision, Strategic Innovation, Entrepreneurship & Economic Development, Sidra Medicine, Ras Laffan Emergency and Safety College, Qatar University, QSTP Tech Venture Fund, Qatar Science & Technology Park, Qatar Shell Research and Technology Center, Qatar Research Development and Innovation Council, Qatar Open Innovation, Qatar Olympic Academy, Qatar National Research Fund, Centre National De Convention Du Qatar, Qatar Mobility Innovation Center, QF's annual Alumni Forum, Qatar Foundation, Qatar Environment and Energy Research Institute, Qatar Computing Research Institute, Q Auto, Qatar Aeronautical College, Product Development Fund Tasmu Digital Valley, Police Institute, Ooredoo Qatar, Northwestern University in Qatar, Mohammed Bin Ghanem Al Ghanem Naval Academy, Nama, Ministry of Municipality and Environment, Ministry of Social Development and Family, Ministry of Environment and Climate Change, MBK Holdings, Mada Innovation Award, Lusail University, KINDI Center For Computing Research, Qatar General Electricity & Water Corporation, Itqan Clinical Simulation and Innovation Centre, Infrastructure Research and Development, Innovation Cafe, International Finance Corporation, Ibtechar, Iberdrola Innovation Middle East, Hamad Medical Corporation, HEC Paris in Qatar, Hamad Bin Khalifa University, Hassad Foods, Georgetown University in Qatar, Global Studies Institute, Gulf Organisation for Research & Development, Global Institute for Strategic Research, Equine Veterinary Medical Center, ExxonMobil Research Qatar, Ministry of Education and Higher Education, Education Development Institute, Doha Institute for Graduate Studies, Ministry of Culture, ConocoPhillips, Carnegie Mellon University Qatar, Community College of Qatar, Commercial Bank of Qatar, Biomedical Research Center, Aspetar Orthopaedic and Sports Medicine Hospital, Al Rayyan International University College, AppLab, Rule of Law and Anti-Corruption Centre, Al Mirqab Capital, Al Jazeera, Agrico for Agricultural Development, Al Faleh Group College with the University of Aberdeen, Joaan Bin Jassim Academy for Defence Studies, Ahmed Bin Mohammed Military College

3	Web Summit Qatar, Wamda, US-Qatar Business Council, TASMU Accelerator, Scale7, Singapore Business Council Qatar, Rowad Conference, Rasmal Ventures, Qatar-Ukraine Business Forum, Qatar Investment Authority, Business Innovation Park, Center of Entrepreneurship Qatar FinTech Hub, Qatar Financial Markets Authority, Qatar Financial Centre, Qatar Finance and Business Academy, Qatar Exchange Venture Market, Qatar Development Bank, Qatari Businesswomen Association, Qatar Business Incubation Center QBIC, Qatari Businessmen Association, Qatar Sports Tech, Accelerator Qatar Business District, Qatar Chamber of Commerce and Industry, Ooredoo Digital & Beyond, Ministry of Commerce and Industry, ISTITHMAR (SME EQUITY), Invest Qatar, Indian Business & Professional Council, Global Entrepreneurship Week Network, Enterprise Qatar, Ministry of Energy and Industry, Doha Women Forum, Doha Venture Capital, Doha Tech Angels, Center for Entrepreneurship & Organizational Excellence, Croatian Business Council, Bangladesh Forum Qatar, Bedaya Center for Entrepreneurship and Career Development, Arab Fintech Forum, Abu Issa Holding
4	The Ministry of Transport, SILA, Qatar IT Conference and Exhibition, Make the Deal, Code Camp
5	StartupGrind, Risin Ventures, Business Start Up Qatar, 360 Nautica
6	Qatar National Bank, Qatar International Islamic Bank, Qatari Investors Group, Qatar Islamic Bank, Masraf Al Rayan, Injaz Qatar, Hongkong and Shanghai Banking Corporation, Doha Bank
7	Qatar Insurance Group, Middle East Venture Partners, MENA Insurtech, McKinsey & Company, KPMG, Ernst & Young, Deloitte, Boston Consulting Group, Bain & Company Doha, Al Khaliji Commercial Bank
8	Philippine Business Council, Pakistan Business Council-Qatar, Portuguese Business Council
9	Ministry of Communications and Information Technology, Mada - Qatar Assistive Technology Center, Idea Camp, Founder Institute, elev8, Digital Incubation Center
10	International Bank of Qatar, Barwa Bank
11	Dutch Business Council Qatar, CCI France Qatar, Canadian Business Council Qatar

Takeaways on both Cluster and Infomap Communities

The Cluster and Infomap analyses reveal Qatar's Entrepreneurial Ecosystem's strengths in innovation, finance, and education, showcasing distinct but interconnected communities. These insights emphasize the ecosystem's diversity, collaboration potential, and targeted sectoral growth, providing a roadmap for strategic investments, policy alignment, and

fostering global competitiveness through tailored entrepreneurship support and resource allocation.

6.4.2. VRIO Analysis: Strategic Community Evaluation

The VRIO framework evaluates **Value, Rarity, Inimitability, and Organization** to identify communities' strategic potential in driving competitive advantages. Applied to Qatar's entrepreneurial ecosystem, it highlights which communities effectively contribute unique, non-replicable resources, fostering innovation and economic development while guiding targeted policy interventions and ecosystem growth.

Table 12: VRIO evaluation of each Infomap Community

Community	Valuable	Rare	Inimitable	Organized
1	0.35 – moderate value contribution	0.06 – low uniqueness in resources	0.46 – moderately hard to imitate	0.50 – adequately organized
2	0.15 – limited value generation	0.05 – rare resource scarcity	0.33 – average imitation difficulty	0.49 – well-structured community
3	0.00 – no direct value contribution	0.00 – resources not rare	0.60 – high imitation difficulty	0.50 – sufficient organization
4	0.00 – lacks value generation	0.00 – lacks rarity	1.00 – uniquely inimitable resources	0.33 – weak organizational capacity
5	0.00 – no inherent value creation	0.07 – low resource uniqueness	1.00 – high imitation resistance	0.50 – strong organization
6	ns	ns	ns	ns

Evaluating Strategic Community Value

The VRIO analysis underscores **Community 1** as moderately valuable and organized, while **Communities 4 and 5** exhibit high inimitability but limited value and rarity. Improving value generation and structural organization across communities would enhance their strategic roles, driving entrepreneurial growth and resource optimization.

6.4.3. Synergy: Identifying Collaboration Patterns

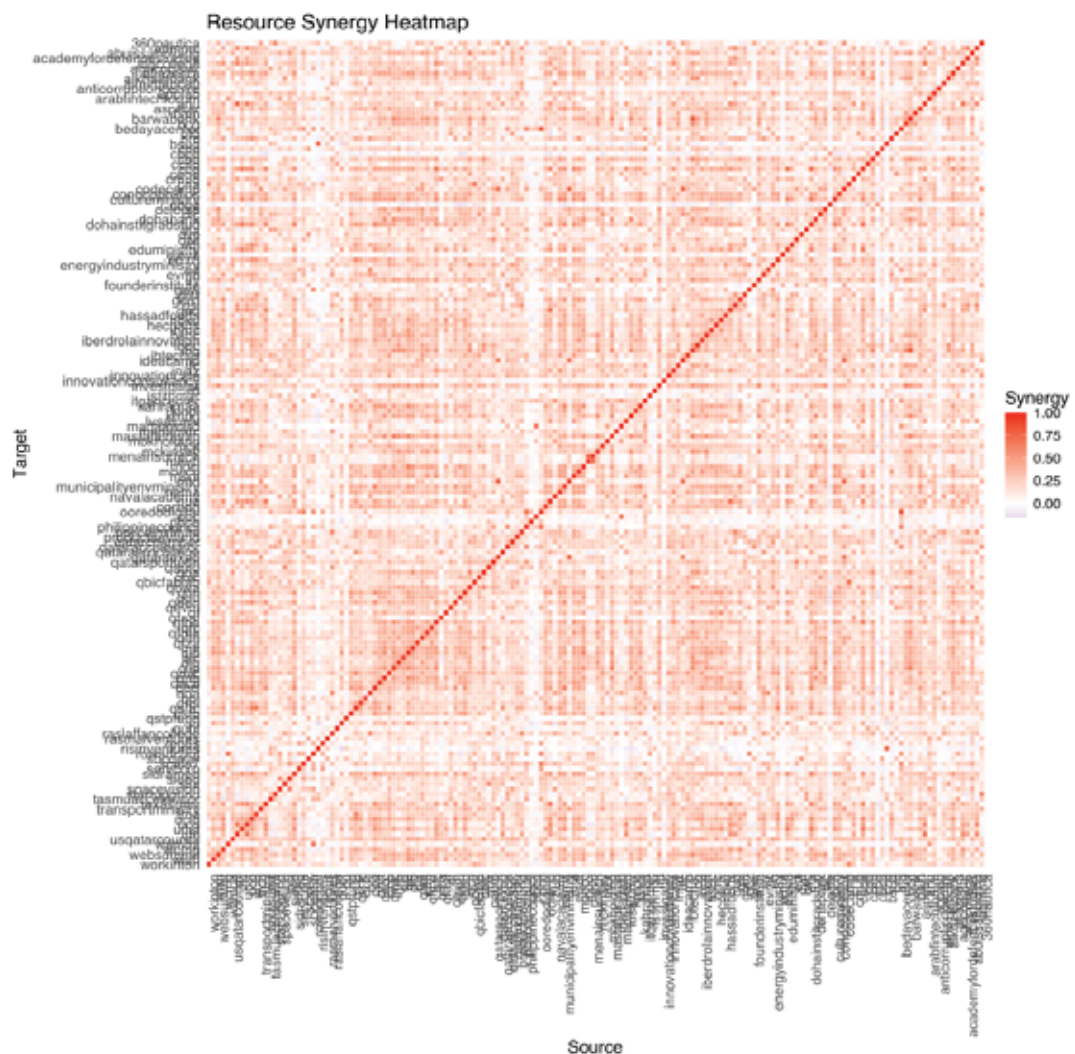
Synergies

Understanding synergies among Qatar's entrepreneurial ecosystem actors is crucial for fostering innovation, collaboration, and sustainable growth, enabling targeted interventions to enhance the ecosystem's efficiency and competitiveness in the global economy.

The strongest synergies are observed between key innovation hubs and governmental bodies, showcasing effective collaboration in policymaking and resource allocation. Conversely, weaker synergies between academia and private investors highlight missed opportunities for research commercialization and startup funding. Bridging these gaps is vital for a robust, interconnected entrepreneurial ecosystem in Qatar.

Table 17 presents a heatmap illustrating the synergies between actors in Qatar's entrepreneurial ecosystem. The color intensity indicates the strength of collaboration, with darker shades reflecting stronger partnerships. This visualization highlights areas of high connectivity and alignment, uncovering key relationships that drive ecosystem dynamics, while also identifying potential gaps where new synergies could be developed.

Figure 17: Synergies between actors



The analysis highlights Qatar's robust collaboration between government and innovation hubs, driving strategic growth. However, weak links between academia and investors indicate untapped potential in research commercialization and funding. Strengthening these connections and fostering inclusivity across ecosystem actors can unlock greater innovation, investment opportunities, and sustainable development for Qatar's entrepreneurial landscape.

6.5 – Key Indicators

Table 13. Key Indicators

Indicators	Measures	Level
Network Connectivity Index	High density (0.1615) and strength of ties (average weight 33.022) foster active resource sharing and innovation across Qatar's ecosystem.	
Diversity of Actor Roles	Diverse actors, from universities to incubators, ensure resilience and the ability to address varied entrepreneurial needs.	
Regional Disparity Index	Doha dominates, but regional gaps (e.g., Al-Khor, Lusail) highlight opportunities for interventions to balance ecosystem development.	
Centrality and Leadership Score	Key influencers like Qatar Foundation and QDB drive strategic collaboration, bolstering ecosystem stability and innovation capabilities.	
Synergy and Collaboration Potential	Strong synergy between government and innovation hubs supports collaborative culture, though academia-investor gaps limit full ecosystem optimization.	
Impact on Entrepreneurship	Robust connectivity, diverse roles, and strong leadership foster innovation, but regional and collaboration gaps constrain holistic entrepreneurial growth.	

Analysis

Qatar's entrepreneurial ecosystem demonstrates significant connectivity, diversity, and leadership, driven by key actors like Qatar Foundation and Qatar University. Strong ties and collaborations, particularly between government bodies and innovation hubs, ensure robust resource sharing and innovation. However, regional disparities, such as the dominance of Doha, and weak academia-investor connections, hinder comprehensive development. Despite these gaps, diverse actor roles and strategic centrality contribute to a resilient and dynamic ecosystem. Addressing disparities and collaboration inefficiencies can enhance inclusivity and resource optimization, supporting Qatar's long-term entrepreneurial growth and global competitiveness.

Recommendations

Address Regional Disparities: Invest in underrepresented cities like Lusail and Al-Khor to create balanced resource distribution and encourage entrepreneurial activity beyond Doha.

Foster Academia-Investor Collaboration: Bridge gaps between universities and private investors through targeted programs, facilitating research commercialization and startup funding.

Enhance Niche Actor Support: Strengthen support for underrepresented actor types, such as FabLabs and business angels, to foster grassroots and creative entrepreneurship.

Promote Inclusive Synergies: Build on strong government-innovation hub synergies by creating platforms that connect academia, startups, and investors, enhancing ecosystem collaboration.

Strengthen Leadership Nodes: Leverage central actors to guide strategic initiatives and mentor smaller players, enhancing system-wide resilience.

These targeted interventions will enhance inclusivity, foster innovation, and position Qatar's entrepreneurial ecosystem for sustainable growth and global leadership.

7. Topic and Thematic Analysis

This chapter delves into the activity themes that fuel Qatar's entrepreneurial ecosystem. We identify, classify, and analyze key topics using an in-depth analytical approach, revealing their impact on innovation, entrepreneurship, and the national economy.

Table 14. Word Clouds



7.1 – Identification of Activity Topics

We enjoy this chapter by investigating the methods used to identify and decrypt the activity themes that define Qatar's entrepreneurial ecosystem. By delving into the results of our analysis, we reveal the major themes that emerge from the data, laying the groundwork for our in-depth investigation.

Table 15: Basic Modeling from 2018 to 2023

2018	2019	2020	2021	2022	2023
0,06091 : qatar invest sheikh minist econom	0,05766 : qatar sector invest busi qatari	0,05987 : student educ programm univers learn	0,05593 : net trade valu volum see	0,07965 : net trade see market volum	0,07568 : net main see index market
0,05806 : develop nation social support centr	0,05402 : technolog digit innov data smart	0,05639 : trade net valu volum sector	0,05585 : qatar minist cooper sheikh state	0,06233 : innov tech-nolog digit qatar busi	0,06294 : win team match player last
0,05612 : student univers educ school graduat	0,05397 : qatar award nation achiev centr	0,05617 : cooper bin sheikh qatar meet	0,0557 : student educ school programm univers	0,05824 : student educ school programm univers	0,06185 : innov technolog digit busi transform
0,05467 : busi technolog innov develop entrepreneur	0,0534 : minist qatar meet sheikh secur	0,05574 : coronavirus case virus covid peopl	0,05459 : depart ministri campaign mme food	0,05603 : rate price market year growth	0,05467 : health hmc care medic patient
0,05164 : qatar confer host event safeti	0,0533 : programm train particip develop manag	0,05465 : qatar sector busi invest compani	0,05409 : innov programm qatar busi develop	0,0535 : invest sector qatar compani busi	0,05451 : educ student programm school univers
0,05095 : have time think see work	0,05266 : student univers educ colleg school	0,05369 : bank financi financ growth rate	0,05341 : growth year bank increas rate	0,05153 : research discuss develop professor scienc	0,0513 : event sport qatar world doha
0,05092 : research health use road scienc	0,05237 : research energi develop project sustain	0,05326 : technolog innov qatar programm startup	0,05243 : event sport organis confer communiti	0,05021 : qatar nation achiev develop support	0,0508 : invest countri qatar econom cooper
0,05073 : say polic peopl offic report	0,05236 : bank financi compani financ market	0,05251 : economi econom rise expect fall	0,05062 : say polic peopl forc hous	0,0502 : minist sheikh bin meet cooper	0,05064 : research discuss confer event forum
0,05067 : expect growth price rate increas	0,05181 : win team play world run	0,05228 : custom servic digit qib mobil	0,05037 : product energi food environ-ment plant	0,04975 : bank custom servic digit financ	0,04909 : say gaza peopl militari pales-tinian

0,05037 : cultur event exhibit museum celebr	0,05154 : woman issu forum role educ	0,05079 : say polic minist forc indian	0,05028 : bank custom qib servic award	0,04721 : qatar cultur museum exhibit event	0,04851 : bank custom servic award group
0,04996 : programm manag skill profession train	0,05077 : have peopl know say get	0,04899 : pandem covid world challeng global	0,05019 : digit servic technolog busi ooredoo	0,0468 : world cup sport fifa qatar	0,04765 : bin sheikh minist moham hamad
0,04969 : energi food product new produc	0,0504 : say attack peopl polic fire	0,0479 : research use water studi develop	0,04902 : compani market invest say investor	0,04673 : health medic care healthcar hmc	0,04763 : council law ministri public issu
0,049 : sport team world aspetar qatar	0,04961 : rate year tax growth increas	0,04751 : woman have peopl time art	0,04828 : global chang world sustain develop	0,04592 : secur ministri issu law order	0,04732 : develop countri intern human world
0,04849 : bank financi financ market award	0,04846 : event cultur qatar doha exhibit	0,04693 : player win play game team	0,04806 : qatar sector countri host chariti	0,04544 : team win have play player	0,0466 : economi year econom polici rate
0,04744 : compani firm say fund group	0,04725 : compani china trade say chines	0,04626 : school ministri public student measur	0,04725 : have say time peopl make	0,04528 : custom new card visa featur	0,04651 : financi bank compani market sector
0,04672 : state china meet trump govern	0,04554 : protest minist say parti govern	0,04563 : health sport medic care healthcar	0,04692 : research develop studi qatar scienc	0,04484 : say peopl tell polic russia	0,04483 : activ social child communiti famili
0,04453 : countri qatar region intern world	0,04465 : say polic case offic drug	0,04538 : hong china kong say hsbc	0,04627 : win club last final game	0,04442 : : sustain energi project product climat	0,04391 : sustain energi project climat product
0,04326 : trade valu net volum sector	0,04408 : sport health medic aspir aspetar	0,04238 : project area road work public	0,0449 : covid vaccin case travel new	0,04422 : peopl arab woman social have	0,04047 : new vehicl design car drive
0,04302 : win top first year second	0,04309 : area road line station doha	0,04196 : trump protest state elect polic	0,04366 : health medic care healthcar patient	0,03974 : servic oper transport network provid	0,03971 : qatar host launch safeti worker
0,04284 : player play match game team	0,04304 : valu trade sector net volum	0,04172 : product food suppli local produc	0,04217 : area park road vehicl car	0,03797 : covid new year report month	0,03538 : peopl say have allah mean

We added similar elements at the bottom of this list by categorizing them by orientation theme.

7.2 – Longitudinal Analysis

The topic modeling data for the collaboration content in the Entrepreneurial Ecosystem Network between 2018 and 2023 reveals significant themes each year that highlight the evolving focus of the entrepreneurial activities and discussions in Qatar. This analysis will explore the thematic evolution and identify key shifts in focus over this period:

2018 – Business and Technological Innovation

Prominent Topics: «busi technolog innov develop entrepreneur»

Analysis: The year 2018 underscores a strong orientation towards business innovation and technological advancements. This reflects an ecosystem maturing with a focus on cutting-edge technologies and innovative business practices, essential for driving forward Qatar's entrepreneurial ambitions in an increasingly digital world.

2019 – Sector-Specific Investments and Digital Transformation

Prominent Topics: «qatar sector invest busi qatari»

Analysis: In 2019, the focus continued on sector-specific investments, particularly in business domains and digital transformation initiatives. This underscores ongoing efforts to tailor the entrepreneurial ecosystem to meet the demands of modern business challenges and opportunities.

2020 – Education and Digital Adaptation

Prominent Topics: «student educ programm univers learn»

Analysis: With a significant shift towards education, 2020 highlights the importance of learning programs and university roles in adapting to digital demands, crucial in a year dominated by global shifts to remote interactions.

2021 – Trade and Economic Adjustments

Prominent Topics: «net trade valu volum see»

Analysis: The emphasis on trade and its value underscores adjustments in economic strategies to navigate post-pandemic recovery, focusing on enhancing trade volumes and market adaptability.

2022 - Continued Focus on Trade and Market Dynamics

Prominent Topics: «net trade see market volum»

Analysis: Continuing from the previous year, the focus remains on strengthening trade and understanding market dynamics, crucial for economic recovery and growth.

2023 - Innovation and Technological Integration

Prominent Topics: «innov technolog digit busi transform»

Analysis: The year 2023 marks a clear pivot towards innovation and technological integration, highlighting a strategic orientation towards embracing digital transformation across business practices to drive future growth.

Conclusion on Topic Analysis from 2018 to 2023

This thematic analysis over the years shows how Qatar's entrepreneurial ecosystem is strategically evolving, emphasizing economic resilience, sector-specific investments, educational enhancements, and technological integration. Each theme builds upon the previous years, reflecting a structured approach to fostering a sustainable and robust environment for entrepreneurs. This progression illustrates the ecosystem's adaptability and its focus on maintaining and extending its competitive edge in a rapidly changing global landscape.

7.3. Thematic Analysis

Based on the analysis of topics from 2018 to 2023 in the Qatar's Entrepreneurial Ecosystem, here are the five main themes of action:

Technological Innovation and Digital Transformation

Overview: This theme includes topics like «technolog digit innov data smart» and «innov technolog digit qatar busi,» indicating a strong focus on digital innovation and the integration of smart technologies into business practices.

Analysis: The recurring emphasis on technology and digital transformation suggests that Qatar is prioritizing the adaptation of new technologies to enhance business operations. This theme is crucial for maintaining competitiveness in a global market and indicates efforts to position Qatar as a hub for technological innovation.

Trade and Market Dynamics

Overview: Highlighted by topics such as «trade net valu volum sector» and «net trade see market volum,» this theme focuses on the international trade dynamics and market fluctuations.

Analysis: This theme underscores Qatar's strategic position in international trade and its responsiveness to global market conditions. The emphasis on trade volumes and values indicates active engagement in enhancing trade capacities and optimizing market opportunities, which is essential for economic diversification.

Educational Focus and Workforce Development

Overview: The topics «student educ programm univers learn» and «educ student programm school univers» reveal a focus on educational programs and learning initiatives.

Analysis: Qatar's emphasis on education in its entrepreneurial ecosystem highlights the importance of human capital development. By investing in education, Qatar ensures a well-prepared workforce capable of supporting and sustaining its economic and entrepreneurial ambitions.that is crucial for the sustainability and resilience of the entrepreneurial ecosystem.

Financial Services and Investments

Overview: Reflected in topics like «qatar invest sheikh minist econom» and «invest sector qatar compani busi,» this theme deals with investments and economic strategies led by key figures and sectors.

Analysis: The focus on investments and financial services suggests a robust effort to attract and manage financial resources effectively. It highlights the role of key stakeholders in shaping the financial landscape, crucial for funding innovation and supporting startups and business expansion.

Health and Public Safety

Prominent Topics: Marked by «coronavirus case virus covid peopl» and «health hmc care medic patient,» this theme reflects the focus on healthcare issues, particularly public health challenges like COVID-19.

Analysis: The emphasis on health and safety, especially in the context of the pandemic, indicates Qatar's commitment to addressing public health crises. It also shows the integration of health innovations and medical research into the broader entrepreneurial ecosystem, which is essential for developing a resilient public health infrastructure.

From 2018 to 2023, Qatar's Entrepreneurial Ecosystem topic modeling reveals five prominent themes: Technological Innovation and Digital Transformation, Trade and Market Dynamics, Educational Focus and Workforce Development, Financial Services and Investments, and Health and Public Safety. These themes highlight a multifaceted approach to fostering a robust economic environment, with a strong emphasis on integrating advanced technologies, enhancing educational programs, optimizing financial services, actively engaging in international trade, and prioritizing health and public safety.

Table 16: Thematic Analysis

Thematic Area	Description	Summative Topic Evaluation	Sum of Evaluations
Technological Innovation and Digital Transformation	Focus on integrating advanced technologies and digital processes to modernize and enhance competitiveness in various sectors.	2019: 0.05766, 2020: 0.05574, 2021: 0.05409, 2022: 0.06233, 2023: 0.06185	0.29167
Trade and Market Dynamics	Emphasis on enhancing Qatar's role in international trade and responding to market dynamics.	2018: 0.06091, 2019: 0.05465, 2020: 0.05639, 2021: 0.07965, 2022: 0.07568	0.32728
Educational Focus and Workforce Development	Prioritization of educational initiatives to develop a skilled workforce for a knowledge-based economy.	2018: 0.05612, 2019: 0.05397, 2020: 0.05987, 2021: 0.0557, 2022: 0.05824, 2023: 0.05467	0.33857
Financial Services and Investments	Enhancement of financial services and investment opportunities to support economic growth and entrepreneurship.	2018: 0.06091, 2019: 0.0535, 2020: 0.05465, 2021: 0.05585, 2022: 0.0535, 2023: 0.0508	0.32921
Health and Public Safety	Commitment to improving health standards and public safety, pivotal during periods such as the COVID-19 pandemic.	2020: 0.05574 (highlighting COVID-19), 2021: 0.05467, 2022: 0.05451, 2023: 0.05467	0.21959

Trends and Insights:

Technological Innovation and Digital Transformation: There is a sustained focus on embedding advanced technologies and digital processes across various business sectors. This trend indicates Qatar's aim to position itself as a leader in digital innovation, crucial for economic modernization and increased global competitiveness.

Trade and Market Dynamics: The emphasis on trade volumes and market dynamics reflects Qatar's strategic efforts to bolster its position in international trade, responding to global economic shifts and capitalizing on new market opportunities.

Educational Focus and Workforce Development: By prioritizing education, Qatar is investing in its human capital to ensure a workforce capable of driving and sustaining its economic growth. This trend is aligned with global movements towards knowledge-based economies where educational attainment correlates strongly with innovation capacity.

Financial Services and Investments: The focus on financial and investment themes underscores efforts to attract investments and manage economic strategies effectively. This trend reveals a proactive approach to economic planning and financial management to support entrepreneurship and business scalability.

Health and Public Safety: The attention to health, particularly highlighted during the COVID-19 pandemic, shows Qatar's commitment to public health as a cornerstone of social and economic stability. The integration of health into the entrepreneurial ecosystem highlights a holistic approach to building a resilient infrastructure that supports public well-being and safety.

These trends and insights collectively demonstrate Qatar's strategic initiatives to enhance its entrepreneurial ecosystem through a balanced focus on technology, education, finance, trade, and health. This comprehensive approach is aimed at creating a dynamic and sustainable environment that supports innovation and facilitates economic diversification.

7.4 – Analysis of Activity Themes

A comprehensive thematic analysis of Qatar's Entrepreneurial Ecosystem regarding entrepreneurship from 2018 to 2023 showcases a well-rounded approach to building a dynamic and sustainable environment for business creators. This analysis spans several critical thematic areas, each underpinning the overarching objective of enhancing and diversifying the entrepreneurial landscape.

Technological Innovation and Digital Transformation (Sum of Evaluations: 0.29167)

Focus: This theme concentrates on incorporating cutting-edge technology and digital strategies to modernize Qatar's various industrial sectors. It includes the adoption of digital technologies in businesses and public services, enhancing efficiencies and opening new avenues for entrepreneurship.

Impact: The emphasis on technological innovation and digital transformation has significantly facilitated entrepreneurial activity by lowering barriers to entry, enabling scalability, and enhancing the global reach of startups. It also encourages a culture of innovation, crucial for the development of a dynamic and competitive economic landscape.

Trade and Market Dynamics (Sum of Evaluations: 0.32728)

Focus: This theme focuses on enhancing Qatar's integration into global markets and adapting to international trade dynamics. It covers initiatives to increase trade volumes, improve market access, and diversify Qatar's economic activities beyond traditional sectors.

Impact: Strengthening trade and market dynamics boosts the entrepreneurial ecosystem by providing broader markets for startups, enhancing competitive practices, and fostering a business environment conducive to foreign and local investment. This global orientation helps local entrepreneurs align with international standards and practices.

Educational Focus and Workforce Development (Sum of Evaluations: 0.33857)

Focus: The focus here is on bolstering educational programs and institutions to develop a skilled workforce that meets the demands of a modern, knowledge-based economy. It includes enhancing higher education, vocational training, and continuous professional development.

Impact: By improving education and workforce capabilities, Qatar is able to supply its industries with highly skilled professionals and innovators, essential for driving forward technological advancements and entrepreneurial ventures. A well-educated workforce attracts higher-quality investments and supports the creation of sophisticated startups.

Financial Services and Investments (Sum of Evaluations: 0.32921)

Focus: This theme addresses the enhancement of financial infrastructure, access to capital, and investment opportunities to support business operations and economic growth. It includes the development of financial markets, banking services, and investment frameworks conducive to entrepreneurship.

Impact: Enhanced financial services and investment opportunities directly impact entrepreneurship by providing necessary capital for startups and growth enterprises. Efficient financial services also ensure that businesses can leverage financial tools to expand and innovate, thereby fostering a robust entrepreneurial ecosystem.

Health and Public Safety (Sum of Evaluations: 0.21959)

Focus: Focuses on improving public health standards and safety, which has been particularly pivotal during health crises like the COVID-19 pandemic. This includes the development of healthcare infrastructure, public health initiatives, and safety regulations.

Impact: The commitment to health and public safety is vital for maintaining a productive workforce and ensuring a stable business environment. Effective health systems and safety measures preserve the well-being of entrepreneurs and employees alike,

fostering a secure environment where businesses can operate and innovate without undue risk.

Conclusion

These themes collectively underline Qatar's strategic approach to nurturing an environment conducive to innovation and business growth, highlighting the interconnectedness of educational advancement, technological prowess, financial robustness, and a stable socio-economic backdrop in fostering a thriving entrepreneurial landscape.

7.5 – Impact on Innovation, Entrepreneurship, and Employment

The themes of action identified within Qatar's Entrepreneurial Ecosystem from 2018 to 2023 significantly impact innovation, entrepreneurship, and employment in several ways:

Impact on Innovation

Technological innovation, trade dynamics, and robust financial services drive Qatar's innovation landscape, fostering advanced solutions and products across sectors. Digital transformation accelerates technological advancements, while education strengthens the workforce's ability to innovate. Enhanced market access encourages businesses to meet global standards, and investments fuel R&D efforts. Public safety advancements further stimulate innovation in healthcare and medical technologies, ensuring the ecosystem remains competitive and adaptive to future challenges.

Impact on Entrepreneurship

Digital transformation and improved market dynamics empower entrepreneurs with new business models, larger markets, and cost-efficient operations. Education equips the workforce with essential entrepreneurial skills, fostering a culture of enterprise among young professionals. Accessible financial services provide startups with the capital needed to launch and scale operations, while a secure public health environment reduces risks, enabling entrepreneurs to focus on growth. Together, these elements create a supportive entrepreneurial ecosystem.

Impact on Employment

Digitalization, educational reforms, and financial investments create jobs in emerging tech and traditional sectors. Trade expansion generates diverse job opportunities, while a skilled workforce enhances employability and stability in competitive markets. Public safety investments ensure a healthy workforce, bolstering productivity and economic growth. Collectively, these factors reduce unemployment and prepare Qatar's workforce for future demands, aligning with the nation's sustainable development goals.

Conclusion

The thematic analysis of Qatar's Entrepreneurial Ecosystem (2018–2023) highlights a comprehensive approach to fostering innovation, entrepreneurship, and employment. Technological advancements, robust financial services, and enhanced education drive innovation and empower entrepreneurs with efficient models and broader market access. Investments in trade dynamics and public safety create a secure and dynamic business environment, while educational reforms and digitalization generate diverse job opportunities, aligning with sustainable development goals. This integrated strategy ensures a competitive, resilient, and inclusive entrepreneurial landscape.

7.6 – Key Indicators

Table 17. Key Indicators

Indicators	Measures	Level
Entrepreneurship oriented Financing	Financial themes highlight robust investment frameworks and funding access, supporting startups and SMEs with essential capital for growth and innovation.	★★★★☆
Action centered on the distribution of other resources	Themes emphasize digital tools, education, and market access, ensuring a broad distribution of resources for ecosystem sustainability.	★★★☆☆
Support-focused action	Strong focus on training, innovation hubs, and public safety creates a supportive environment for entrepreneurial activities and resilience.	★★★★★
Action centered on the creation of opportunities	Trade dynamics, market diversification, and digital transformation drive the creation of entrepreneurial opportunities across various sectors.	★★★★★
Action centered on Ecosystem Governance	Stable governance themes ensure policy alignment, sector-specific support, and effective coordination across ecosystem actors for sustained growth.	★★★★★
Impact on Entrepreneurship	Enhanced innovation, education, and trade integration foster a dynamic entrepreneurial environment with scalable opportunities and broad ecosystem impact.	★★★★☆

Analysis

Chapter 7 illustrates Qatar's strategic focus on fostering a thriving entrepreneurial ecosystem through key thematic areas. The emphasis on financial services provides startups and SMEs with access to critical funding. Education and workforce development ensure a skilled labor force aligned with knowledge-economy demands. Digital transformation and innovation hubs lower barriers for new ventures, while trade dynamics expand market opportunities. Strong governance, supported by health and safety initiatives, creates a stable environment for entrepreneurial activities. This cohesive framework not only promotes innovation and entrepreneurship but also aligns with Qatar's Vision 2030 goals of economic diversification and sustainability.

Recommendations

To enhance the entrepreneurial ecosystem, Qatar should continue to invest in financial services tailored to SMEs and startups, ensuring easy access to capital for innovation. Expanding training initiatives focused on emerging tech fields like AI and CleanTech can address workforce gaps. Increased emphasis on underrepresented niches like AgriTech and FinTech could diversify economic activities further. Strengthening public-private partnerships will optimize governance and resource distribution. Finally, developing platforms for global trade and collaboration can boost market access for entrepreneurs. This balanced approach will create a resilient, innovative, and inclusive entrepreneurial landscape.

Very High Trophic Level Actors**Examples: Founder Institute (1.97), MEVP (1.86), Menainurtech (1.86)**

Contribution: These actors are pivotal drivers of the ecosystem, offering strategic resources, expertise, and leadership. Their influence fosters innovation, promotes ecosystem integration, and attracts global investments. They serve as innovation hubs and knowledge-sharing platforms, ensuring resource allocation aligns with entrepreneurial demands.

High Trophic Level Actors**Examples: Servcorp (1.42), Qatar IT Expo (1.43), Bain (1.72)**

Contribution: High-level actors significantly enhance ecosystem maturity by providing funding, marketing strategies, mentorship, and innovation support. Their impact is evident in scaling startups, strengthening market entry, and creating synergies between emerging and established enterprises.

Moderate Trophic Level Actors**Examples: Scale7 (0.30), Qatar Business Incubation Center (0.21), QDB (-0.49)**

Contribution: These actors act as intermediaries, bridging gaps between high-level influencers and emerging businesses. They facilitate niche services like incubation, prototyping, and resource distribution, ensuring adaptability and operational support across the ecosystem.

Low Trophic Level Actors**Examples: Aspetar (-1.02), ConocoPhillips (-0.98), Lusail University (-0.25)**

Contribution: Actors at low trophic levels focus on specialized or sectoral support, such as healthcare innovation, academia, and regional education. While their broader ecosystem influence is limited, their expertise fosters diversity in underrepresented sectors.

Very Low Trophic Level Actors**Examples: Qatar Foundation (-3.55), HBKU (-3.15), QNFR (-2.01)**

Contribution: These actors are foundational institutions focusing on long-term research, policy advocacy, and academic excellence. Their role in creating intellectual capital and influencing systemic policies is vital, though they depend on collaborations with higher trophic actors for execution.

Overall Contribution to Ecosystem Efficiency

Strategic Integration: Very high and high trophic actors lead the ecosystem by fostering innovation and ensuring resource flow.

Collaborative Bridging: Moderate-level actors play critical roles in connecting different ecosystem strata, ensuring inclusivity and adaptability.

Sectoral Diversity: Low and very low-level actors contribute niche expertise, enhancing ecosystem diversity and creating a robust base for specialized entrepreneurship.

Challenges: The reliance on a few high-level actors and the uneven influence of lower-level actors suggest potential risks during disruptions. Addressing these gaps by fostering mid-tier collaborations and enhancing low-level contributions is crucial for sustained ecosystem growth.

Table 18: Trophic Level analysis

Trophic level	Notable Actors	Analysis
Very High	Founder Institute, Middle East Venture Partners, MENA Insurtech, Dutch Business Council Qatar, Pakistan Business Council-Qatar, Bain & Company Doha, Doha Venture Capital, Canadian Business Council Qatar, International Bank of Qatar, Barwa Bank, Portuguese Business Council, 360 Nautica, Business Start Up Qatar, Al Khaliji Commercial Bank, Idea Camp, Make the Deal, Deloitte, Risin Ventures	Very high trophic level actors, such as Founder Institute and Middle East Venture Partners, are ecosystem drivers. They provide strategic resources, thought leadership, and innovation-focused initiatives. Their contributions establish benchmarks for excellence, enabling high-value collaboration, resource mobilization, and entrepreneurial growth. These actors significantly influence the ecosystem by fostering partnerships, funding, and policy shaping, creating an environment conducive to sustainable entrepreneurial ventures and scalable business success.

High

Qatar IT Conference and Exhibition, ServCorp, Innovation Consultancy, Singapore Business Council Qatar, elev8, Bangladesh Forum Qatar, StartupGrind, Qatar-Ukraine Business Forum, Qatari Investors Group, KPMG, Wamda, Qatar Insurance Group, Qatari Businesswomen Association, Digital Incubation Center, Ernst & Young, Center for Entrepreneurship & Organizational Excellence, Philippine Business Council, Rasmal Ventures, Regus, Business Innovation Park, Qatari Businessmen Association

High trophic level actors, like ServCorp and Singapore Business Council Qatar, contribute vital resources and operational support. They bridge strategic gaps by enabling startups and SMEs to access mentorship, funding, and business expertise. These actors play an essential role in ensuring ecosystem maturity and resilience through their collaborative efforts, scaling opportunities, and fostering cross-sectoral synergies. Their contributions amplify innovation and create a robust entrepreneurial foundation.

Average

Accelerator Qatar Business District, Hongkong and Shanghai Banking Corporation, Boston Consulting Group, Croatian Business Council, Arab Fintech Forum, Doha Tech Angels, Qatar Chamber of Commerce and Industry, US-Qatar Business Council, Qatar Financial Markets Authority, ISTITHMAR (SME EQUITY), Code Camp, Workinton, Masraf Al Rayan, Hassad Foods, Abu Issa Holding, Ooredoo Digital & Beyond, Qatar Exchange Venture Market, McKinsey & Company, Mada - Qatar Assistive Technology Center, Global Entrepreneurship Week Network, Doha Bank, The Ministry of Transport, Invest Qatar, TASMU Accelerator, College of the North Atlantic Qatar, Qatar Investment Authority, Web Summit Qatar

Average trophic level actors, including McKinsey & Company and TASMU Accelerator, function as intermediaries connecting emerging businesses with influential leaders. They support operational processes, niche innovation, and sectoral diversity. These actors help maintain ecosystem adaptability by addressing specific entrepreneurial needs and fostering collaboration. While their influence is moderate, their focus on resource alignment and operational support ensures consistent progress within the ecosystem.

Low

Ministry of Communications and Information Technology, Rule of Law and Anti-Corruption Centre, Rowad Conference, Bedaya Center for Entrepreneurship and Career Development, Qatar Finance and Business Academy, Enterprise Qatar, Qatar International Islamic Bank, Qatar National Bank, Agrico for Agricultural Development, Qatar Open Innovation, Scale7, Center of Entrepreneurship Qatar FinTech Hub, QBIC Fab Lab, Ministry of Energy and Industry, Qatar Business Incubation Center QBIC, QSTP Tech Venture Fund, Qatar Sports Tech, Ministry of Commerce and Industry, Commercial Bank of Qatar, Al Mirqab Capital, CCI France Qatar

Low trophic level actors, such as Qatar Business Incubation Center and Qatar Financial Centre, provide specialized expertise and niche services. Their contributions, while limited in influence, add diversity to the ecosystem by addressing sector-specific challenges like R&D, education, and technical development. These actors highlight gaps in broader resource distribution and collaboration, emphasizing the need for enhanced integration and targeted support.

Very Low

Qatar Financial Centre, Doha Women Forum, Qatar Islamic Bank, Injaz Qatar, Centre National De Convention Du Qatar, Ooredoo Qatar, Mada Innovation Award, Lusail University, Al Rayyan International University College, Police Institute, Innovation Cafe, Q Auto, International Finance Corporation, University Foundation College, Joaan Bin Jassim Academy for Defence Studies, Al Faleh Group College with the University of Aberdeen, Qatar Research Development and Innovation Council, Ras Laffan Emergency and Safety College, Global Studies Institute, Infrastructure Research and Development, Ibtchar, Qatar Development Bank, Gulf Organisation for Research & Development, SILA, Al Jazeera, Ministry of Environment and Climate Change, AppLab, Ministry of Culture, Iberdrola Innovation Middle East, University of Doha for Science and Technology, Qatar General Electricity & Water Corporation, MBK Holdings, Ministry of Municipality and Environment, Ministry of Social Development and Family, Qatar Olympic Academy, Doha Institute for Graduate Studies, Community College of Qatar, Itqan Clinical Simulation and Innovation Centre, Strategic Innovation, Entrepreneurship & Economic Development, ConocoPhillips, Carnegie Mellon University Qatar, Spacevision, Qatar Mobility Innovation Center, Aspetar Orthopaedic and Sports Medicine Hospital, University of Calgary in Qatar, Nama, Ministry of Education and Higher Education, KINDI Center For Computing Research, ExxonMobil Research Qatar, World Innovation Summit for Health, Ahmed Bin Mohammed Military College, TotalEnergies Research Center, Product Development Fund Tasmu Digital Valley, Biomedical Research Center, Qatar Aeronautical College, Qatar Shell Research and Technology Center, Weill Cornell Medicine-Qatar, HEC Paris in Qatar, Mohammed Bin Ghanem Al Ghanem Naval Academy, Georgetown University in Qatar, Sidra Medicine, Texas A&M University at Qatar, Northwestern University in Qatar, University College London, Equine Veterinary Medical Center, Hamad Medical Corporation, Qatar Science & Technology Park, Qatar National Research Fund, QF's annual Alumni Forum, Qatar Computing Research Institute, Education Development Institute, Qatar Environment and Energy Research Institute, Global Institute for Strategic Research, Qatar University, Hamad Bin Khalifa University, Qatar Foundation

Very low trophic level actors, including Qatar Foundation and Hamad Bin Khalifa University, depend heavily on external resources and partnerships to sustain their operations. Their contributions are typically sector-specific, focusing on research, education, or policy areas with limited ecosystem-wide influence. Despite their challenges, these actors remain crucial for ecosystem stability, offering unique insights and efforts that, if supported adequately, could scale their impact.

Qatar's entrepreneurial ecosystem thrives on a balanced trophic structure. High and very high actors drive innovation and resource flow, while moderate and lower actors ensure sectoral diversity and specialized expertise, creating a dynamic and interconnected ecosystem.

8.1.2. Compartmentalization: Collaborative Substructures and Communities

Compartmentalization analysis identifies functional substructures within Qatar's entrepreneurial ecosystem, providing insight into collaborative networks. These compartments reveal distinct sectors, interdependencies, and specialization, guiding policymakers and stakeholders to create targeted interventions that enhance ecosystem connectivity and resilience.

Table 19: Compartmentalization

Compartment	Members
1	Workinton, Web Summit Qatar, Wamda, US-Qatar Business Council, The Ministry of Transport, TASMU Accelerator, StartupGrind, SILA, ServCorp, Scale7, Singapore Business Council Qatar, Rowad Conference, Risin Ventures, Regus, Rasmal Ventures, Qatar-Ukraine Business Forum, Qatar Insurance Group, Qatar Investment Authority, Business Innovation Park, Center of Entrepreneurship Qatar FinTech Hub, Qatar Financial Markets Authority, Qatar Financial Centre, Qatar Finance and Business Academy, Qatar Exchange Venture Market, Qatar Development Bank, Qatari Businesswomen Association, QBIC Fab Lab, Qatar Business Incubation Center QBIC, Qatari Businessmen Association, Qatar Sports Tech, Qatar IT Conference and Exhibition, Accelerator Qatar Business District, Qatar Chamber of Commerce and Industry, Ooredoo Digital & Beyond, Make the Deal, Ministry of Commerce and Industry, Microsoft for Startups Founders Hub, Middle East Venture Partners, MENA Insurtech, McKinsey & Company, Ministry of Communications and Information Technology, Mada - Qatar Assistive Technology Center, ISTITHMAR (SME EQUITY), Invest Qatar, Innovation Consultancy, Idea Camp, Indian Business & Professional Council, Hyperthink, Global Entrepreneurship Week Network, Founder Institute, Ernst & Young, elev8, Doha Women Forum, Doha Venture Capital, Doha Tech Angels, Digital Incubation Center, Deloitte, Code Camp, College of the North Atlantic Qatar, Center for Entrepreneurship & Organizational Excellence, Croatian Business Council, Business Start Up Qatar, Bangladesh Forum Qatar, Bedaya Center for Entrepreneurship and Career Development, Boston Consulting Group, Bain & Company Doha, Arab Fintech Forum, Abu Issa Holding, 360 Nautica

2	World Innovation Summit for Health, Virginia Commonwealth University School of the Arts Qatar, University Foundation College, University of Doha for Science and Technology, University of Calgary in Qatar, Spacevision, Strategic Innovation, Entrepreneurship & Economic Development, Sidra Medicine, Ras Laffan Emergency and Safety College, Qatar University, Qatar Olympic Academy, Centre National De Convention Du Qatar, Q Auto, Police Institute, Mohammed Bin Ghanem Al Ghanem Naval Academy, Nama, Ministry of Municipality and Environment, Ministry of Social Development and Family, Ministry of Environment and Climate Change, MBK Holdings, Mada Innovation Award, Lusail University, KINDI Center For Computing Research, Itqan Clinical Simulation and Innovation Centre, Ibtechar, Hamad Medical Corporation, Global Studies Institute, Enterprise Qatar, ExxonMobil Research Qatar, Ministry of Education and Higher Education, Doha Institute for Graduate Studies, Ministry of Culture, Community College of Qatar, Biomedical Research Center, Aspetar Orthopaedic and Sports Medicine Hospital, Al Rayyan International University College, Rule of Law and Anti-Corruption Centre, Al Mirqab Capital, Al Jazeera, Agrico for Agricultural Development, Al Faleh Group College with the University of Aberdeen, Joaan Bin Jassim Academy for Defence Studies, Ahmed Bin Mohammed Military College
3	Weill Cornell Medicine-Qatar, University College London, TotalEnergies Research Center, Texas A&M University at Qatar, QSTP Tech Venture Fund, Qatar Science & Technology Park, Qatar Shell Research and Technology Center, Qatar National Research Fund, Qatar Mobility Innovation Center, QF's annual Alumni Forum, Qatar Foundation, Qatar Environment and Energy Research Institute, Qatar Computing Research Institute, Qatar Aeronautical College, Product Development Fund Tasmu Digital Valley, Ooredoo Qatar, Northwestern University in Qatar, Qatar General Electricity & Water Corporation, Infrastructure Research and Development, Innovation Cafe, International Finance Corporation, Iberdrola Innovation Middle East, HEC Paris in Qatar, Hamad Bin Khalifa University, Georgetown University in Qatar, Global Institute for Strategic Research, Equine Veterinary Medical Center, Ministry of Energy and Industry, Education Development Institute, Dutch Business Council Qatar, ConocoPhillips, Carnegie Mellon University Qatar, CCI France Qatar, AppLab
4	Qatar Research Development and Innovation Council, Qatar Open Innovation, Hassad Foods
5	Qatar National Bank, Qatar International Islamic Bank, Qatari Investors Group, Qatar Islamic Bank, Masraf Al Rayan, KPMG, Injaz Qatar, International Bank of Qatar, Hongkong and Shanghai Banking Corporation, Gulf Organisation for Research & Development, Doha Bank, Commercial Bank of Qatar, Canadian Business Council Qatar, Barwa Bank, Al Khaliji Commercial Bank
6	Philippine Business Council, Pakistan Business Council-Qatar, Portuguese Business Council

Compartmentalization reveals Qatar’s entrepreneurial ecosystem diversity, emphasizing tailored policies. Supporting highly connected compartments amplifies innovation, while nurturing smaller communities enhances inclusivity. Strategic collaboration across compartments fosters knowledge-sharing and strengthens the ecosystem’s resilience and global competitiveness.

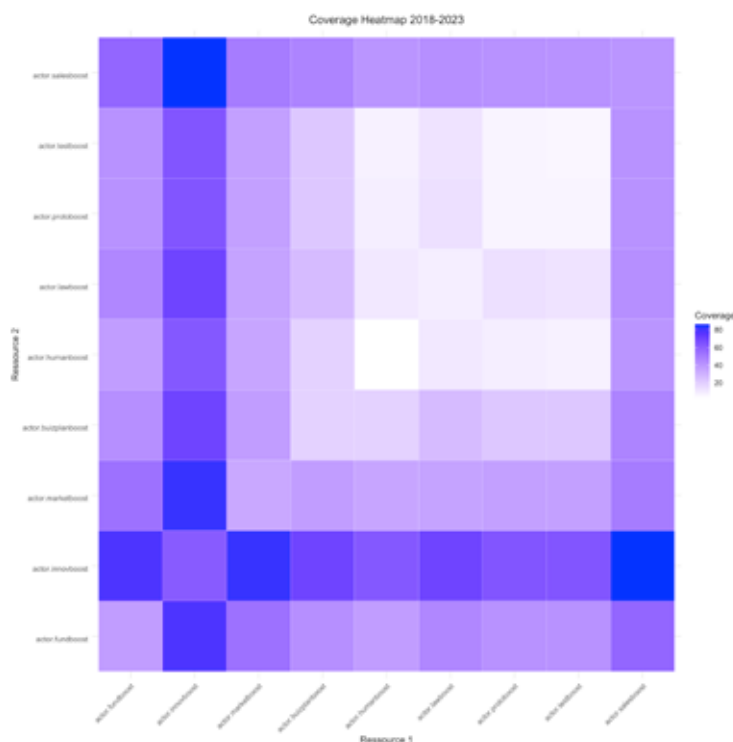
8.1.3. Coverage Measures: Availability and Completeness of Services

The coverage analysis evaluates the availability and completeness of critical services for Qatar’s entrepreneurial ecosystem, highlighting service interconnections and uncovering gaps in support crucial for sustainable ecosystem development.

The analysis reveals a strong interlinkage between innovation and funding services (81% coverage) and high collaboration between innovation and sales support (88%). However, weaker connections exist in areas such as business planning with human resources (19%) and legal support (27%). The weakest service coverage is observed in human resources interconnections, underscoring a significant gap. Addressing these deficiencies, particularly in human resource and legal support, is vital to fostering a more cohesive and well-rounded entrepreneurial ecosystem in Qatar.

Figure 19 depicts a **Service Coverage Heatmap**, where color intensity represents the extent to which different services are provided across the entrepreneurial ecosystem. Darker shades indicate higher service availability or concentration, while lighter areas highlight gaps in coverage. This visualization identifies well-supported areas and uncovers underserved segments, offering actionable insights for improving service distribution and accessibility.

Figure 19: Service Coverage



Strengthening under represented connections, like human resources and legal support, is critical for comprehensive service delivery. Prioritizing investments in underserved areas will improve resource inclusivity and ensure balanced development across ecosystem services, supporting sustainable entrepreneurship and innovation-driven growth in Qatar's evolving entrepreneurial landscape.

8.2. Dynamic Measures

Flow Diversity and Interaction Strength measures offer valuable insights into Qatar's entrepreneurial ecosystem by revealing inclusiveness and adaptability. These metrics highlight how actors connect, share resources, and maintain dynamic exchanges, fostering ecosystem robustness and ensuring resilience in a competitive global environment.

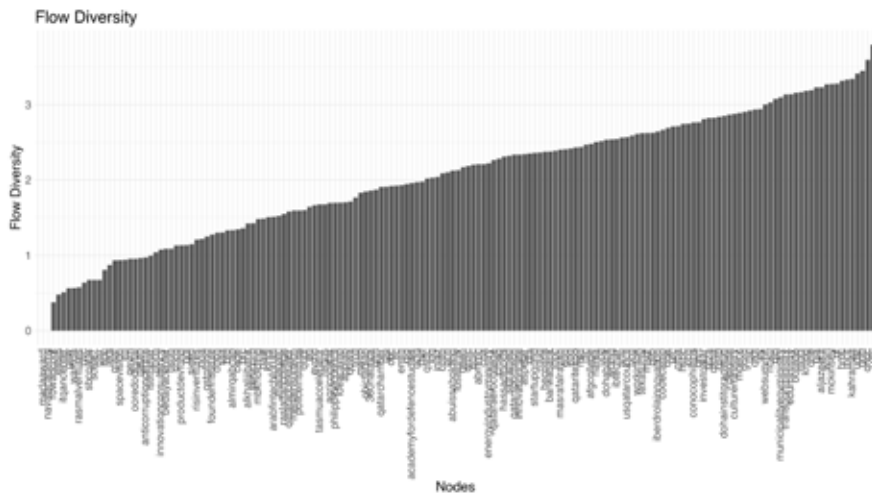
8.2.1. Flow Diversity: Measuring Inclusiveness and Fostering Adaptability

Flow Diversity quantifies the range and variability of interactions between ecosystem actors. It reflects inclusiveness, adaptability, and the extent to which actors engage in diverse, impactful collaborations.

High-flow diversity actors such as **Qatar Foundation (3.27)**, **Hamad Medical Corporation (3.28)**, and **QNB (3.18)** indicate strong inclusiveness, engaging in multi-sectoral collaborations that foster knowledge and resource sharing. Mid-tier actors like **Deloitte (2.85)** and **Qatar Chamber (1.87)** demonstrate focused engagement within specific domains, contributing niche expertise. Conversely, actors with low-flow diversity, such as **Scale7 (0.67)** and **Innovation Café (2.33)**, reveal limited ecosystem integration, suggesting opportunities to enhance their collaborative capacity to strengthen ecosystem resilience.

Figure 20 presents a **Flow Diversity Diagram**, showcasing the variety and distribution of resource and information flows among actors in the entrepreneurial ecosystem. The diagram highlights the balance between concentrated and dispersed connections, emphasizing areas with high diversity of interactions. This visualization provides insights into ecosystem adaptability and resilience, revealing opportunities to enhance collaborative networks.

Figure 20: Flow Diversity



Qatar's entrepreneurial ecosystem thrives on diverse interactions, but fostering broader inclusivity is essential for under-connected actors. Policymakers should encourage cross-sectoral collaborations, capacity-building initiatives, and engagement platforms to enhance flow diversity, creating a robust and adaptable ecosystem capable of sustaining long-term growth and innovation.

8.2.3. Interaction Strength: Collaboration Intensity and Systemic Dependencies

The interaction strength analysis identifies collaboration intensity between Qatar's ecosystem actors, crucial for understanding the network's dynamics, fostering synergies, and addressing weak links to optimize ecosystem development and health.

Qatar's strongest interactions involve **Qatar Foundation (QF)** with major institutions like **HBKU** and **QNRIF**, reflecting robust academic and innovation linkages. Conversely, weak interactions are scattered across smaller entities like **municipal ministries** and lesser-known actors. Enhancing weak ties while leveraging strong collaborations is vital for a more cohesive entrepreneurial ecosystem.

Table 20: Interaction strengths analysis

Strength	Number of Actors	Actors	Analysis
Very High	3	QF, HBKU, QNRF	Exceptional synergies in innovation and research foster ecosystem leadership and innovation development.
High	5	HMC, QCRI, QEERI, QBIC, QDB	Strong partnerships boost innovation, healthcare, and business incubation with strategic outputs.
Medium	7	Injaz, QFBA, QFC, NUQ, ConocoPhillips, CMUQ	Moderate but impactful collaboration in education, financial support, and specialized areas.
Low	10	Qatar Chamber, Aspetar, Energy Ministry, KMPG	Limited yet improving ties suggest potential for targeted initiatives to enhance connectivity.
Very Low	15	Municipalities, Smaller Banks, Non-Core Entities	Weak links imply significant gaps in collaboration requiring strategic integration and policy intervention.

Policy Significance: Strengthening weak connections and leveraging high-impact collaborations are pivotal to Qatar’s entrepreneurial ecosystem. Policies must prioritize integrating underrepresented actors and expanding strategic ties to foster balanced, inclusive growth and ensure resilience in Qatar’s innovation network.

8.3. Risks

Introduction on risk analysis on functions and services using bottleneck analysis Mitigating Systemic Vulnerabilities

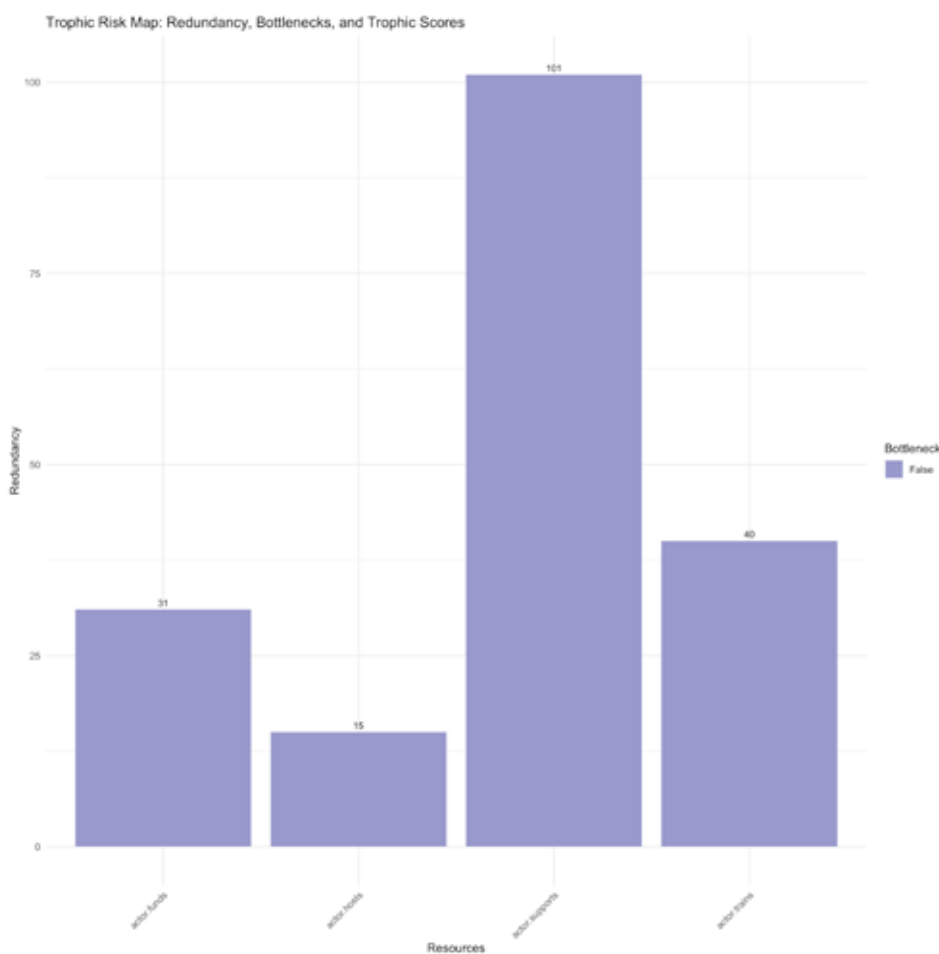
8.3.1. Risk Diagram of Functions

The functional risk analysis highlights **actor.supports** as the most redundant function with 101 actors, ensuring ecosystem stability through diverse support services. **Actor.trains** (40 actors) and **actor.funds** (31 actors) exhibit moderate redundancy, reducing reliance on individual contributors while maintaining sufficient capacity. **Actor.hosts**, with only 15 actors, is the least redundant, posing a potential risk during disruptions or

capacity shortages. None of these functions are identified as bottlenecks, but enhancing redundancy in hosting could further strengthen the resilience of the ecosystem.

Figure 21 illustrates the **Functional Risk Diagram**, highlighting areas of vulnerability within the entrepreneurial ecosystem. Nodes represent actors, while connections indicate functional dependencies. Risk levels are visualized through color or size variations, emphasizing actors or functions critical to ecosystem stability. This diagram identifies potential points of failure and informs strategies to mitigate risks and strengthen resilience.

Figure 21: Functional Risk Diagram



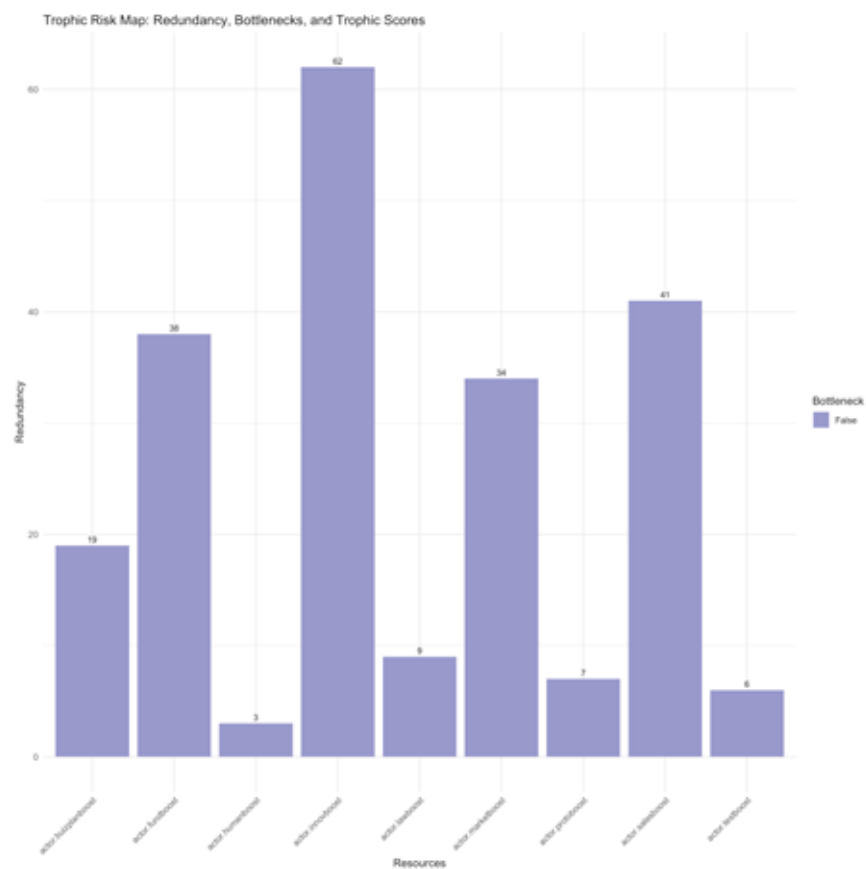
8.3.2. Risk Diagram of Services

The services risk analysis underscores **actor.humanboost** (3 actors), **actor.testboost** (6 actors), and **actor.protoboost** (7 actors) as the least redundant services, representing significant vulnerabilities. While **actor.buizplanboost** (19 actors) and **actor.lawboost** (9 actors) show moderate redundancy, they may still require capacity enhancements. Conversely, **actor.salesboost** (41 actors), **actor.fundboost** (38 actors), and **actor.marketboost** (34 actors) offer greater resilience. **Actor.innovboost**, with 62 actors, is the

most robust. Although no bottlenecks are identified, addressing low redundancy services is critical for ecosystem stability and efficiency.

Figure 22 presents the **Service Risk Diagram**, illustrating the vulnerabilities within the ecosystem's service framework. Nodes represent specific services, while connections indicate their interdependencies. Risk levels are depicted through variations in color or size, highlighting critical or fragile services. This visualization helps identify potential disruptions and supports strategic planning to enhance service resilience and continuity.

Figure 22: Service Risk Diagram



8.3.3. Risk Analysis

The risk analysis highlights the need for **diversification** to address low redundancy in critical functions and services, particularly **actor.humanboost**, **actor.testboost**, and **actor.protoboost**, which have minimal actor involvement. While no bottlenecks currently exist, fostering greater participation in these areas is essential to mitigate vulnerabilities and ensure a resilient ecosystem. Policymakers should prioritize resource allocation and support programs that enhance redundancy in underrepresented services, promoting a balanced and diversified entrepreneurial landscape to sustain Qatar's innovation and economic growth.

8.5 – Key Indicators

Table 21. Key Indicators

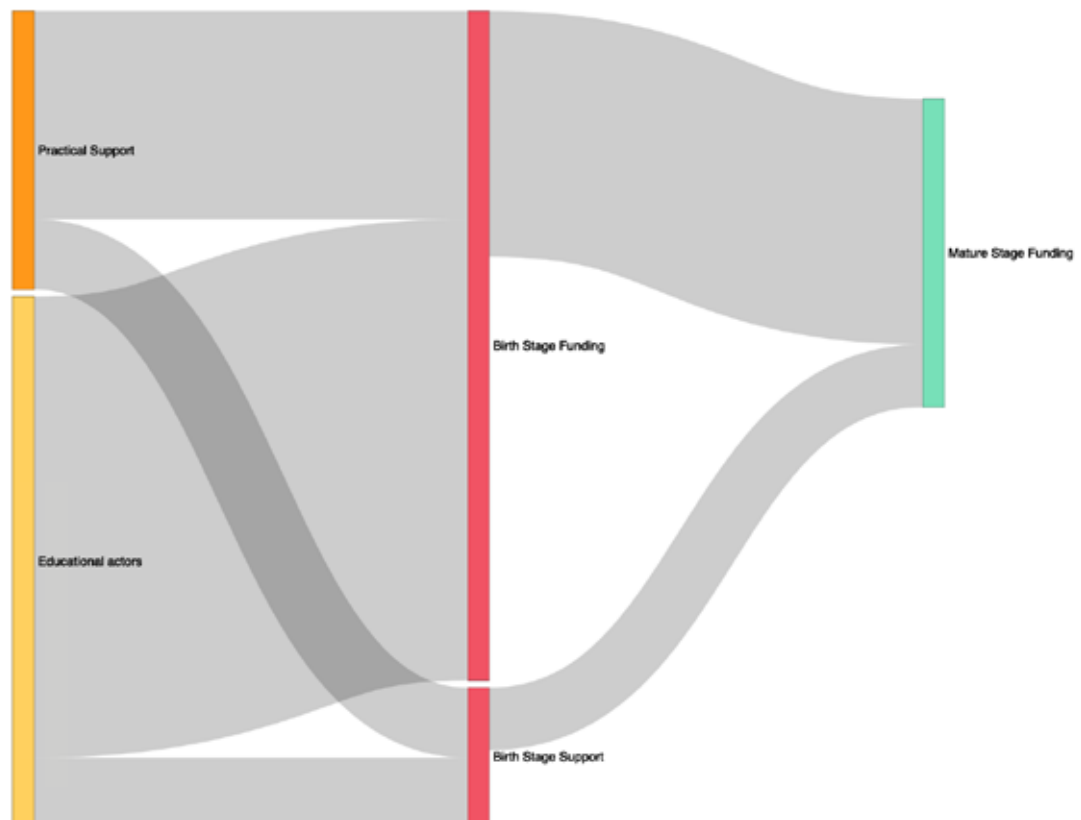
Indicators	Measures	Level
Resource Disponibility	Value Chain analysis reveals substantial entrepreneurship-oriented financing, indicating high readiness to mobilize resources.	★★★★☆
Resource Availability	Resource distribution focuses on action-oriented frameworks, ensuring effective allocation across the ecosystem.	★★★★☆
Resource Complementarity	Entrepreneurial support services highlight moderate complementarity, requiring improved alignment of existing resources.	★★★☆☆
Resource Risk	Governance-centric actions are essential to mitigate risks in underrepresented services and functions.	★★★☆☆
Impact on Entrepreneurship	Analysis indicates significant resource-driven impact on entrepreneurship, particularly through innovation-focused initiatives.	★★★★☆
Analysis		
The analysis underscores a strong entrepreneurial orientation in Qatar's resource availability, with robust financial mobilization and action-focused distribution strategies. However, gaps remain in resource complementarity and redundancy, notably in low-redundancy services such as actor, humanboost and actor.testboost. While governance-focused actions have mitigated immediate bottlenecks, long-term stability requires better resource alignment. This ensures an adaptable and interconnected ecosystem, fostering entrepreneurship and innovation. Policymakers should bolster resource integration and address weak areas, supporting a balanced ecosystem and driving sustainable entrepreneurial growth.		
Recommendations		
Enhance Resource Diversification: Develop strategies to increase redundancy in underrepresented services, such as humanboost, testboost, and protoboost, to improve ecosystem resilience. Promote Resource Integration: Encourage better alignment between entrepreneurial support services to enhance complementarity, addressing resource distribution imbalances. Focus on Governance: Strengthen ecosystem governance by implementing monitoring mechanisms to identify and preempt emerging bottlenecks. Support Cross-Sector Collaboration: Facilitate multi-sectoral initiatives to improve flow diversity and interaction strength, expanding the ecosystem's capacity to innovate. Invest in Capacity-Building: Allocate funding and technical support to low-impact actors, encouraging broader participation and fostering inclusivity in the ecosystem. By addressing these recommendations, Qatar can strengthen its entrepreneurial ecosystem, ensuring long-term sustainability and competitiveness.		

9. Visual Network Analytics

9.1. Sankey Diagram: Visualizing Actor Ties by Role

Figure 23 visualizes the connections between different types of actors in the entrepreneurial ecosystem using a **Sankey Diagram**. Nodes represent actor categories, while the flows between them illustrate the volume and strength of their interactions. The thickness of the flows indicates the intensity of collaboration, providing insights into the ecosystem's structure and highlighting key relationships and imbalances.

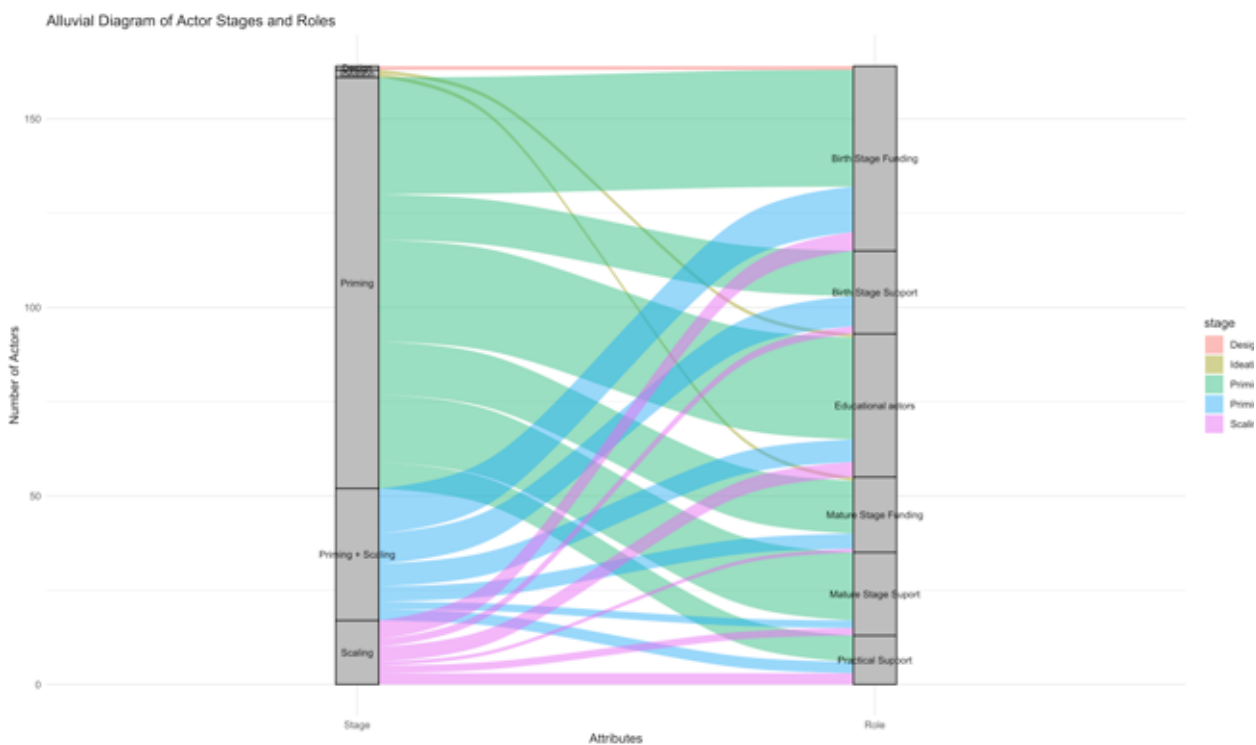
Figure 23: Sankey Diagram between type of actors



9.2. Alluvial Diagrams: Actor Specialties and Roles

Figure 24 presents an **Alluvial Bipartite Diagram** linking actors' roles to their stages of intervention within the ecosystem. Nodes represent roles and intervention stages, while flows illustrate their connections. The width of each flow reflects the intensity of involvement, highlighting how actors contribute across different stages. This visualization uncovers patterns, gaps, and overlaps in ecosystem engagement.

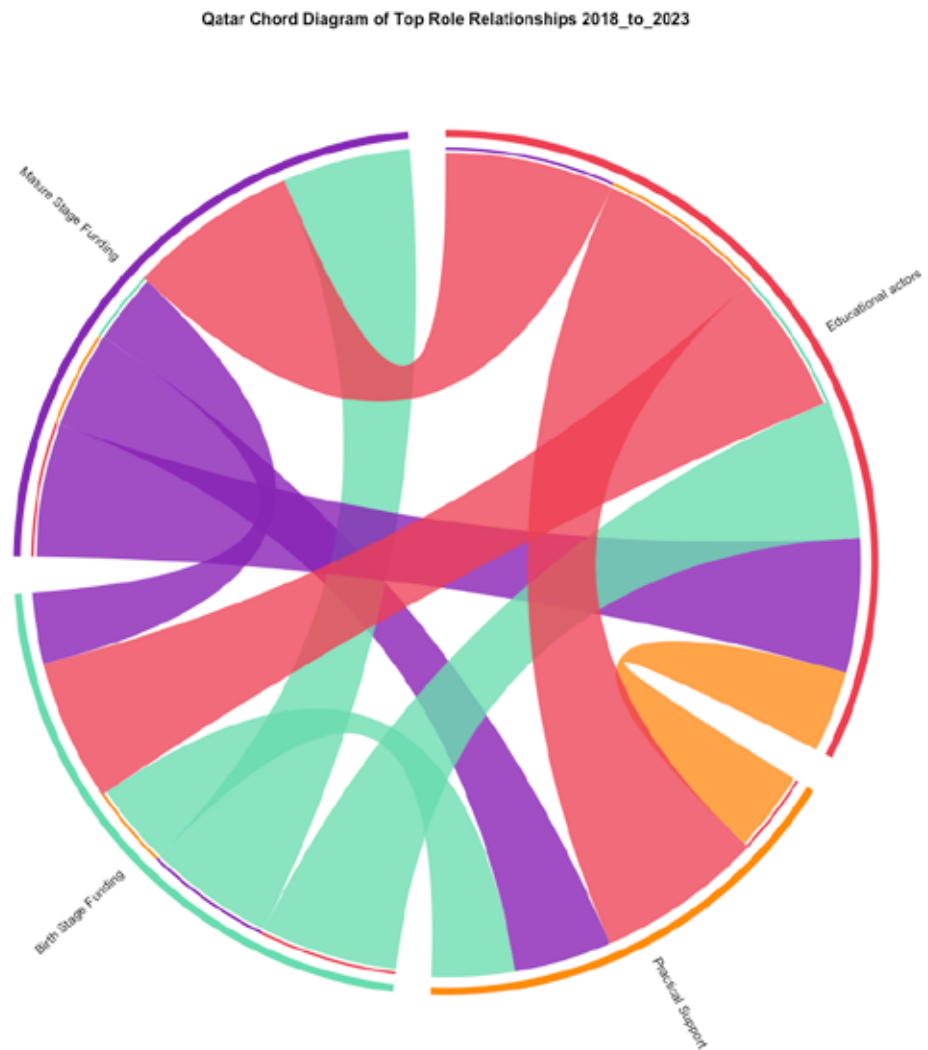
Figure 24: Alluvial Bipartite Diagram between actors roles and stage of intervention



9.3. Chord Plot: Mapping Actor Roles

Figure 25 illustrates a **Chord Diagram** representing the relationships between actors' roles in the entrepreneurial ecosystem. Each segment represents a role, with arcs connecting related roles. The thickness of the arcs indicates the strength or frequency of interactions. This visualization highlights interdependencies, collaboration dynamics, and the distribution of roles, offering insights into the ecosystem's functional cohesion.

Figure 25: Chord diagram



10. Holistic Evaluation

10.1. Factorized Metrics Summary: Cohesiveness, Interconnectedness, Proximity, and Entrepreneurial Capacity

Introduction

Qatar's entrepreneurial ecosystem from 2018 to 2023 demonstrates growth in collaborative culture but faces challenges in cohesiveness and resource distribution. These metrics underline its capacity to foster innovation, collaboration, and opportunity generation. By addressing gaps in proximity and cohesiveness, the ecosystem can optimize entrepreneurial outcomes and sustain long-term growth.

Table 22. Qatar's Entrepreneurial Ecosystem measures summary

	2018-2023	vs 2017-2022
Capacity to Inform Entrepreneurs factor - Cohesiveness factor (a)	0,0034	-0,0002
Collaborative Culture factor - Interconnectedness factor (b)	24,2851	1,6673
Capacity to distribute resources to entrepreneurs factor - Proximity factor (c)	16,48	-1,7452
Entrepreneurial Opportunities generation factor (a*b*c)	1,35	-0,11

10.1.1. Capacity to Inform Entrepreneurs (Cohesiveness)

Presentation: The cohesiveness factor (0.0034) reflects a slight decline (-0.0002) compared to the previous period, signaling reduced clustering and localized collaboration among actors.

Analysis: While marginally cohesive, this drop suggests challenges in maintaining strong, tightly-knit communities. Enhancing communication channels and localized support networks can foster a more cohesive ecosystem, ensuring better knowledge dissemination to entrepreneurs.

10.1.2. – Collaborative Culture (Interconnectedness)

Presentation: The interconnectedness factor (24.2851) grew by 1.6673, reflecting a stronger collaborative culture within the ecosystem.

Analysis: This increase highlights improved relationships and active engagement among actors. Stronger ties and synergies enhance ecosystem resilience, supporting innovation and shared resources. Leveraging this momentum can address gaps in inclusivity and expand collaborative platforms.

10.1.3. – Capability to distribute resources to entrepreneurs (Proximity)

Presentation: The proximity factor (16.48) declined by -1.7452, indicating a drop in the efficiency of resource distribution across the ecosystem.

Analysis: Reduced proximity suggests challenges in connecting resources with entrepreneurs, particularly outside Doha. Targeted initiatives, such as regional hubs, can address this gap, ensuring more equitable resource access and broader ecosystem participation.

10.1.4. – Diagnostic of Capability to Generate Productive Entrepreneurial Opportunities

Presentation: The opportunity generation factor (1.35) fell by -0.11, reflecting a decrease in the ecosystem's ability to create impactful entrepreneurial opportunities.

Analysis: Despite strong interconnectedness, declining proximity and cohesiveness limit opportunity generation. Strengthening regional networks and enhancing cohesiveness can bolster this metric, ensuring a more vibrant and productive entrepreneurial landscape.

Conclusion

Qatar's entrepreneurial ecosystem shows promising growth in collaborative culture, fostering stronger synergies among actors. However, declining cohesiveness and resource proximity hinder equitable opportunity distribution and generation. By enhancing localized support, improving proximity, and addressing regional disparities, Qatar can unlock its full entrepreneurial potential, ensuring sustainable innovation and ecosystem resilience. These interventions will solidify Qatar's position as a global leader in entrepreneurship and innovation.

10.2. Key Indicators : Holistic Insights for Policy Design

Holistic Assessment and Trends in 100 words about all the previously provided data

Table 23. Chapter 2 Key Indicators

Indicators	Measures	Level
Entrepreneurial Information Diffusion	The ecosystem is moderately connected (0.0034 cohesiveness), limiting information flow. Entrepreneurs face challenges accessing critical guidance and resources.	
Collaborative Culture	Strong interconnectedness (24.2851) reflects a robust collaborative culture that fosters innovation, resource sharing, and collective problem-solving.	
Resource Complementarity	Reduced proximity (16.48) shows inefficiency in resource distribution, creating barriers for entrepreneurs, especially in underserved regions.	
Entrepreneurial Opportunity Generation	The low opportunity factor (1.35) highlights limited entrepreneurial activity and market adaptability, impeding ecosystem vibrancy and investor appeal.	
Impact on Entrepreneurship	While collaboration is strong, weak resource allocation and opportunity generation reduce ecosystem inclusivity and effectiveness in fostering entrepreneurial growth.	
Analysis		

Qatar's entrepreneurial ecosystem reflects strengths in collaboration and interconnectedness, showcasing its capacity for innovation and collective resource utilization. However, challenges in cohesiveness, proximity, and opportunity generation reveal structural inefficiencies that limit ecosystem inclusivity and resilience. Regional disparities exacerbate these challenges, hindering equitable access to critical resources and entrepreneurial support. The current ecosystem's impact on entrepreneurship is moderately positive but requires strategic improvements to address underrepresented actors and regions. Bridging these gaps will enhance Qatar's competitiveness and sustainability as a global entrepreneurial hub.

Recommendations

To strengthen Qatar's entrepreneurial ecosystem, focus on three key areas:

Enhancing Cohesiveness: Invest in localized support networks and digital platforms to improve knowledge dissemination and enable better connectivity among actors.

Improving Resource Distribution: Develop regional hubs and targeted policies to address proximity challenges and ensure equitable access to financial, human, and technical resources.

Boosting Opportunity Generation: Foster sectoral diversity by supporting emerging industries (e.g., CleanTech, FinTech) through incentives, research funding, and market adaptability initiatives.

Additionally, encourages cross-sectoral collaboration, particularly between academia and private investors, to commercialize research and foster innovation. Strengthening these areas will reduce barriers, enhance inclusivity, and solidify Qatar's position as a leader in entrepreneurship and innovation.

Qatar's entrepreneurial ecosystem demonstrates a dynamic interplay of strengths and challenges, reflecting its journey toward becoming a global innovation hub. Anchored by robust network dynamics with 223 active actors and over 2,000 collaborations, the ecosystem thrives on strong partnerships, particularly between government bodies and innovation hubs. Doha dominates as the central hub, hosting 65% of activities, yet regional imbalances highlight untapped potential in areas like Lusail and Al-Khor. Key institutions such as Qatar Foundation and Qatar University play pivotal roles in driving policy alignment and innovation, fostering stability and growth. However, weaknesses like limited academia-investor collaboration, low resource redundancy, and regional disparities hinder inclusivity and efficiency. Opportunities lie in addressing these gaps through targeted interventions in governance, service complementarity, and thematic development. By leveraging its strengths in innovation-driven growth and governance, Qatar is well-positioned to align its entrepreneurial ecosystem with Vision 2030, ensuring sustainable and diversified economic progress.

11. Strategic Insights

Table 24. Key Indicators summary

Structure (Chapter 6)	★★★★☆
Orientation (Chapter 7)	★★★☆☆
Value Chain (Chapter 8)	★★☆☆☆
Holistic Analysis (Chapter 10)	★★☆☆☆

Qatar's entrepreneurial ecosystem demonstrates a dynamic interplay of strengths and challenges, reflecting its journey toward becoming a global innovation hub. Anchored by robust network dynamics with 223 active actors and over 2,000 collaborations, the ecosystem thrives on strong partnerships, particularly between government bodies and innovation hubs. Doha dominates as the central hub, hosting 65% of activities, yet regional imbalances highlight untapped potential in areas like Lusail and Al-Khor. Key institutions such as Qatar Foundation and Qatar University play pivotal roles in driving policy alignment and innovation, fostering stability and growth. However, weaknesses like limited academia-investor collaboration, low resource redundancy, and regional disparities hinder inclusivity and efficiency. Opportunities lie in addressing these gaps through targeted interventions in governance, service complementarity, and thematic development. By leveraging its strengths in innovation-driven growth and governance, Qatar is well-positioned to align its entrepreneurial ecosystem with Vision 2030, ensuring sustainable and diversified economic progress.

11.1. Key Factors for Entrepreneurial Growth

Qatar's entrepreneurial ecosystem is driven by a synergy of factors:

Network Dynamics: With 223 active actors and 2158 collaborations, Qatar's ecosystem boasts robust connectivity and collaboration. High average tie strength (33.022) reflects impactful partnerships, particularly between government entities and innovation hubs.

Territorial Focus: Doha remains the dominant hub, hosting 65% of ecosystem activity. However, underrepresented cities like Lusail and Al-Khor indicate potential for balanced geographic distribution.

Leadership Centrality: Institutions like Qatar Foundation and Qatar University anchor the ecosystem, driving innovation and policy alignment. Their strategic roles ensure ecosystem stability and sustained growth.

Community and Synergy Analysis: Community segmentation reveals strong innovation clusters, academic hubs, and financial networks. However, weak academia-investor collaborations indicate opportunities for improved research commercialization and funding access.

Thematic and Value Chain Analysis: Qatar's ecosystem is highly aligned with financial support, innovation-driven growth, and digital transformation. However, challenges in resource complementarity (e.g., low redundancy in humanboost and testboost services) require attention.

Risk and Structure Analysis: The ecosystem's structural resilience is high, yet gaps in service redundancy and regional disparities pose potential risks. Governance-centric measures mitigate systemic vulnerabilities but demand continuous improvement.

Strengths

Robust Connectivity: High-density networks and strong collaboration foster resource sharing and innovation.

Leadership Anchors: Institutions like Qatar Foundation and QDB ensure stability and strategic direction.

Innovation-Driven Growth: Strong focus on innovation hubs, digital transformation, and funding access for startups and SMEs.

Community Specialization: Distinct clusters cater to education, finance, and innovation, enhancing ecosystem adaptability.

Governance and Resilience: Stable policies ensure ecosystem alignment with Vision 2030 goals, promoting sustainability and diversification.

Weaknesses

Regional Disparities: Overconcentration in Doha limits inclusivity and access in smaller regions.

Resource Inefficiency: Low redundancy in critical services like humanboost and testboost impedes system resilience.

Limited Academia-Investor Collaboration: Missed opportunities for research commercialization hinder innovation scaling.

Weak Resource Complementarity: Inefficiencies in resource distribution create barriers for entrepreneurs in underserved sectors.

Opportunity Generation Gaps: Limited market adaptability and low entrepreneurial activity reduce the ecosystem's vibrancy and attractiveness to investors.

Strategic interventions targeting these weaknesses can unlock Qatar's full entrepreneurial potential and drive sustained economic growth.

11.2. Policy Recommendations for Ecosystem Development

For Government Officials and Policy Makers

Consequences: Regional disparities and weak resource distribution limit ecosystem inclusivity, hindering entrepreneurial growth in underrepresented areas and reducing Qatar's potential for balanced economic development and innovation-driven competitiveness.

Recommendations: Invest in regional hubs, improve resource redundancy, and foster academia-investor collaborations to commercialize research and diversify the entrepreneurial landscape.

Implications: A balanced ecosystem with equitable access will reduce regional gaps, enhance inclusivity, and strengthen Qatar's global standing as an innovation hub aligned with Vision 2030.

For Educational and Mentorship Actors

Consequences: Limited academia-investor collaboration and resource misalignment hinder the commercialization of research and the development of entrepreneurial talent, reducing the ecosystem's capacity to innovate and sustain long-term growth.

Recommendations: Strengthen academia-industry partnerships and develop mentorship programs in emerging fields like AI, CleanTech, and FinTech to address skill and innovation gaps.

Implications: Improved education-industry alignment will nurture entrepreneurial talent, foster innovation, and ensure a sustainable pipeline of skilled professionals for Qatar's growing economy.

For Non-Financial Support Actors

Consequences: Low redundancy in services like humanboost and testboost creates vulnerabilities, limiting ecosystem resilience and entrepreneurs' access to critical support for product development and skill-building.

Recommendations: Expand underrepresented services and create integrated support systems to ensure comprehensive resource availability for entrepreneurs.

Implications: Strengthened non-financial support structures will boost entrepreneurial readiness, enhance resilience, and enable Qatar's ecosystem to scale sustainably while fostering innovation.

For Financial Support Actors

Consequences: Insufficient funding diversity and limited accessibility in underserved regions constrain entrepreneurial opportunities and reduce inclusivity in the ecosystem.

Recommendations: Develop funding mechanisms for niche sectors like AgriTech and FinTech, and expand regional financial access through tailored SME programs.

Implications: Enhanced financial accessibility and sectoral diversity will empower entrepreneurs, attract investments, and solidify Qatar's position as a leader in entrepreneurial finance and innovation.'

The diagnosis of Qatar's entrepreneurial ecosystem, combined with World Bank data from 2018 to 2023, reveals a nuanced interaction between structural dynamics and economic indicators, highlighting both strengths and challenges. The high GDP per capita, recovering from \$52,316 in 2020 to \$87,661 in 2022, reflects Qatar's success in leveraging innovation and diversification strategies. However, the fluctuating FDI inflows, peaking at \$7.26 billion in 2019 and declining sharply to \$1.25 billion in 2021, expose vulnerabilities in attracting sustained foreign investment, despite regulatory advancements.

Ecosystem connectivity, with 223 active actors and 2,158 collaborations, aligns with strong network dynamics observed in Chapter 6, fostering innovation hubs and policy alignment. Yet, regional disparities remain stark, as Doha hosts 65% of activities, while areas like Lusail and Al-Khor remain underdeveloped, consistent with World Bank unemployment figures showing rates peaking at 14% in 2020 and 2021. This overcentralization stifles inclusivity and limits geographic balance.

Thematic and value chain analyses from Chapters 7 and 8 reveal opportunities for resource alignment and service redundancy. The limited academia-investor collaboration correlates with inefficiencies in commercializing research, as noted in stagnation in new business density post-2018. Similarly, the absence of redundancy in critical services like «humanboost» and «testboost» affects resilience, a critical issue during periods of economic uncertainty reflected in FDI volatility.

Holistically, as explored in Chapter 10, Qatar's governance-focused measures provide a solid foundation for mitigating risks, yet strategic interventions are required. Enhancing regional inclusivity, improving academia-industry partnerships, and addressing sectoral inefficiencies are essential. Aligning these efforts with Vision 2030 ensures sustainable economic growth and positions Qatar as a globally competitive innovation hub.

12. Conclusion

12.1. Summary of Findings

Qatar's entrepreneurial ecosystem demonstrates strong collaboration, leadership, and innovation capacity but faces challenges in regional disparities, resource alignment, and underrepresented services. Key actors like Qatar Foundation and Qatar Development Bank drive ecosystem resilience, while gaps in academia-investor collaboration and niche sector support hinder holistic growth. Addressing these challenges is vital for sustainable entrepreneurship and global competitiveness.

Table 25. Strengths, Challenges and Actionable Insights

	Strengths	Challenges	Actionable Insight
Government Officials and Policy Makers	Strong leadership and collaborative culture enable resource sharing and innovation. Government-innovation hub synergies are key drivers of ecosystem vibrancy.	Regional disparities (e.g., dominance of Doha) and weak proximity hinder equitable access and inclusivity.	Policymakers must prioritize regional hub development, governance improvements, and cross-sector collaboration to reduce disparities and foster balanced growth.
Educational and Mentorship Actors	Universities and research institutions provide a solid foundation for talent development and innovation.	Limited collaboration with private investors and underrepresentation in niche fields like CleanTech and FinTech.	Strengthen academia-industry partnerships and develop mentorship programs in emerging sectors to enhance talent pipelines and innovation potential.
Non-Financial Support Actors	Innovation hubs and incubators are critical enablers, supporting startups with infrastructure and guidance.	Low redundancy in services like humanboost and testboost creates systemic vulnerabilities.	Expand underrepresented services and create integrated support networks to ensure resilience and comprehensive entrepreneurial readiness.

Financial Support Actors	Strong financial frameworks and investment mechanisms support startups and SMEs effectively.	Limited funding access in underserved regions and insufficient diversity in niche sector investments.	Tailor financial programs for emerging sectors and underserved areas to broaden opportunities and enhance inclusivity.
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12.2. Future Directions: Enhancing Analysis and Impact

Key Conclusions for the 2018 to 2023 Period

Qatar’s entrepreneurial ecosystem has demonstrated significant growth and resilience during the 2018 to 2023 period, reflecting a robust foundation for entrepreneurship and innovation. Key conclusions from the analysis reveal both strengths and areas for improvement that will shape future policy and development.

Connectivity and Collaboration:

The ecosystem’s high connectivity index (0.1615) and strong collaborative culture (interconnectedness factor 24.2851) underscore robust ties among actors. Government bodies and innovation hubs emerged as central players, driving resource-sharing, innovation, and policy alignment.

However, gaps in academia-investor relationships and limited synergies between private and public sectors have constrained the full optimization of the ecosystem’s potential.

Diversity and Inclusivity:

The ecosystem hosts a diverse set of actors, from universities to incubators, ensuring adaptability and resilience. Yet, underrepresented sectors like CleanTech, FinTech, and AgriTech remain underserved, limiting niche sector innovation.

Regional disparities, such as Doha’s dominance, highlight the need for more inclusive resource distribution across cities like Lusail and Al-Khor.

Resource Allocation and Efficiency:

Strong frameworks for financing and support services, such as InnovBoost and FundBoost, drive innovation and growth. Nevertheless, services like humanboost, testboost, and protoboost face low redundancy, increasing ecosystem vulnerability.

Reduced proximity (16.48) indicates inefficiency in resource distribution, creating barriers for entrepreneurs in less connected regions.

Opportunity Generation and Ecosystem Governance:

Entrepreneurial opportunity generation (1.35) lags, suggesting limited adaptability to market dynamics and constrained avenues for new ventures.

Effective governance has ensured policy alignment and strategic coordination among key players, but further improvements are needed to sustain long-term resilience.

Overall, the 2018 to 2023 period demonstrates a maturing ecosystem with strong leadership, collaboration, and innovation capacity. However, addressing disparities in resource distribution, fostering niche sector growth, and enhancing inclusivity will be critical for future progress.

Perspectives for the Next Period Study (2019-2024)

The upcoming 2019-2024 study should focus on refining strategies and monitoring key elements to enhance ecosystem impact and adaptability. Recommendations for policymakers and stakeholders include:

Enhancing Regional Inclusivity:

Develop regional hubs in underserved areas like Lusail and Al-Khor to balance resource distribution.

Introduce incentives for businesses and investors to operate in these regions, ensuring more equitable ecosystem growth.

Fostering Sectoral Diversity:

Support niche sectors like CleanTech, FinTech, and AgriTech through targeted incentives, research funding, and public-private partnerships.

Promote collaboration among universities, industry, and government to create tailored initiatives for emerging industries.

Improving Academia-Investor Collaboration:

Establish platforms to facilitate research commercialization and startup funding, bridging gaps between academic institutions and private investors.

Develop mentorship programs that link academic talent with seasoned entrepreneurs and investors.

Strengthening Resource Redundancy:

Increase redundancy in underrepresented services such as humanboost, testboost, and protoboost by engaging more actors in these areas.

Implement a monitoring system to identify and address resource bottlenecks in real time.

Leveraging Digital Transformation:

Expand digital platforms to improve knowledge dissemination, actor connectivity, and resource accessibility.

Focus on integrating AI-driven tools for data analysis, network optimization, and ecosystem insights.

Key Metrics to Monitor:

Connectivity and Synergy: Regularly measure tie strength, collaboration patterns, and synergy across sectors.

Opportunity Generation: Track the emergence of new ventures, investment flows, and sectoral diversification.

Regional Growth: Monitor activity levels in underserved regions to evaluate the effectiveness of interventions.

Resource Allocation: Assess the efficiency and redundancy of services to ensure balanced ecosystem development.

By focusing on these strategic areas and monitoring critical metrics, Qatar can strengthen its entrepreneurial ecosystem, ensuring resilience, inclusivity, and global competitiveness in the coming years.

13. Appendices

13.1. About OSE in Africa

Vision

The Observatory of Support for Entrepreneurship (OSE) seeks to stimulate entrepreneurial productivity in Africa, thereby promoting social well-being, economic and democratic development, and growth.

Ambition

Our goal at the Observatory of Support for Entrepreneurship (OSE) is to become the primary driver of entrepreneurial transformation on the African continent. We hope to play a critical role in the strengthening and expansion of entrepreneurial ecosystems through in-depth analysis based on network theory. By focusing our efforts on the mapping of relationships, communities, opportunities, and challenges within these ecosystems, we hope to inspire positive changes, foster innovation, and propel long-term economic development in Africa.

Missions

Extensive studies: Using network theory and content modeling, we conduct extensive studies on 33 entrepreneurial ecosystems in Africa. Our goal is to analyze collaborative dynamics, identify key stakeholders and communities, and detect emerging trends in order to provide actionable data and strategic recommendations.

Interaction mapping: We map the relationships and interactions between the various actors in the entrepreneurial ecosystem and its sub-communities, from the start-up phase to the growth phase, passing through financing, training, and support. This map identifies opportunities for collaboration and innovation, as well as measures to improve the performance of these ecosystems.

Identification of opportunities: By analyzing the collected data, we identify gaps and opportunities in African entrepreneurial ecosystems. These data influence our recommendations for increasing entrepreneurial support and promoting long-term economic development.

Values

Engagement in innovation: We are firm believers in innovation as a critical driver of economic and social growth. We are committed to promoting novel approaches to stimulating entrepreneurship in Africa and encouraging creativity among ecosystem actors.

Collaboration and partnership: We value collaboration with all stakeholders, including entrepreneurs, investors, government agencies, universities, and civil society organizations. We believe that building a strong network of partners is critical to achieving our common goals.

Analytical rigor: Our approach is founded on rigorous and methodical analysis. We place a high value on rigorous data collection and objective, scientific interpretation to ensure that our recommendations are founded on strong foundations.

Long-term impact: We hope to have a long-term impact on African entrepreneurship. Our initiatives aim to generate positive long-term change by identifying and resolving structural challenges that entrepreneurs face across the continent.

Finally, we hope to shape a prosperous entrepreneurial future for Africa by harnessing the incredible power of network analysis to analyze, link, and catalyze entrepreneurial ecosystems for better resource allocation to entrepreneurs. Our missions are focused on in-depth research, mapping, and identifying opportunities, while our core values include innovation, collaboration, analytical rigor, and long-term effect.

13.1. Glossary of Terms

1. fundboost

Definition:

actor.fundboost refers to whether an actor in the entrepreneurial ecosystem provides **fundraising assistance** to startups and entrepreneurs. This includes services like connecting entrepreneurs to venture capitalists, angel investors, or crowdfunding platforms, as well as offering financial advisory services to help secure the necessary capital for growth.

2. innovboost

Definition:

actor.innovboost assesses if an actor offers **support for innovation**. This involves fostering creativity, providing access to R&D facilities, offering innovation workshops, or facilitating partnerships with research institutions. The goal is to enhance product development, technological advancements, or innovative business models.

3. marketboost

Definition:

actor.marketboost identifies actors who assist with **marketing strategy** development. This includes help with branding, market analysis, identifying target audiences, designing marketing campaigns, and optimizing digital presence to drive business growth and customer acquisition.

4. buizplanboost

Definition:

actor.buizplanboost indicates whether an actor provides **business plan assistance**. This involves helping entrepreneurs develop comprehensive, viable business plans that outline goals, strategies, market analyses, operational structures, and financial projections to attract investors or stakeholders.

5. humanboost

Definition:

actor.humanboost measures whether an actor offers **help with building the team**. This includes recruitment, talent development, team-building workshops, or HR support to ensure that startups have the right skills and personnel to succeed.

6. lawboost

Definition:

actor.lawboost refers to whether an actor provides **help with legal protection**. This includes support with intellectual property (IP) protection, compliance with regulations, drafting legal documents, and navigating legal frameworks to safeguard business interests.

7. protoboost

Definition:

actor.protoboost evaluates if an actor offers **prototype development assistance**. This includes technical expertise, access to fabrication labs (FabLabs), 3D printing services, or workshops that help entrepreneurs create and refine product prototypes.

8. testboost

Definition:

actor.testboost determines if an actor provides **help with full-scale testing** of products or services. This includes pilot program management, user feedback collection, and assistance in testing functionality, usability, and market fit before a full-scale launch.

9. salesboost

Definition:

actor.salesboost describes whether an actor provides **sales and growth assistance**. This encompasses strategies for scaling operations, improving sales funnels, establishing partnerships, expanding into new markets, and increasing revenue generation.

These terms represent the various types of support provided by actors in Qatar's entrepreneurial ecosystem, highlighting the diverse services essential for fostering innovation, scaling businesses, and driving sustainable growth.

13.2. Methodological Details

The improvement of the performance of African entrepreneurial ecosystems has a significant impact on societal, economic, and managerial goals. Here's a rundown of the issues raised by this enhancement:

The Social Issues

Job creation: Dynamic entrepreneurial ecosystems stimulate job creation, lowering unemployment and contributing to economic inclusion of local populations.

The reduction of poverty: A growing performance of entrepreneurial ecosystems allows for the creation of economic opportunities for underserved communities, hence contributing to poverty reduction and more equitable resource distribution.

Social innovation: Performance-driven entrepreneurial ecosystems foster the emergence of novel solutions to critical social needs such as health, education, access to safe drinking water, and other major challenges.

Autonomy of the young: By strengthening entrepreneurial ecosystems, young people have more opportunities to learn new skills, realize their potential, and positively contribute to the development of their country.

Economic challenges

Economic growth: Effective entrepreneurial ecosystems stimulate economic growth by encouraging the formation and development of new businesses, hence increasing output and national income.

Attractiveness of investments: Dynamic ecosystems attract domestic and foreign investment, strengthening the country's ability to mobilize financial resources to support development.

Economic diversification: The development of entrepreneurial ecosystems promotes economic diversification by encouraging innovation in a variety of sectors, reducing reliance on a single sector.

Commerce and exports: Emerging businesses boost international competitiveness by producing exportable goods and services, hence increasing export revenue and improving commercial balance.

Practical Issues

Access to funding: The development of entrepreneurial ecosystems makes it easier for entrepreneurs to obtain the capital they need to start and grow their businesses, reducing financial constraints.

Support Infrastructures and services: A well-developed ecosystem provides infrastructure, business support services, and professional resources that promote the growth and sustainability of businesses.

Competence development and training: The development of entrepreneurial ecosystems promotes the establishment and development of entrepreneurial skills, enhancing entrepreneurs' ability to manage their businesses effectively.

Regulatory and legislative framework: Policies that benefit businesses and a favorable regulatory environment are critical for encouraging entrepreneurship and stimulating innovation.

The improvement of the performance of entrepreneurial ecosystems in Africa has the potential to generate significant benefits on societal, economic, and practical levels, contributing significantly to the region's long-term development and prosperity.

Spillovers

The Observatory of Support for Entrepreneurship (OSE) plays a critical strategic role in improving the performance of Africa's entrepreneurial ecosystems by acting as a transformational and catalytic tool. Here's how he contributes strategically to this improvement:

Data collection objectives: The Observatory collects factual and objective data on the interactions, relationships, and flows within entrepreneurial ecosystems. These data form a solid foundation for identifying forces, weaknesses, and opportunities, assisting decision-makers and key actors in making clear decisions.

Extensive analysis: The Observatoire conducts a thorough analysis of the dynamics inside entrepreneurial ecosystems using network methodology. This allows for the identification of key actors, conflict hotspots, and emerging trends, as well as the provision of critical information for strategic intervention.

A map of the links : The Observatoire identifies opportunities for collaboration and mutual reinforcement by mapping the relationships and interactions between the many actors in the entrepreneurial ecosystem. This may encourage the formation of fruitful partnerships, the exchange of knowledge, the efficient distribution of resources, and the encouragement of innovation.

Strategic recommendations: Taking a cue from its in-depth analysis, the Observatoire issues strategic recommendations to strengthen support for entrepreneurship in Africa. These recommendations have the potential to stimulate public policies, private investments, and development activities aimed at improving the entrepreneurial ecosystem.

Plaidoyer and sensitization: The Observatory serves as a platform for awareness and advocacy by highlighting the challenges and opportunities associated with entrepreneurship in Africa. It can help to create a more favorable business environment by raising awareness among stakeholders and advocating for positive changes.

Capabilities enhancement: The Observatory can help to strengthen the capacities of entrepreneurial ecosystem actors by providing data-driven training, resources, and best practices.

Impact monitoring: The Observatory can measure progress in improving the performance of entrepreneurial ecosystems by tracking changes over time and evaluating the impact of initiatives, allowing for continuous strategic adjustment.

Engaging with the Observatory of Support for Entrepreneurship (OSE) is crucial for Qatar as it aligns with the nation's Vision 2030 goals of economic diversification and innovation-driven growth. OSE provides critical insights into entrepreneurial ecosystems, identifying gaps in resources, collaboration, and inclusivity. By leveraging OSE's data-driven analyses, Qatar can address regional disparities, optimize resource allocation, and foster innovation across emerging sectors like FinTech and CleanTech. OSE also strengthens global competitiveness by bridging academia, government, and private sector collaborations, ensuring sustained growth. Engagement with OSE positions Qatar as a leader in entrepreneurship, fostering resilience and adaptability in a dynamic economy.

For entrepreneurs, the list, geographical distribution, roles and functions, and forms of activity of Moroccan entrepreneurs offer a variety of significant and practical benefits:

1. **Access to Resources:** Entrepreneurs can quickly identify key players who can provide critical resources such as funding, mentoring, training, legal advice, and business development services.
2. **Networking and Collaboration:** Knowing the players in the entrepreneurial network opens up opportunities for networking and collaboration. Entrepreneurs have the ability to form strategic alliances, find co-founders, clients, suppliers, and mentors.
3. **Understanding of the Ecosystem:** A thorough understanding of the geographical distribution and roles of the actors allows entrepreneurs to better understand how the ecosystem works and how they may best integrate into it.
4. **Strategic Direction:** By identifying the actors who correspond to their specific needs, entrepreneurs may more precisely direct their growth, innovation, and product development strategies.
5. **Access to Financing Opportunities:** By identifying potential investors, entrepreneurs can more effectively target the sources of financing that are best suited to their needs, whether it is risk capital, loans, grants, or other forms of financing.
6. **Access to Markets:** A list of participants and their activities can assist entrepreneurs in identifying potential commercial partners, distributors, or customers, allowing them to expand their operations into other markets.
7. **Clarified decision-making:** The information about the actors enables entrepreneurs to make clear decisions about their growth, product development, and networking strategies.
8. **Promotion and visibility:** By connecting with ecosystem actors, entrepreneurs can increase their visibility, promote their products and services, and have access to promotional opportunities.

- 9. Evolution and adaptation:** Entrepreneurs may track the evolution of the ecosystem over time, identifying new opportunities and emerging trends that may shape their business.
- 10. Continuing Education:** Interactions with various ecosystem actors provide entrepreneurs with opportunities to learn, share experiences, and improve their skills as business leaders.

List the actors in the Moroccan entrepreneurial ecosystem, specifying their geographical distribution, role distribution, and distributions by functions and type of activity, provides several significant advantages for a comprehensive study of this ecosystem. Here's why this procedure is necessary:

- 11. Ecosystem Mapping:** By compiling an exhaustive list of actors, you can obtain a detailed overview of the Moroccan entrepreneurial ecosystem. This allows for a better understanding of the scope and diversity of the parties involved.
- 12. Identification of key players:** By identifying key actors such as incubators, accelerators, investors, universities, established businesses, and entrepreneurs, you may identify the individuals and entities that have a significant impact on the ecosystem.
- 13. Geographic Analysis:** The geographical distribution of actors allows for the visualization of concentrations of entrepreneurial activity in various regions of Morocco. This can reveal entrepreneurial centers or clusters, highlighting areas where innovation and creativity are particularly active.
- 14. Comprehension of Roles:** Predicting the roles of each actor (for example, venture capitalist, mentor, trainer, service provider, etc.) allows for a more in-depth understanding of how each contributes to the development of the ecosystem. This may reveal the strengths and weaknesses in entrepreneur support.
- 15. Collaboration and Networking:** By identifying the actors based on their roles and types of activities, you can identify opportunities for collaboration and partnership among various parties involved. Entrepreneurs will be able to find the resources they need to grow their businesses more easily as a result.
- 16. Influence evaluation:** By measuring the distribution and engagement of actors by role and function, you may assess the overall influence of the ecosystem on entrepreneurial growth. This allows us to determine which parts of the ecosystem contribute the most to job creation, innovation, and economic development.
- 17. Strategies Adaptation:** Once you have a complete picture of the actors, their roles, and their responsibilities, you can fine-tune the strategies and entrepreneurship support policies to better meet the needs of each group of actors.

OSE Conceptual Model

The Observatory of Support for Entrepreneurship (OSE) Conceptual Model is based on scientific research published in peer-reviewed journals as well as popular publications such as Harvard Business Review France® by the authors of this report and is based on the capture of big data on the internet using automated scraping techniques.

To that end, we will embark on an initial phase of identifying and qualifying key players in the targeted entrepreneurial ecosystem, using a variety of reference sources such as bilateral cooperation organizations, patronal conferences, professional associations, institutions, researchers, and universities. We were able to identify more than 4500 entrepreneurial support actors in 33 African countries, making this the most comprehensive and exhaustive repository to date.

To enable our Observatory to provide its full value, we qualified the entire database of over 1500 actors by specifying the kind and role of each actor, using a consistent model of distribution of these types and roles. The following types of actors are listed: Schools, Universities, Training Centers, Incubators, Accelerators, Social Economy Actor, Co-Working Spaces, Startup Nurseries, Third Places, FabLabs, Cultural Space for Startups, Media with an Entrepreneurial Focus, Startup Event, Clusters, Competitiveness poles, Business Angel Clubs, Venture Capital, National Institutions, International Institutions, Local Institutions, Banks.

Figure 26: Critical Roles in Entrepreneurial Ecosystems

Actors Roles

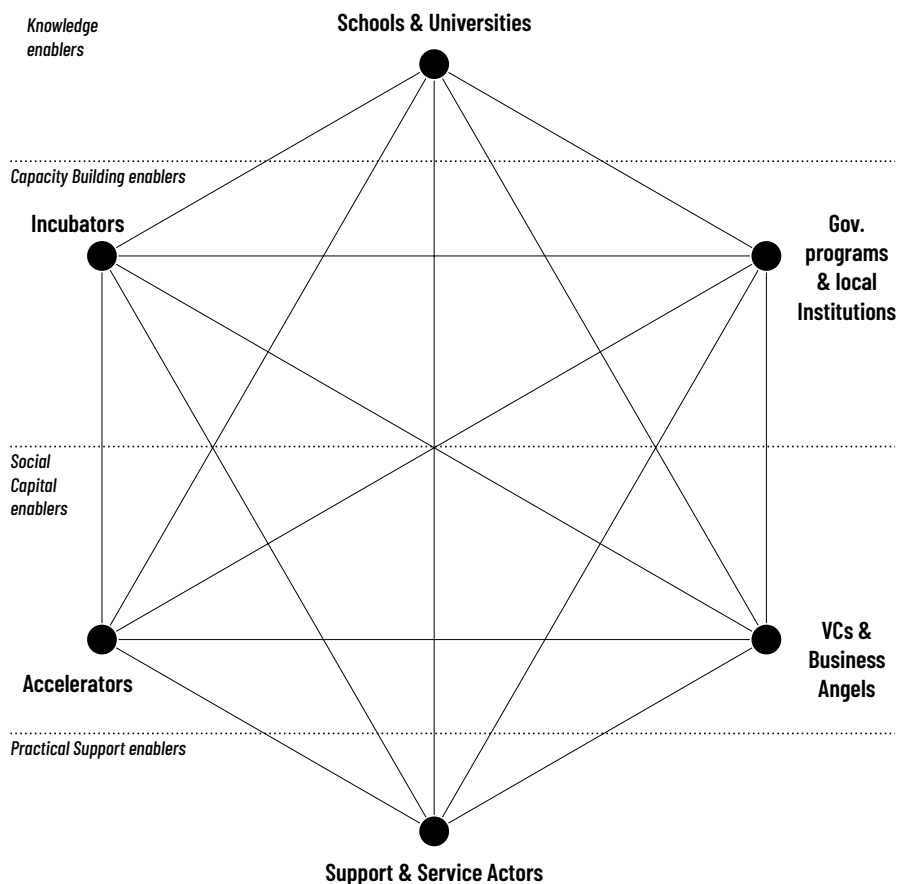


Table 26. shows the actual number of identified actors in 33 African ecosystems.

Country	# Actors
Saudi Arabia	391
Egypt	314
Oman	267
Tunisia	256
Bahrain	248
South Africa	240
Qatar	238
Morocco	210
United Arab Emirates	196
Algeria	191
Nigeria	185
Kuwait	175
Iran	150
Senegal	138
Kenya	107
Burkina Faso	103
Cameroon	103
Mauritania	100
Ghana	94
Congo Republic	80
Democratic Republic of Congo	80
Gabon	80
Benin	80
Jordan	80
Tanzania	76
Mozambique	75
Madagascar	73
Rwanda	72
Ivory Coast	71
Uganda	70
Togo	56
Mali	54
Libya	30

Following the hexagonal model, each of these categories of actors has been assigned a key role: schools and universities, incubators, accelerators, practice support, private equity investors (VCs and business angels), and government actors.

Based on our observations of their activity, we identified the following elements to qualify the base:

Structure : public, private, associative, and public-private partnerships.

Intervention Stage : Ideation, Conception, Debt financing, Scaling, Expansion.

Demanded Contreparties: Fees (service), Shares, logo or label display, refund (loan).

Supporting Topics: Generalist, FinTech, BioTech, EdTech, CleanTech, HealthTech, GreenTech, InsurTech, AdTech, RetailTech, AgriTech.

Scale : International, National, Local Mono Site, Multi Site.

Support function: Fundraising assistance (FundBoost), innovation assistance (InnovBoost), marketing strategy assistance (MarketBoost), business plan assistance (BuizPlanBoost), team formation assistance (HumanBoost), legal assistance (LawBoost), prototype assistance (ProtoBoost), large-scale testing assistance (TestoBoost), and commercialization assistance (SalesBoost).

Implantation city : Abia, Abuja, Accra, Ad-Dahirah, Agadir, Akwa Ibom, Al Wakrah, Alger, Amersfoort, Ar-Rayyan, Ash-Shahaniyah, Baam, Bauchi, Bayelsa, Beni Mellal, Berlin, Bethesda, Borno, Bryanston, CapeTown, Casablanca, Delta, Doha, Driebergen-Rijsenburg, Dubai, Ekiti, El Jadida, Enugu, Essaouira, Fès, Frankfurt, Ifrane, Ikeja, Johannesburg, Kaduna, Kano, Kénitra, Kumasi, Kwara, Lagos, Londres, Lusail, Luxembourg, Marrakech, Marseille, Meknès, Nairobi, Nassarawa, New York City, Ogun, Ogundana, Ondo, Osun, Ouagadougou, Oujda, Oyo, Paris, Plateau, Pretoria, Rabat, Rivers, Rome, Seattle, Settat, Tamale, Tanger, Temara, Thiais, Dakar, Mwanza, Arusha, Dar es Salaam, Tunis, Sfax, Sousse, Ariana, Douala, Sandton, Athlone, Cobham, Mohammadia, Bobo Dioulasso, Abidjan, kampala, Masindi, Mukono, Kigali, Lomé, Kpalimè, yaoundé, Buea, Stellenbosch, Durban, Sandton, Volta, Berekuso, Cotonou, Kempton, Fransville, Arabie, Tshawne, Alton, Oslo, Nelspruit, Mpumalanga, Port Elizabeth, Midrand, Bohicon, Parakou, Abomey-Calavi, Lyon.

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The sum of this data constitutes the most comprehensive and complete data set for Africa. We then developed a specific software to collect all feasible information and observations for each country, which we then deduplicated and recorded. Using name co-occurrence approaches, we identified all forms of collaboration and partnership among the actors in these ecosystems, using a technique known as association dyadic identification. These data are graph matrices that can be used to identify networks. We can perform network

studies that are powerful and significant when we use the other qualification data mentioned previously.

Furthermore, the data we've gathered allows us to use incredibly sophisticated «Natural Language Processing» tools, as well as «topic modeling.» The application of topic modeling to the Moroccan entrepreneurial ecosystem from 2018 to 2023 provides a number of benefits and significant insights for a comprehensive and nuanced knowledge of this time period. Here are some of the reasons why this method is useful:

Emerging Themes Identification: Topic modeling automatically extracts the important themes covered in textual materials such as press stories, reports, blogs, and so on. This would aid in identifying emergent themes that have piqued the interest of the Moroccan entrepreneurial ecosystem throughout this time period.

Trend analysis: Topic analysis can identify patterns that have influenced the entrepreneurial environment, such as rising company sectors, unique issues faced by entrepreneurs, or policy talks.

Major Event Impact: Topic modeling can assist in understanding how important events such as the global financial crisis, governmental reforms, or other economic developments have affected discussions and concerns within the ecosystem.

Actor perception: Topic analysis can illustrate how various ecosystem participants (entrepreneurs, investors, institutions, media, and so on) have viewed and reacted to changes and opportunities over time.

Evolving Priorities: You may analyze changes in ecosystem priorities and examine how they have influenced entrepreneurs' decisions and activities by examining how subjects have developed from year to year.

Resource targeting: Topic modeling can help you more effectively target resources, support programs, and policies based on the real needs highlighted in ecosystem talks.

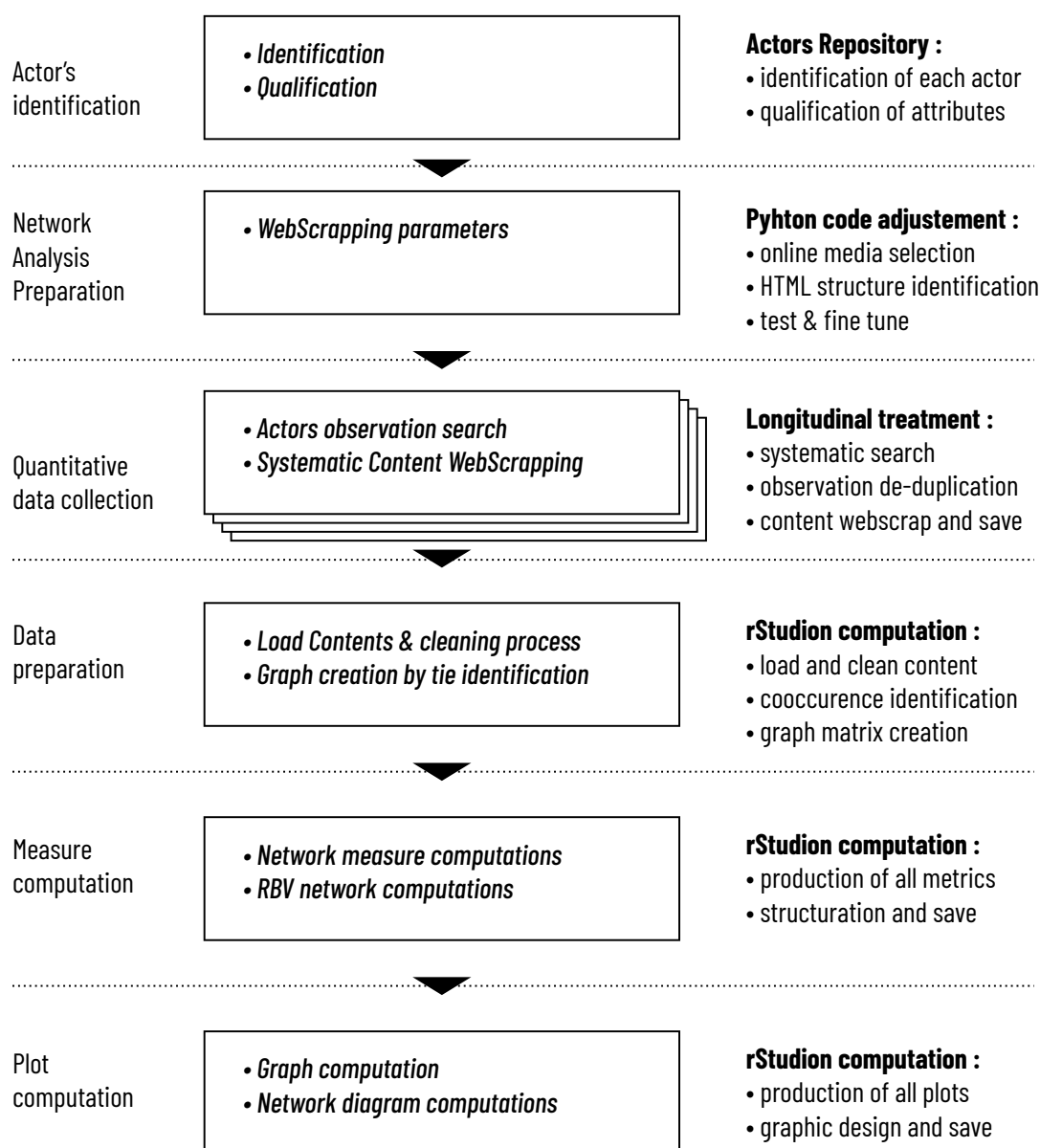
Inspiration for Research: The outcomes of topic modeling can be used as a starting point for additional research into specific themes that emerge, leading to new ideas, discoveries, and analyses.

Impact evaluation: By identifying the most discussed subjects and analyzing their evolution, you may determine the true impact of specific projects, changes, or events on the entrepreneurial environment.

Recognition of Actors: Topic analysis can help to understand how various actors have contributed to conversations and projects within the ecosystem, providing a more nuanced view of their roles and contributions.

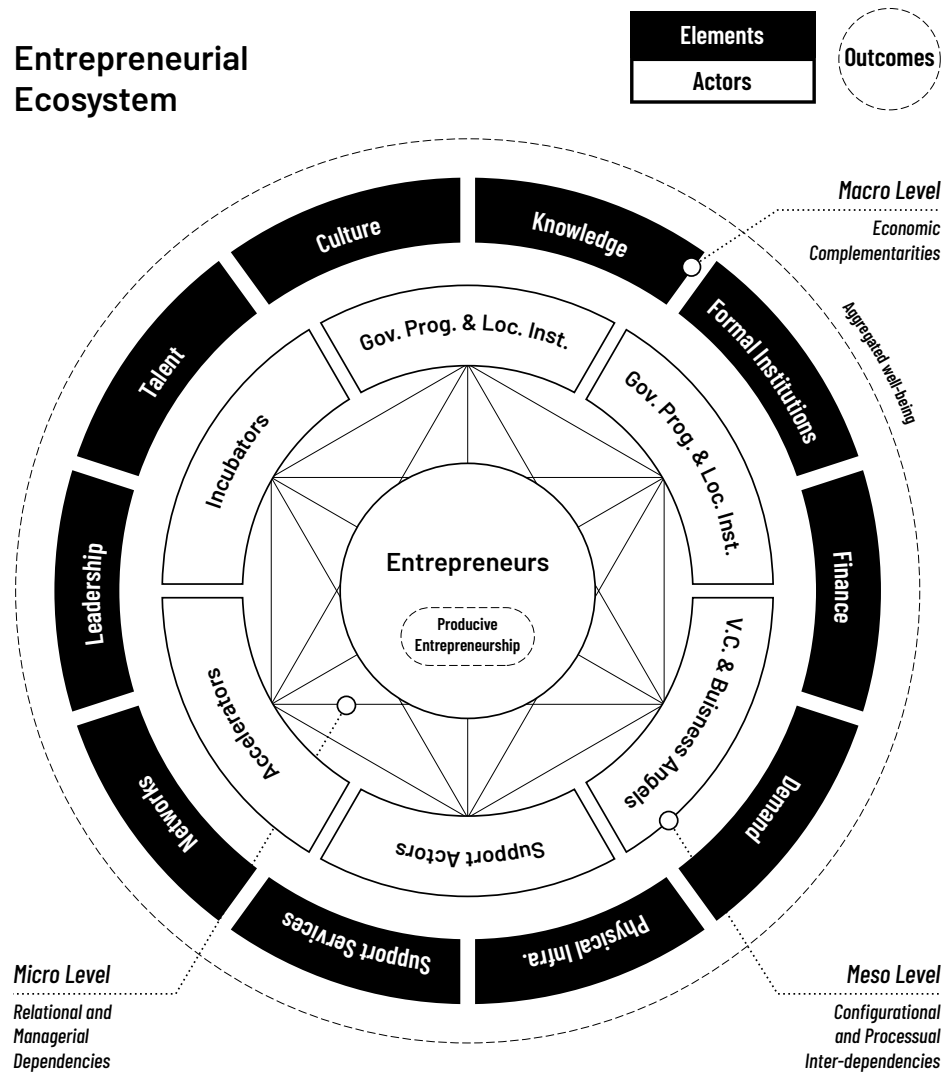
Figure 27. Data Processing Protocol

Entrepreneurial Ecosystem Inter-organizational network analysis protocol



The combination of these data allows us to implement our general conceptual model, which is depicted below. Organizations such as the «Global Entrepreneurship Monitor» can conduct a «Macro-Level» analysis of the 10 economic components of entrepreneurial ecosystems. This very relevant analysis has the disadvantage of causing public policies that are costly and slow to implement, as well as slow to have meaningful results; we're talking about a time scale of 15 to 20 years.

Figure 28. General Conceptual Model



The Observatory of Support for Entrepreneurship (OSE) Model focuses on a «Meso-Level» analysis that allows for a configurational and processual examination of the organizational interdependence of entrepreneurs' actors.

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